CEBUTECH-SPACE: A WEB-BASED SYSTEM FOR FACULTY CLASS SCHEDULING IN CEBU TECHNOLOGICAL UNIVERSITY - CONSOLACION CAMPUS

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ABSTRACT

The **CebuTech Space: A Web-Based Faculty Class Scheduling System** is a web-based application developed to automate and streamline the class scheduling process at Cebu Technological University – Consolacion Campus. The system is designed to address issues associated with manual scheduling, such as time conflicts, overlapping room assignments, unbalanced faculty workloads, and delayed dissemination of schedule information.

CebuTech Space supports six (6) primary user roles: Administrator, Program Head, Faculty, Student, Dean, and Campus Director. The Administrator manages institutional data, including colleges, programs, subjects, rooms, and user accounts. The Program Head is responsible for creating, managing, and submitting class schedules while ensuring conflict-free assignments and balanced faculty workloads. Faculty members can view, accept, or respond to assigned schedules, while Students are provided with real-time access to their class timetables. The Dean and Campus Director oversee and review finalized schedules, validate faculty loads, and approve official academic reports through the system.

The system provides features such as automated conflict detection, centralized schedule management, real-time notifications, and report generation to support informed decision-making. It is developed using modern web technologies including HTML5, CSS3, JavaScript, ReactJS, and Node.js, with a secure database for data management. Development tools such as Visual Studio Code, Git, and GitHub were utilized for collaboration and version control.

The proponents conclude that the implementation of CebuTech Space significantly improves scheduling efficiency, reduces administrative workload, enhances communication among stakeholders, and supports a more organized and transparent academic environment. Overall, the system contributes to the institution's digital transformation and long-term operational sustainability.

Keywords: *faculty scheduling, web-based system, academic management, classroom scheduling, innovatio*

DEDICATION

researchers would like to sincerely thank everyone who helped and supported us in completing this capstone project, “CebuTech-Space: A Web-Based System for Faculty Class Scheduling in Cebu Technological University – Consolacion Campus.”

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CHAPTER I

INTRODUCTION

Rationale of the Study

The efficient management of faculty class schedules plays a crucial role in the seamless operation of any academic institution. At Cebu Technological University – Consolacion Campus, the current reliance on manual scheduling methods has led to various recurring challenges, including class conflicts, and difficulties in adapting to schedule changes. These issues not only place a heavy burden on administrative personnel but also disrupt academic continuity and negatively affect both faculty and student experiences. Recognizing the need for a more efficient and accurate system, this capstone project presents the development of A Web-Based Faculty Class Scheduling System specifically designed for the needs of Cebu Technological University - Consolacion Campus.

The implementation of this system is specifically designed to support the administrative oversight of the university leadership. For the Campus Director, the system serves as a strategic tool to ensure institutional operational efficiency and provide a secure, digitized infrastructure for campus-wide resource management. Simultaneously, the system assists the Dean of the College of Management and Entrepreneurship by providing a high-level monitoring platform to oversee academic program workflows, ensuring that faculty loads are balanced and institutional policies regarding room and time allocations are strictly maintained.

The primary objective of the system is to transform the current labor-intensive process into an automated, streamlined, and easily accessible platform that can generate conflict-free schedules and assign faculty members based on their qualifications, availability, and preferred time slots. Leveraging modern web technologies, the system will allow real-time updates, provide centralized access to up-to-date schedules, and enable dynamic adjustments in response to changes such as faculty leave or classroom unavailability. Additionally, it will offer features such as automated conflict detection, and load balancing

tools to support better decision-making and resource management. Stakeholders—including program heads, faculty, and students—will benefit from improved coordination, greater transparency, and enhanced accessibility to schedules through an intuitive user interface.

Ultimately, this project aims to contribute to the university's broader efforts toward digital transformation by promoting operational efficiency, improving communication among stakeholders, and supporting the delivery of a more organized and responsive academic environment.

General Problem

Many educational institutions, including Cebu Technological University –Consolacion Campus, continue to rely on traditional, manual methods for faculty class scheduling. This outdated approach often leads to scheduling conflicts, inefficient utilization of faculty resources, and time-consuming manual adjustments. These issues hinder the smooth operation of academic activities, reduce administrative productivity, and compromise the overall quality of education delivery

General Objective

This project aims to develop A Web-Based Faculty Class Scheduling System designed to automate the scheduling process, reduce conflicts, optimize the assignment of faculty workloads, and improve accessibility to schedule-related information. The system will serve as a centralized platform for schedule management, enabling faster decision-making and better coordination among stakeholders.

Specific Problems

1. The process of creating and managing conflict-free class schedules is time-consuming. This includes manual challenges in setting class times, assigning classrooms, and avoiding overlaps in both rooms and instructor schedules, often resulting in inefficient timetables and last-minute adjustments.

2. Challenges in effectively assigning instructors to classes. Balancing teaching loads, considering faculty availability, and arranging substitutes or rescheduling sessions when necessary are often tedious and error-prone when done manually.

3. Lack of quick and reliable access for instructors to their assigned schedules. Faculty members often experience delays or confusion in retrieving accurate schedule information, leading to miscommunication and reduced teaching preparedness.

4. Faculty members and students currently lack a quick and reliable way to access and receive notifications about schedule changes, which can lead to miscommunication and a lack of preparedness. The manual process of communicating these changes is inefficient and may cause confusion.

5. The manual method of scheduling does not provide an easy way for program head to manage and balance faculty teaching loads, making it difficult to ensure equitable distribution of work. This can result in some faculty being overloaded while others are underutilized, and it is a tedious and error-prone process when done manually

Specific Objectives

1. To develop a scheduling system that enables program heads to efficiently create, manage, and update class schedules. The system will include automated conflict- checking, visual timetable design, and flexibility for real-time adjustments.

2. To design a faculty assignment system that assists administrators in allocating instructors based on their qualifications, teaching load, availability, and subject expertise. The system will support automated suggestions and workload tracking for better distribution.

3. To implement a user-friendly interface that allows instructors to securely and conveniently access their updated class schedules, including class times, subject codes, and room assignments, using laptop and desktop computer devices with internet access.

4. To implement a real-time notification system that automatically alerts instructors and students about schedule updates and changes, ensuring all stakeholders are informed promptly and accurately.

5. To design a faculty assignment system that assists program chairs/coordinators in allocating instructors based on their qualifications, teaching load, availability, and subject expertise. The system will include tools for workload tracking to support better distribution and will also be able to suggest automated assignments.

Scope and Limitation Scope

This project focuses on the development of CebuTech – Space: A Web-Based Faculty Class Scheduling System for Cebu Technological University – Consolacion Campus. The system will streamline the creation, assignment, and distribution of class schedules. It is designed primarily for Program Heads (for managing schedules and assigning instructors), Faculty (for accessing their assigned schedules), and Students (for viewing class timetables).

Key features of the system include:

- Conflict-free timetable creation, with automatic detection of scheduling conflicts such as overlapping instructor assignments or double-booked classrooms.
- Instructor workload management, allowing program heads to monitor and balance teaching loads based on course assignments and availability.
- Classroom resource management, including the assignment of rooms according to class size and room capacity.
- Real-time schedule updates and notifications, ensuring that all users are informed of changes as they occur. The system will also provide an option for schedules to be saved as a draft or finalized for official upload.
- Web-based accessibility, allowing users to access the system via web browsers on desktop computers. The system is not open to mobile access.

The system is intended to enhance scheduling efficiency, reduce manual errors, and provide easy access to up-to-date schedule information for all stakeholders.

Limitations

While the system addresses core scheduling needs, it will have the following limitations:

- It will not support payroll processing, student enrollment, or attendance tracking functions.
- Integration with other institutional systems such as ERP (Enterprise Resource Planning) or SIS (Student Information System) is outside the scope of this capstone project.
- The system does not support scheduling for irregular students; due to the unique and often non-standard course combinations they take.
- Advanced features such as AI-driven optimization for auto-scheduling, mobile application access, and multi-campus integration are considered future enhancements and are not included in the initial version.
- Testing and implementation will be limited to Cebu Technological University – Consolacion Campus during the capstone phase.
- The system requires a stable internet connection and will only be accessible through online platforms using desktop browsers; offline access or mobile support will not be available in the initial release.

Significance of the Study

- **Program Head:** The system will streamline the complex process of creating and managing class schedules, reducing the time and effort required to design timetables and to avoid conflicts. It will provide tools for efficient instructor assignment, ensuring balanced workloads and accommodating availability, thereby improving overall academic planning.

- **Faculty:** They will benefit from immediate and easy access to their schedules, including class timings, course details, and assigned rooms, minimizing confusion and enhancing productivity. The system will facilitate timely communication of schedule changes, allowing instructors to adjust their plans accordingly.

- **Students:** The system will provide students with immediate and user-friendly access to class schedules, enabling them to effectively plan their studies and manage their time, ensuring they can attend all necessary classes for their academic progress.

- **Administrator:** Serves as the primary authority over the system's foundational data. The system provides a centralized management module that allows for the efficient registration of academic programs, colleges, and user accounts. By automating administrative oversight, the system ensures data integrity and security across the entire campus infrastructure.

- **College Dean**: The system streamlines the review and certification of faculty workloads and section schedules, ensuring all program metrics like units and preparations are verified accurately before approval

- **Campus Director**: It provides high-level oversight and final authorization of campus resources, facilitating a secure digital audit trail for the official approval of all academic schedules

Flow of the Study

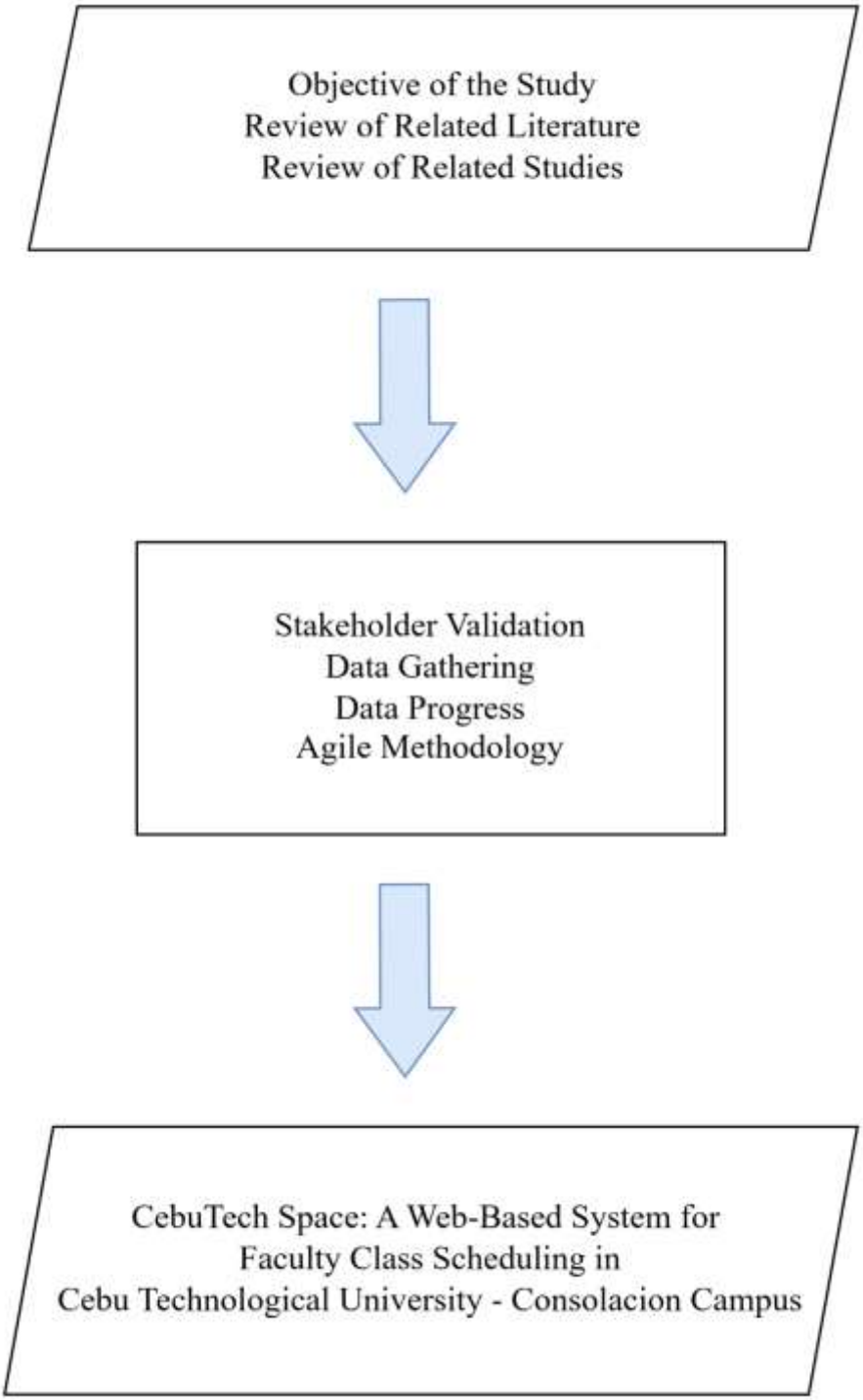

Figure 1: **Flow of the Study**

Definition of Terms

The terms are conceptually and operationally defined for a better understanding of the readers

- **Timetable** – a schedule that shows the planned times at which events, tasks, or activities are intended to happen.
- **Web-Based** – refers to any application, service, or resource that runs on a web browser over the internet rather than being installed locally on a computer.
- **Agile Methodology** – a way of managing and developing projects, especially software, in an iterative, flexible, and collaborative manner.
- **Overlapping** – when two or more things cover, extend over, or coincide with each other partially or completely.
- **Streamline** – to design or organize something in a way that makes it more efficient, smooth, and simple.
- **Susceptibility** – the state or quality of being easily affected, influenced, or harmed by something.
- **Time fold** – a software platform/AI-powered planning engine that optimizes complex scheduling and routing problems (like employee shift scheduling, vehicle routing, maintenance, etc.) using constraint solving.
- **CebuTech Space** – is a web-based faculty class scheduling system designed to automate and streamline the creation of conflict-free academic timetables for Cebu Technological University - Consolacion Campus.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Review Related Literature

The Review of Related Literature and Studies (RRL) establishes the critical need for a new system by confirming that the current reliance on manual scheduling at Cebu Technological University – Consolacion Campus creates common, industry-recognized problems like scheduling conflicts and unbalanced faculty loads ; consequently, this chapter analyzes technological solutions found in Local Literature—specifically the Decision Support Framework by Ortega et al. (2015), alongside the National Literature models such as the Integer Linear Programming approach by Cruz & DeLos Reyes (2023) and the scheduling system by Botangen & Khan (2019) —as well as international studies leveraging advanced search methods (like Evolutionary Algorithms by Diallo & Tudose, 2024) and the visual-interface approach of Bushman (2021) , to inform the development of a web-based, automated system that prioritizes conflict resolution, efficient resource allocation, and real-time information access

Local Literature

Ortega et al. (2015) conducted a study on an "Online Class Scheduling and Faculty Loading System within a Decision Support Framework." This research is particularly relevant as it directly addresses the development of a system to aid in the complexities of class scheduling and faculty management, which aligns with the goals of the proposed web-based system. The study emphasizes the importance of creating tools to support academic administrators in making informed decisions regarding scheduling, demonstrating the need for and value of such systems within the local context.

National Literature

Querubin and Contillo (2024) investigated optimizing class scheduling specifically for the University of Northern Philippines through the application of a Greedy Algorithm. This is a computer science approach where the algorithm makes the best choice at each step to find a good, though not always perfect, solution. In scheduling, a greedy algorithm might

assign classes to rooms one by one, always picking the best available room at that moment. Their research highlights the potential of using algorithmic solutions to enhance the efficiency and accuracy of class scheduling, particularly in addressing the complexities and constraints inherent in a university environment. This work underscores how algorithms can be effective in finding optimal solutions for resource allocation within academic scheduling.

Botangen and Khan (2019) developed a class-scheduling system tailored for Central Luzon State University (CLSU) to address persistent issues in manual scheduling processes, such as conflicts in room assignments, overlapping time slots, and uneven distribution of faculty loads. The proposed system employed computerized scheduling to minimize human error and reduce time consumption. The study highlighted the challenges faced by academic institutions when relying on traditional scheduling methods, particularly the lack of efficiency and susceptibility to inconsistencies. By using algorithm-based logic, the system aimed to optimize class and room allocations while taking into account various constraints such as course requirements, faculty preferences, and room capacities. The implementation demonstrated significant improvements in generating conflict-free schedules and streamlining administrative tasks. This research supports the value of adopting technology-based scheduling systems in higher education to improve academic planning and operational efficiency.

Cruz and Reyes (2023) developed an integer linear programming (ILP) model aimed at optimizing the class scheduling process for the College of Science at De La Salle University. Their research focused on creating a system that assigns classes to appropriate time slots, instructors, and classrooms while satisfying various constraints such as instructor availability, room capacity, and the need to avoid scheduling conflicts. Using real institutional data—including faculty schedules, classroom availability, and student course preferences—the model was designed to minimize overlaps and maximize efficiency in resource usage. Hard constraints ensured the feasibility of the schedules (e.g., no double-booking of instructors or rooms), while soft constraints addressed preferences, such as accommodating student course selections. The ILP formulation allowed for a structured and automated approach to scheduling, with the potential to be solved using optimization tools like CPLEX or Gurobi. The results showed that the model could significantly reduce manual effort,

prevent common scheduling issues, and serve as a scalable framework for other departments or institutions seeking to improve their academic scheduling systems.

International Literature

The study by Francis Patrick Diallo and Cătălin Tudose (2024), published in Applied Sciences, tackle the problem of efficiently organizing teaching schedules in a university setting. They present an automated scheduling system that applies evolutionary algorithms and the Time fold constraint solver to create effective timetables. This system takes into account various factors such as room capacity, faculty availability, student course choices, and institutional policies. By using a multi-objective optimization approach, it seeks to balance different goals— like reducing scheduling conflicts and making the best use of available resources. The system proved to be effective, leading to fewer conflicts, better allocation of rooms and instructor time, and increased satisfaction among both faculty and students. With a flexible, scalable design and an intuitive interface, the tool allows administrators to easily manage and fine-tune schedules. Overall, the study highlights how AI and optimization techniques can significantly improve the academic scheduling process.

Diallo and Tudose (2024) developed an automated system aimed at improving the scheduling of teaching activities within academic institutions. Using evolutionary algorithms combined with multi-objective optimization techniques, the system effectively manages various constraints, including classroom capacity, instructor availability, and student needs. The researchers designed a complete scheduling framework—from mathematical modeling to algorithm development and performance testing. Their results showed that the system could generate efficient, conflict-free schedules while maximizing the use of available resources. Additionally, the model proved to be flexible and scalable, making it suitable for different types of educational institutions and adaptable for future enhancements.

Khorbotly and White (2024) introduced a faculty assignment system that improves the efficiency of course scheduling by prioritizing instructor preferences and qualifications. The tool uses linear programming to match faculty members to available course sections while ensuring that each class is staffed and teaching loads are not exceeded. By focusing solely on faculty assignments—separate from room and time scheduling—the model

simplifies a complex part of the scheduling process. It provides an objective and transparent method that helps minimize bias and increases fairness in assignments. Delivered as a user-friendly web application, the tool also highlights areas where faculty expertise may be lacking, supporting better staffing decisions in the future. This system ultimately streamlines scheduling tasks while promoting faculty satisfaction.

Chen et al. (2022) introduced an innovative method for classroom scheduling that integrates mixed-integer programming (MIP) with a genetic algorithm (GA) to produce effective and preference-sensitive timetables. Their model takes into account strict requirements such as room availability, teacher schedules, and class capacity, alongside more flexible factors like student and teacher preferences, minimizing gaps between classes, and ensuring evenly distributed schedules. The GA approach uses a sequence-based structure where each element represents a course assigned to a specific time slot, and applies selection, crossover, and mutation techniques to improve the schedule. Their experiments, conducted on a small-scale dataset, showed that the system successfully generated conflict-free schedules with maximum efficiency. The results highlight how even a basic GA can effectively handle the complexity of classroom scheduling while accommodating stakeholder preferences. This approach offers a useful solution for schools and universities aiming to streamline their scheduling processes through automation.

Bushman (2021) introduced a flexible scheduling tool that combines visual interaction with integer linear programming (ILP) to improve how university class schedules are created. This web based system allows users to clearly view class assignments by time, room, and instructor, while automatically identifying conflicts such as overlapping schedules or double-booked classrooms. A notable aspect of the tool is its ability to incorporate instructor preferences—such as preferred time slots and courses—which are factored into the ILP model to produce optimized schedules. The system enables both manual adjustments and automatic scheduling, giving users the ability to explore different options by updating preferences and re-running the optimization. When tested with real scheduling data from the University of Pittsburgh's Computer Science department, the tool showed improvements in reducing conflicts, maximizing classroom use, and distributing teaching loads more fairly. Overall, Bushman's study highlights how integrating optimization models with visual tools

can make the class scheduling process more efficient, user-friendly, and responsive to faculty needs.

The reviewed literature consistently indicates that automated scheduling systems enhance accuracy, reduce conflicts, and streamline administrative tasks within academic institutions. Ortega et al. (2015) demonstrated that online scheduling improves decision-making and decreases manual workload, while Botangen and Khan (2019) highlighted its effectiveness in minimizing room conflicts and instructor overlaps. Studies utilizing optimization techniques—such as Greedy Algorithms (Querubin & Contillo, 2024) and Integer Linear Programming (Cruz & Reyes, 2023)—further validate that algorithmic approaches generate reliable, conflict-free timetables and support efficient resource allocation. International research, including Diallo and Tudose (2024), Chen et al. (2022), and Bushman (2021), likewise emphasizes that combining optimization with intuitive interfaces enhances flexibility, reduces errors, and improves stakeholder satisfaction. Collectively, these findings justify the integration of automated conflict detection, load balancing, and effective visualization into the development of the Cebu Technological University– Consolacion Faculty Scheduling System.

Despite these shared objectives, meaningful distinctions exist between earlier studies and the system currently being developed. Several works—such as Cruz and Reyes (2023) and Chen et al. (2022)—prioritize mathematically intensive optimization models, while others, such as Khorbotly and White (2024), focus primarily on faculty-preference matching independent of room and time allocation. Large-scale frameworks like those of Diallo and Tudose (2024) are designed for broader university environments. In contrast, the system under development for Cebu Technological University–Consolacion directly addresses localized challenges such as manual schedule preparation, limited real-time updates, delayed dissemination, and difficulties in balancing faculty workloads. Its emphasis on web-based accessibility, real-time notifications, and straightforward conflict-management tools makes it well-aligned with the campus's existing technological resources and operational practices. Consequently, while guided by prior literature, the system is intentionally designed to meet the specific needs and constraints of Cebu Technological University–Consolacion.

Comparative Matrix

A Comparative Matrix is a side-by-side comparison chart that helps us quickly see the similarities and differences between different things. It is a straightforward way to check and contrast information without going into too much detail.

Features	Description	Online Class Scheduling and Faculty Loading System within a Decision Support Framework	Using Integer Linear Programming for University Class Scheduling	The Faculty Assignment Tool	CebuTech-Space
Login & Logout	Secure access for users.	✓	✗	✓	✓
User Management	Manage accounts and roles.	✓	✗	✓	✓
Faculty Management	Manage faculty details and load.	✓	✓	✗	✓
Subject Management	Add and organize subjects.	✗	✓	✓	✓

Section/Class Management	Manage sections and classes.	✕	✓	✓	✓
Room/Location Management	Manage rooms and locations.	✕	✓	✕	✓
Schedule Generation	Create class schedules automatically.	✕	✓	✕	✓
Conflict Detection	Prevent time/room/faculty conflicts.	✓	✓	✓	✓
Schedule Viewing & Reports	View, filter, and print schedules.	✓	✕	✓	✓
Notifications	Send alerts for updates/changes.	✕	✕	✕	✓
TOTAL	100%	50%	60%	60%	100%

Table 1: **Comparative Matrix**

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the systematic approach and technical framework utilized to execute the study effectively. It establishes the procedures for design, development, and evaluation to ensure the system meets all specified project goals. The content includes the Software Engineering Methodology, Research Environment, Planning/Concept Initiation Phase, and the Analysis-Design Phase. It details the Development/Construction Phase, Program Specification, and the Testing/Quality Assurance Phase.

Environment

Figure 2: **Cebu Technological University- Consolacion Campus**

The researchers utilized a Quantitative approach to assess the requirements and challenges of the current manual scheduling process at Cebu Technological University - Consolacion. Data was gathered via a web-based survey using Google Forms from a total of 55 respondents, consisting of 50 students and 5 faculty members. The researchers employed Purposive Sampling to select these participants, ensuring that the data collected was relevant and technically sound. By limiting the respondents to the faculty and students of Cebu Technological University-Consolacion, the study gathered insights from individuals with direct experience regarding manual scheduling challenges, thereby ensuring the proposed system's requirements are based on actual user needs and the primary stakeholders' functionality and effectiveness.

Software Engineering Methodology

Software Engineering Methodology refers to a systematic approach used in the analysis, design, development, testing, and maintenance of software systems. It provides a structured framework that guides the entire software development process, ensuring that the resulting system is functional, efficient, and meets user requirements. There are types and models such as waterfall, lean, agile, and other methodologies. In this study, the research decided to adopt the Agile Methodology for the development of the Web-Based Faculty Class Scheduling System. Agile is a modern software development approach that emphasizes iterative development, flexibility, and continuous feedback. It allows software to be developed in small, incremental modules, which are regularly tested, evaluated, and refined to meet user requirements effectively.

Figure 3: **Agile Methodology Model**

Agile is particularly suitable for this system because a web-based faculty scheduling application requires frequent updates, real-time adjustments, and user-centered features. By using Agile, the system can progressively implement core functionalities such as timetable generation, classroom allocation, and faculty assignments, while allowing changes or enhancements to be incorporated based on feedback from administrators and faculty. This ensures that the system remains accurate, reliable, and aligned with the dynamic scheduling needs of the university.

Planning Phase

Business Model Canvas

The Business Model Canvas is a strategic management tool that provides a visual, one page overview of how a business creates, delivers, and captures value.

<table>
<tr><th>Key Partners</th><th>Key Attributes</th><th>Value Propositions</th><th>Customer Relationships</th><th>Customer Segments</th></tr>
<tr><td rowspan="3">• CTU-Consolacion Campus Administration
• Faculty
• Students</td><td>• Application Development
• Platform Management
• Design and Planning
• User Support and Training</td><td rowspan="3">• Conflict-Free Automated Scheduling
• Efficient Faculty Load Balancing
• Real-time Schedule Access
• Minimized Scheduling Confusion</td><td>• Service-Oriented Support
• Self-Service & Notification</td><td rowspan="3">• Program Heads/Coordinators
• Faculty
• Students</td></tr>
<tr><td>Key Resources</td><td>Channels</td></tr>
<tr><td>• Technical Platform</td><td>• Web Browser Access</td></tr>
<tr><td colspan="2">Cost Structure</td><td colspan="3">Revenue Streams</td></tr>
<tr><td colspan="2">• Development Cost
• Maintenance Cos</td><td colspan="3">• Not Applicable</td></tr>
</table>

Figure 4: **<u>Business Model Canvas</u>**

Program Workflow

Is a systematic representation of the sequence of operations, processes, and data movements that occur within a system to achieve a specific outcome. It serves as a visual or structured guide that outlines how a user interacts with the software and how the system responds at each step.

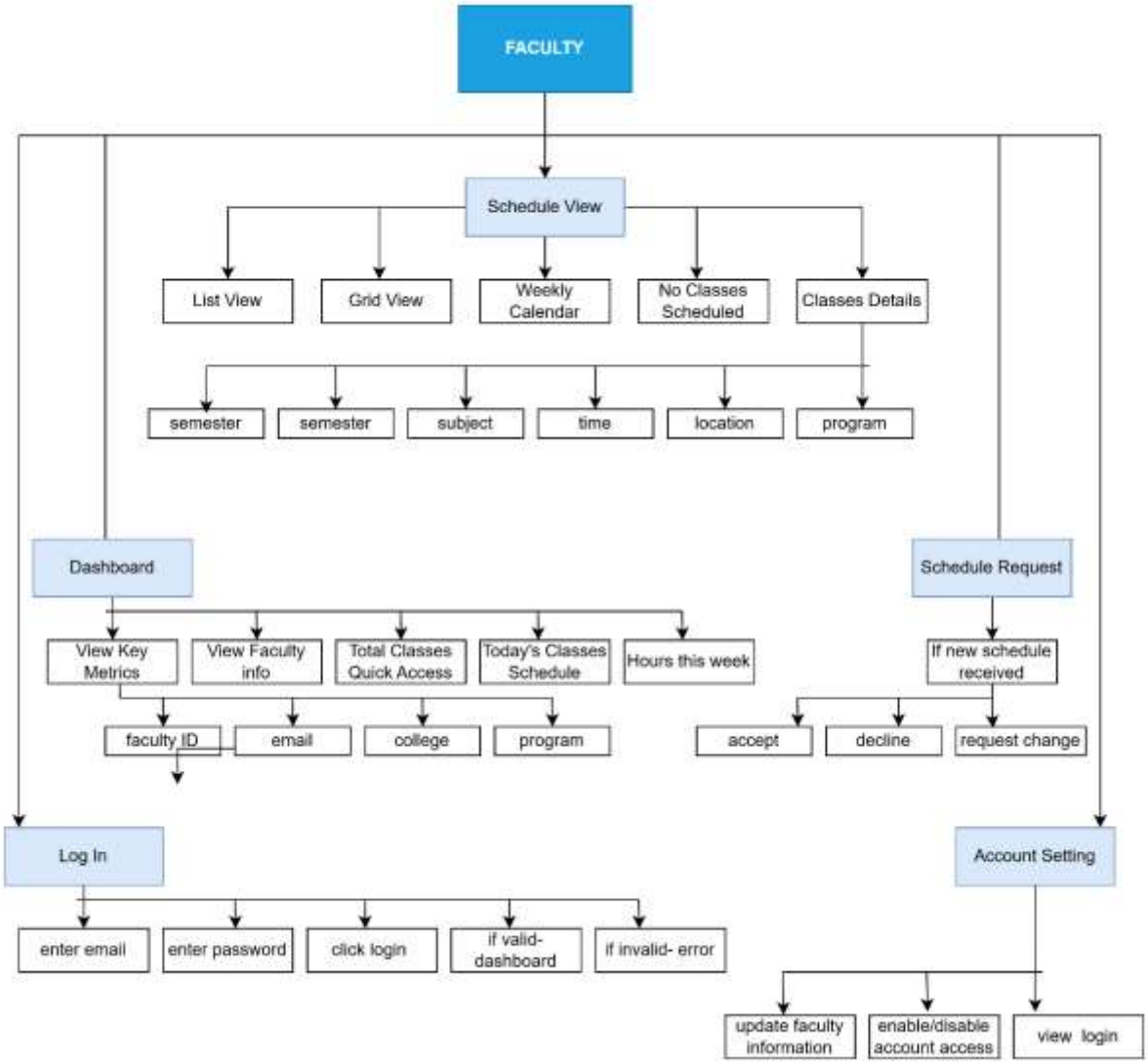

Figure 5: **Program Workflow for Faculty**

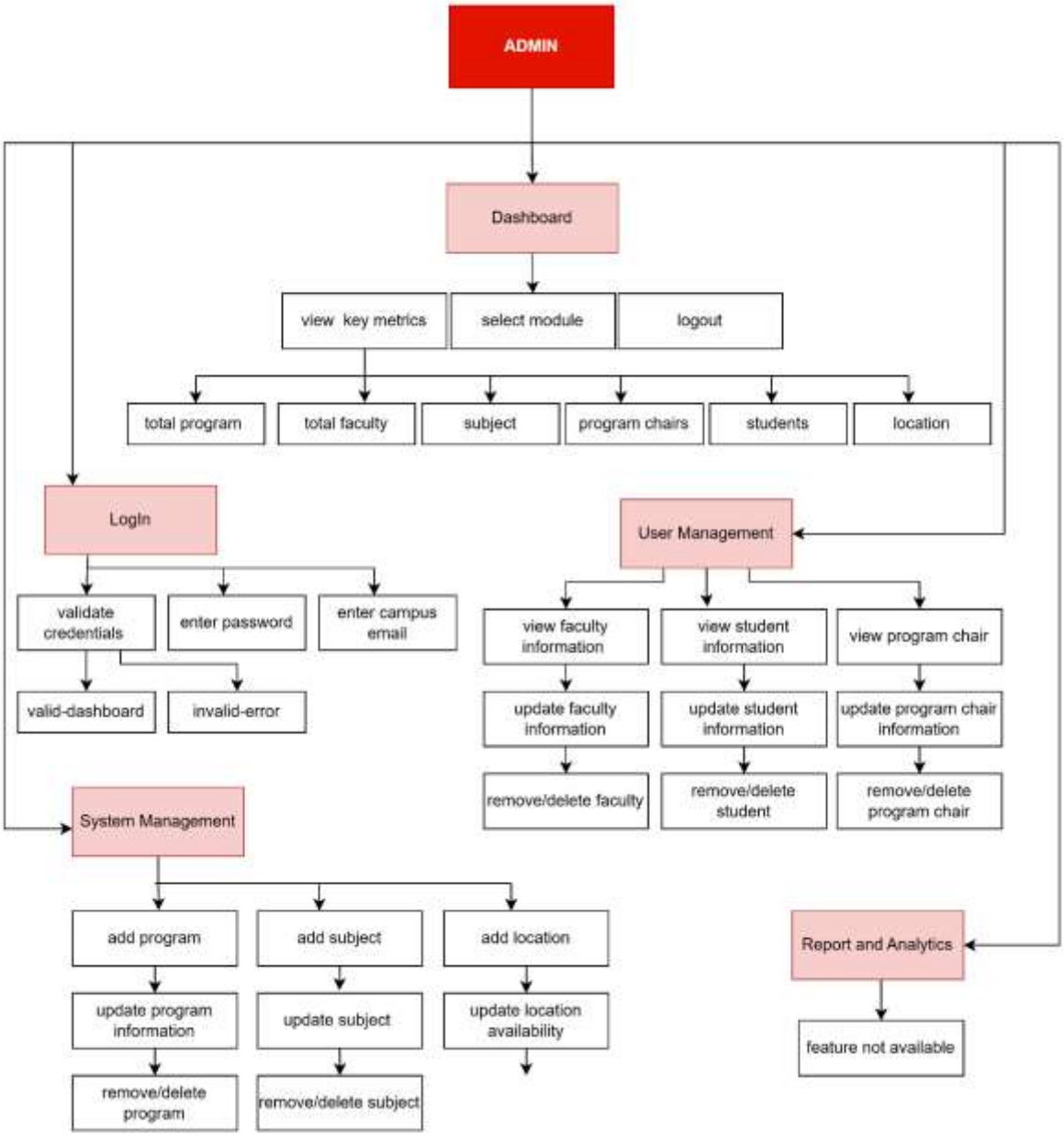

Figure 6: **Program Workflow for Administrator**

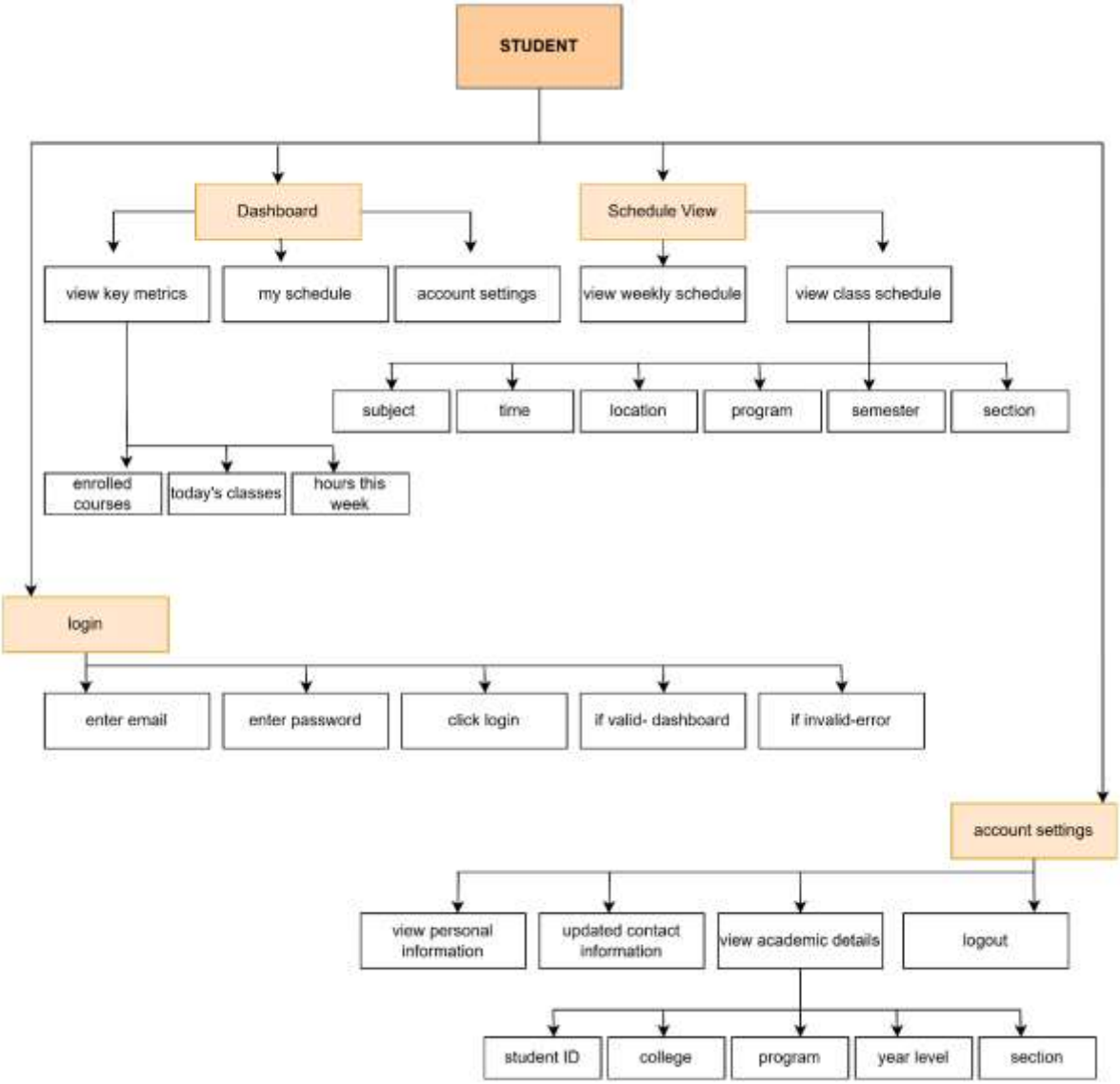

Figure 7: **Program Workflow for Student**

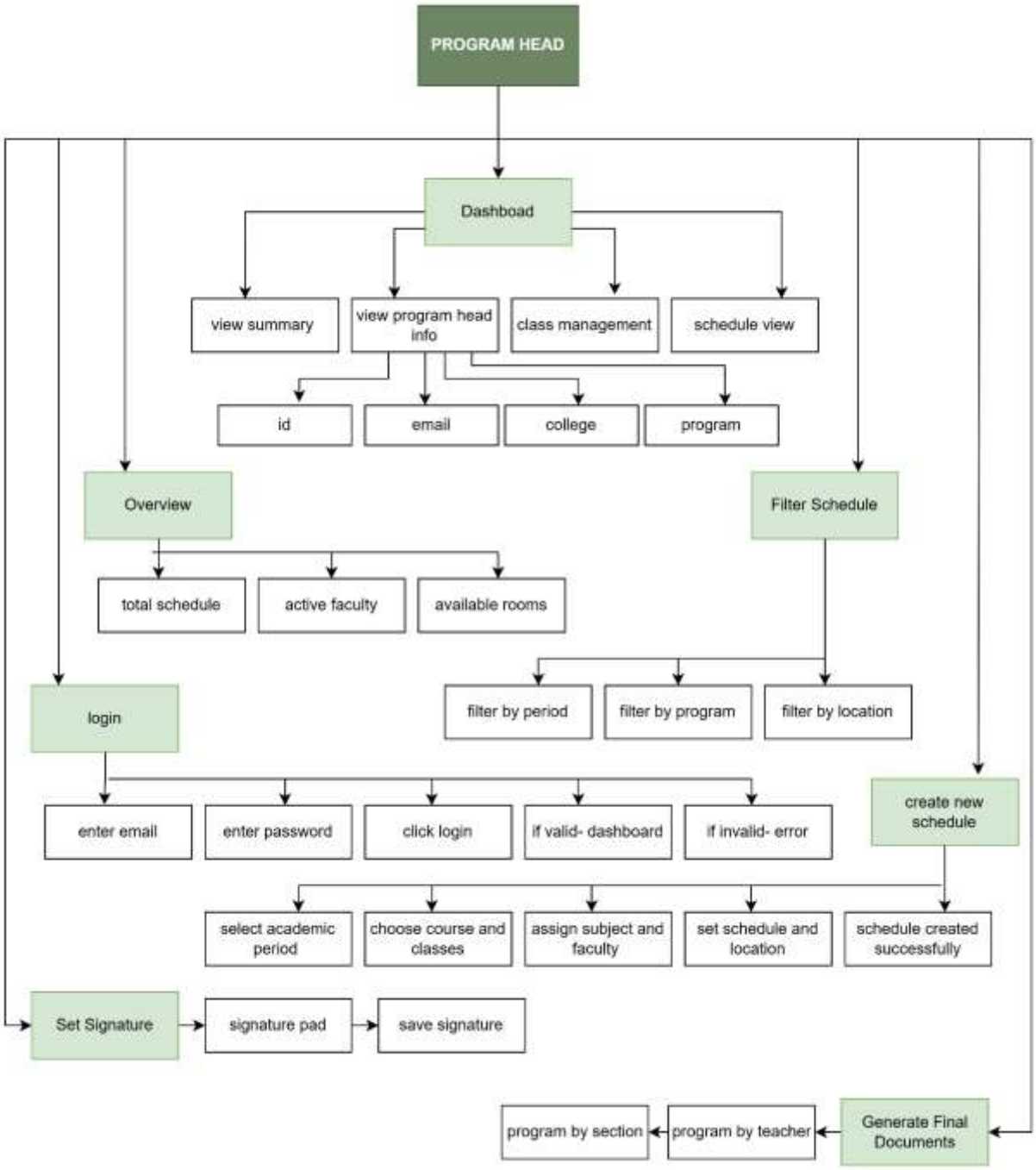

Figure 8: **Program Workflow for Program Head**

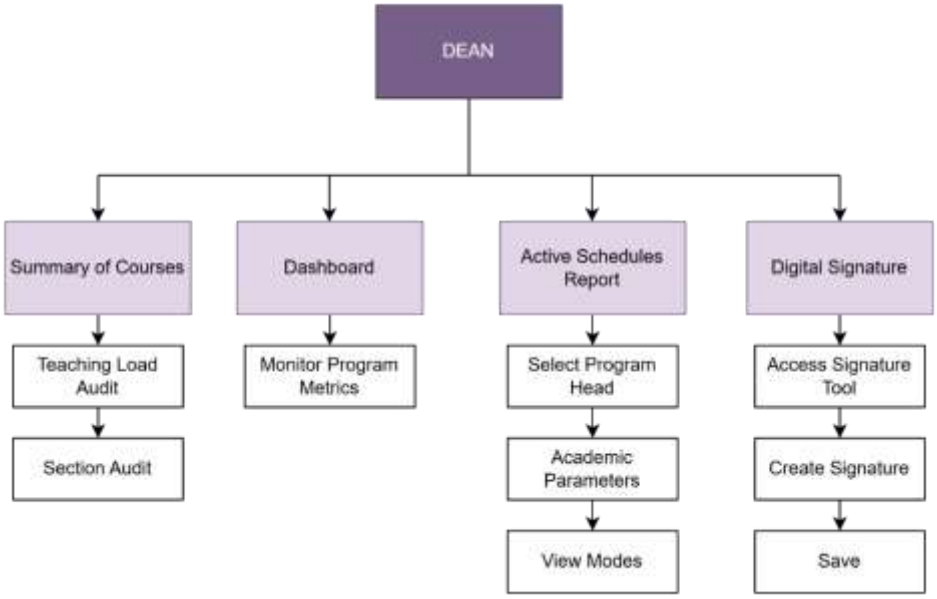

Figure 9: **Program Workflow for Dean**

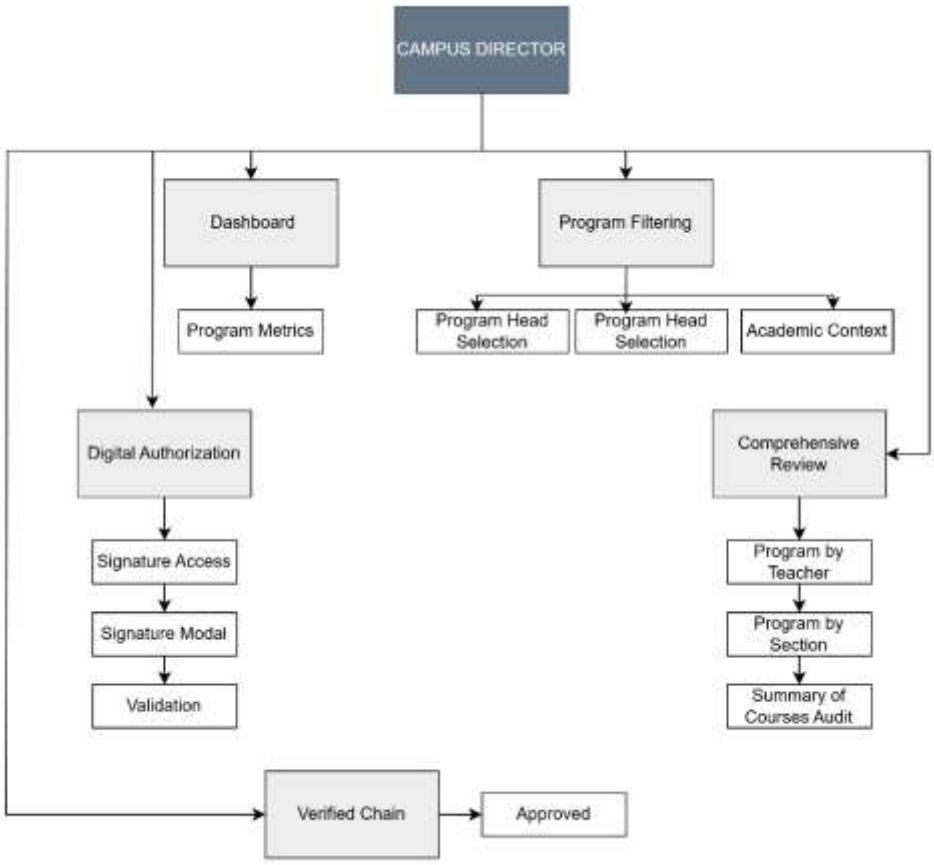

Figure 10: **Program Workflow for Campus Director**

Validation Board

The figure below presents the validation board, which illustrates the identified problems, proposed solutions, assumptions, and success criteria of the study based on stakeholder perspectives. It serves as a guide for the researchers in ensuring that the developed system addresses actual needs and aligns with the objectives of the research.

Stakeholders	**Program Chair/Coordinator (Admin)**	**Instructors/Faculty**
Problem	The manual scheduling process is time-consuming, tedious, and error-prone, leading to scheduling conflicts (room/time overlaps) and difficulty in equitably balancing faculty teaching loads.	Lack of quick, reliable, and centralized access to personal class schedules and delayed or confusing communication regarding last-minute schedule changes.
Solution	Develop a scheduling system with automated conflict-checking and workload tracking tools to efficiently create, manage, and update conflict-free schedules and ensure equitable load distribution.	Implement a user-friendly interface for secure, convenient access to updated schedules on desktop browsers, and a real-time notification system for instant alerts on changes.
Riskiest Assumption	That program heads will readily trust the automated load balancing and conflict-checking outputs and that the system's logic can effectively handle all complex institutional rules and constraints without manual override.	That instructors will consistently use the Web-Based System as the primary source for schedule information and notifications, and that a desktop-only solution meets all urgent communication needs.

Success Criteria	95% reduction in time spent on initial schedule creation and a 100% conflict-free schedule output, validated by program heads.	90% of faculty members report the system is their *primary* and *most accurate* source for schedule information and that all schedule changes are received via the system within 5 minutes of being finalized.
Results	Placeholder for post-implementation data	Placeholder for post-implementation data
Learnings	Need to include a robust audit log and an easy-to-understand visualization of the load-balancing logic to build trust and promote adoption among administrators.	Future iterations must explore the feasibility of a responsive design or a mobile application to ensure urgent schedule updates are accessible anytime, anywhere.

Table 2: **Validation Board**

The researchers observed that the existing manual faculty scheduling process is time-consuming and prone to errors, resulting in scheduling conflicts, unbalanced teaching loads, and delayed dissemination of schedule information. To address these concerns, the researchers proposed a web-based faculty class scheduling system that automates conflict detection, supports equitable workload distribution, and provides real-time access to updated schedules. This solution aims to enhance administrative efficiency and improve coordination among program heads and faculty.

Gantt Chart

Is visual project management tool that shows tasks, their durations, and timelines to help track and manage project progress.

Projected Completed

CebuTech Space					Duration in Weeks											
Capstone 1					March				April				May			
	Task Name	Task Lead	Start Date	End Date	Week 1	Week 2	Week 3	Week 4	Week 1	Week 2	Week 3	Week 4	Week 1	Week 2	Week 3	Week 4
1	Title Hearing	All	3/14/25	3/14/25												
2	Rationale of the Study	Ceniza	3/15/25	3/21/25												
3	Objectives of the Study	Ceniza	3/15/25	3/21/25												
4	Scope and Limitations	Ceniza	3/15/25	3/21/25												
5	Significance of the Study	Ceniza	3/15/25	3/21/25												
6	Flow of the Study	Ceniza	3/22/25	3/26/25												
7	Definition of Terms	Cortes	3/22/25	3/26/25												
8	Related Literatures	Ceniza	3/27/25	4/11/25												
9	Related Studies	Ceniza	3/27/25	4/11/25												
10	Comparative Matrix	Ceniza	3/27/25	4/11/25												
11	Software Engineering Methodology	Ceniza	4/12/25	4/18/25												

12	Business Model Canvas	Ceniza	4/12/25	4/18/25												
13	Program Workflow	Ceniza /Rivera	4/19/25	4/25/25												
14	Gantt Chart	Ceniza /Amosora	4/19/25	4/25/25												
15	Functional Decomposit-ion Diagram	Ceniza	4/26/25	5/9/25												
16	Use case Diagram	Ceniza	4/26/25	5/9/25												
17	Storyboard	Ceniza	4/26/25	5/9/25												

18	Entity Relationship Diagram	Ceniza	5/10/25	5/30/25												
19	Data Dictionary	Ceniza	5/10/25	5/30/25												
20	Network Design	Ceniza /Amo sora	5/10/25	5/30/25												
21	Network Model	Ceniza /Amo sora	5/10/25	5/30/25												
22	Network Topology	Ceniza /Amo sora	5/10/25	5/30/25												
23	Technology Stack diagram	Salac	5/10/25	5/30/25												
24	Software Specification	Salac	5/10/25	5/30/25												
25	Hardware Specification	Salac	5/10/25	5/30/25												
26	Program Specification	Cortes	5/10/25	5/30/25												
27	List of Modules	Cortes	5/10/25	5/30/25												
28	Testing Phase	Cortes	5/10/25	5/30/25												

Table 2: **Gantt Chart (Manuscript)**

Web System

Target Completion			Actual Completion		Duration in Weeks																			
Task #	Modules				Aug				Sep				Oct				Nov				Dec			
		Task Lead	Start Date	End Date	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
	Manage Account/Authentication																							
1	Login	Denziel	10/10/2025	10/12/2025																				
2	Logout	Denziel	10/12/2025	10/12/2025																				
3	Registration	Denziel	10/12/2025	10/12/2025																				
4	Reset Password	Denziel	10/12/2025	10/12/2025																				
5	Account Settings	Denziel	10/12/2025	10/13/2025																				
	Dashboard Module																							
1	Dashboard	Denziel	10/15/2025	10/15/2025																				
	College & Program Management																							
1	College Management	Denziel	10/16/2025	10/16/2025																				
2	Program/Course Management	Denziel	10/16/2025	10/16/2025																				
3	Subject Management	Denziel	10/17/2025	10/17/2025																				
4	Course Subject - Offering	Denziel	10/17/2025	10/19/2025																				
	User Management																							
1	Progrm Head Management	Denziel	10/23/2025	10/25/2025																				
2	Faculty Management	Denziel	10/27/2025	10/30/2025																				
3	Student Management	Denziel	10/30/2025	11/5/2025																				
	Location/Room Management																							
1	Location Management	Denziel	11/3/2025	11/4/2025																				
	Scheduling Module																							
1	Create Schedule	Denziel	11/6/2025	11/6/2025																				
2	View Schedule	Denziel	11/6/2025	11/6/2025																				
3	Update/Edit Schedule	Denziel	11/6/2025	11/8/2025																				
4	Reset Password	Denziel	11/8/2025	11/8/2025																				
5	Remove/Cancel Schedule	Denziel	11/9/2025	11/10/2025																				
	Schedule Approval Workflow																							
1	Submit Schedule for Approval	Denziel	11/6/2025	11/10/2025																				
2	Approve/Disapprove Schedule	Denziel	11/7/2025	11/9/2025																				
3	Track Approval Status	Denziel	11/9/2025	11/9/2025																				
	Notification Module																							
1	View Notification	Denziel	11/10/2025	11/12/2025																				
2	Schedule Related Notification	Denziel	11/12/2025	11/14/2025																				

Table 3**: Gantt Chart (Web System)**

Functional Decomposition Diagram

Is a visual representation that breaks down a system into its major functions and sub-functions to clearly show how the overall system operates.

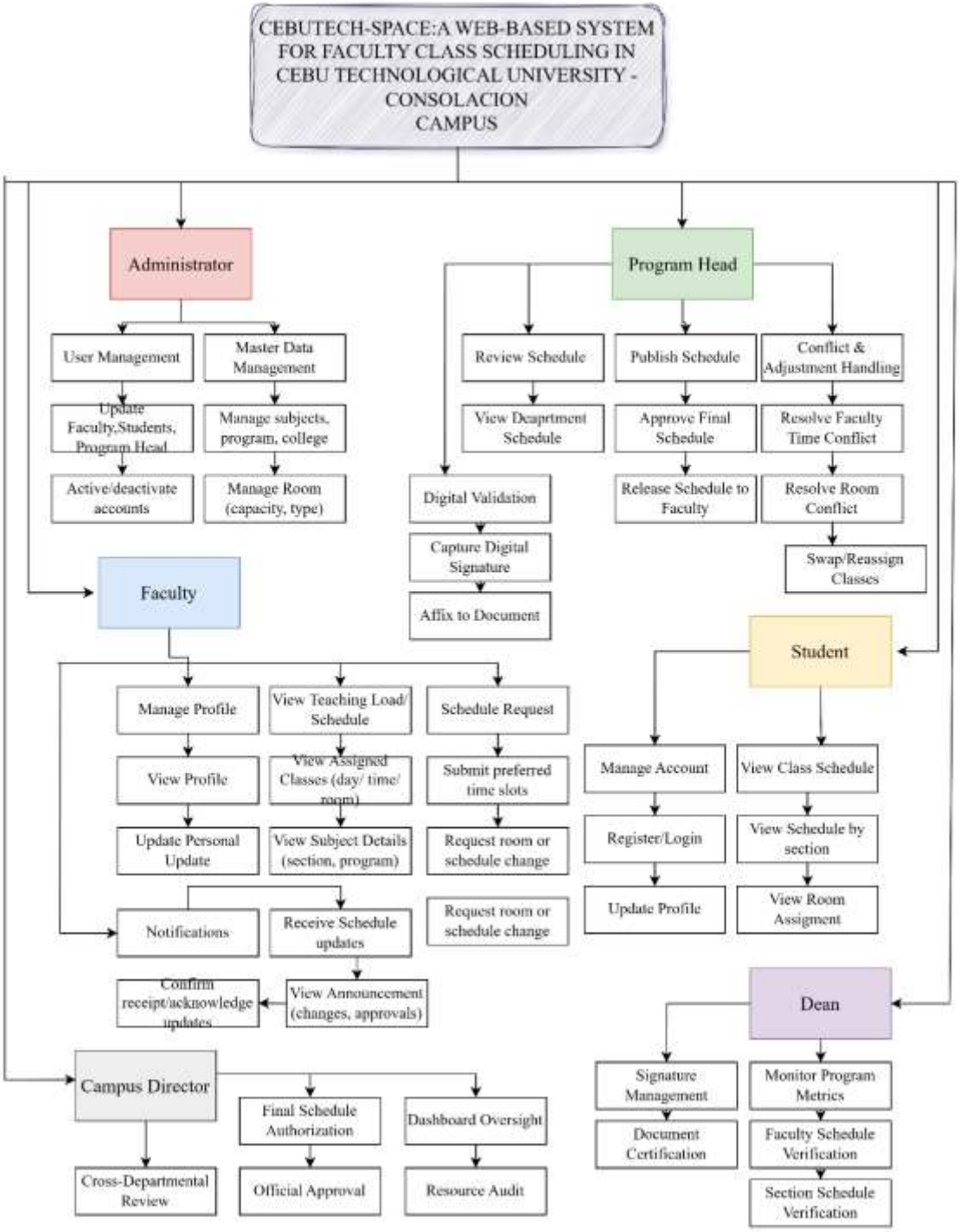

Figure 11**: Functional Decomposition Diagram**

Analysis Design Phase

The Analysis–Design Phase is a stage in system development where requirements are analyzed and transformed into detailed system designs, defining how the system will function and be structured.

Use Case Diagram

A use case diagram illustrates the interactions between the system and its users—Administrator, Program Head, Faculty, Student, Dean, and Campus Director—by showing the specific functions each user can perform within the system.

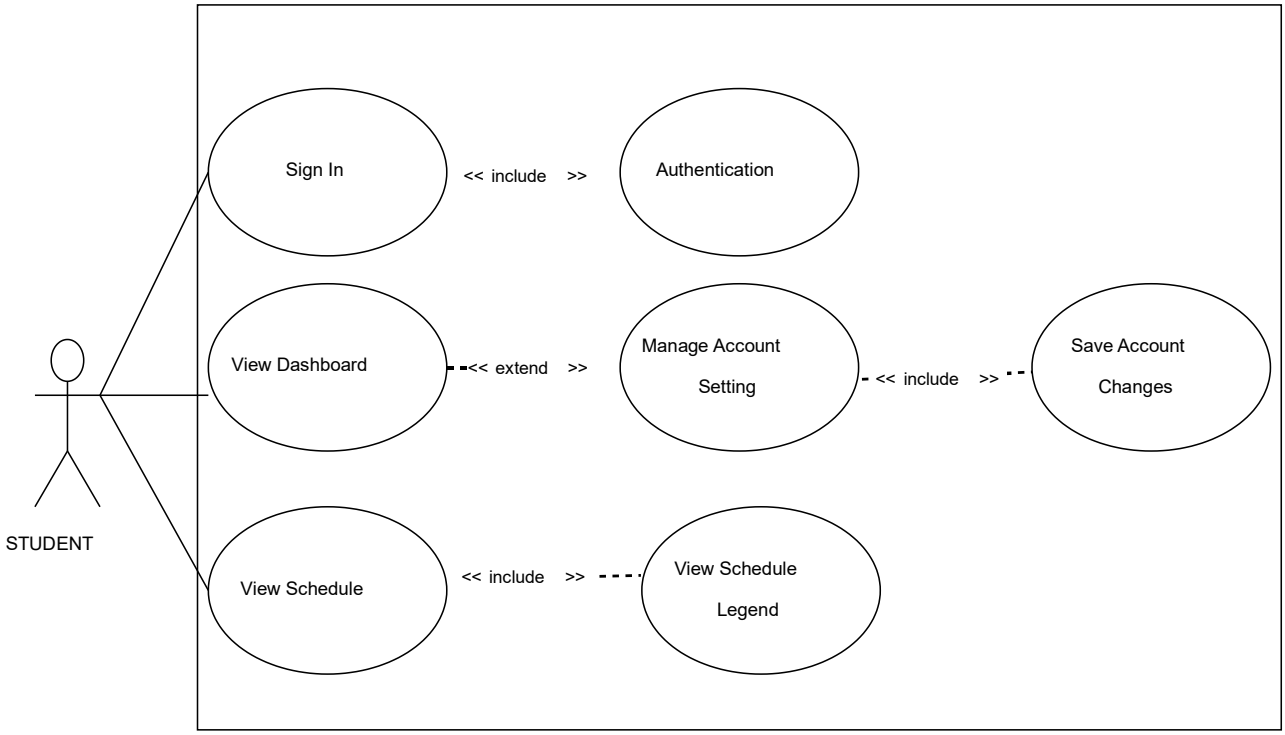

Figure 12: **<u>Use Case Diagram for Student</u>**

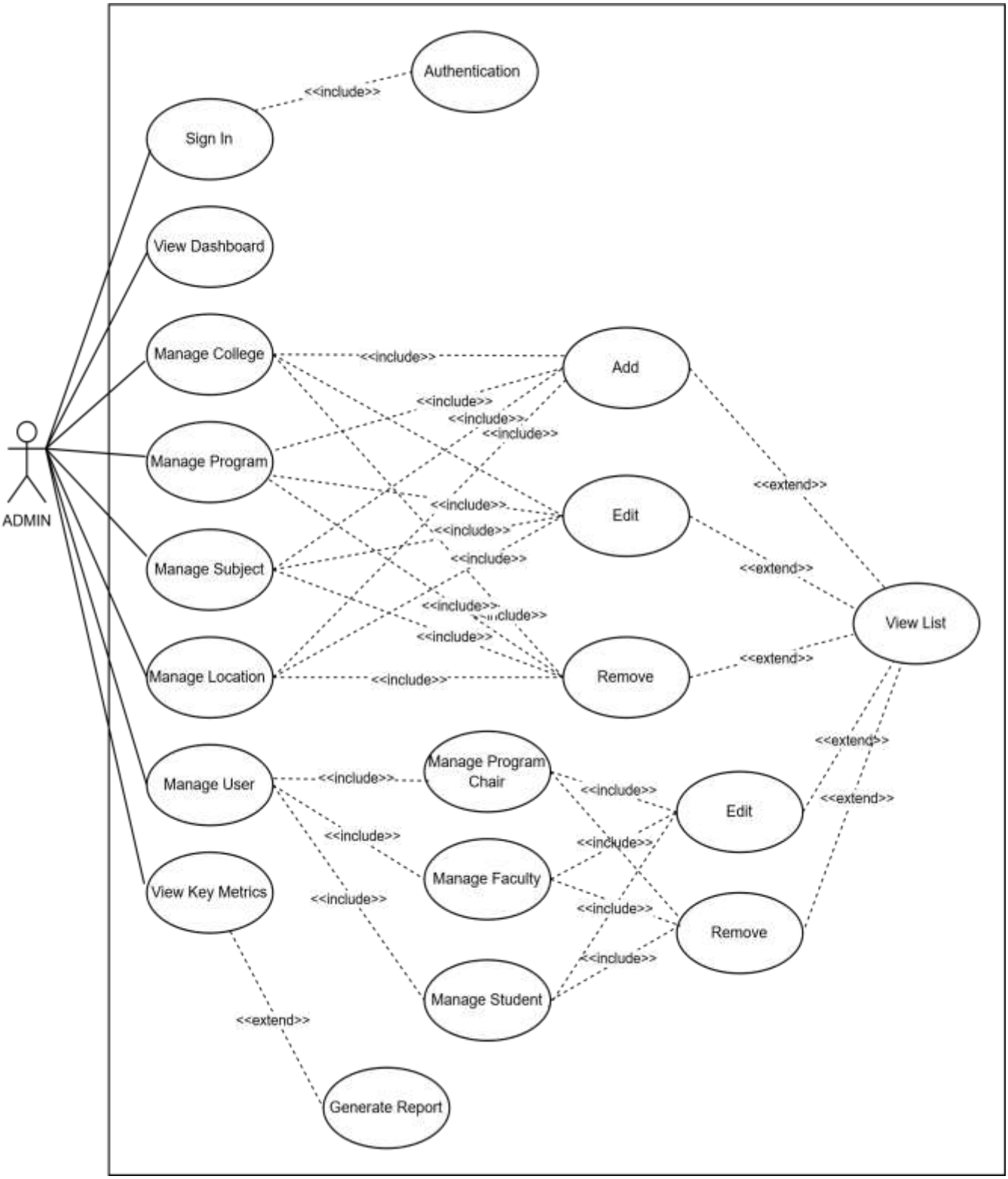

Figure 13: **Use Case Diagram for Administrator**

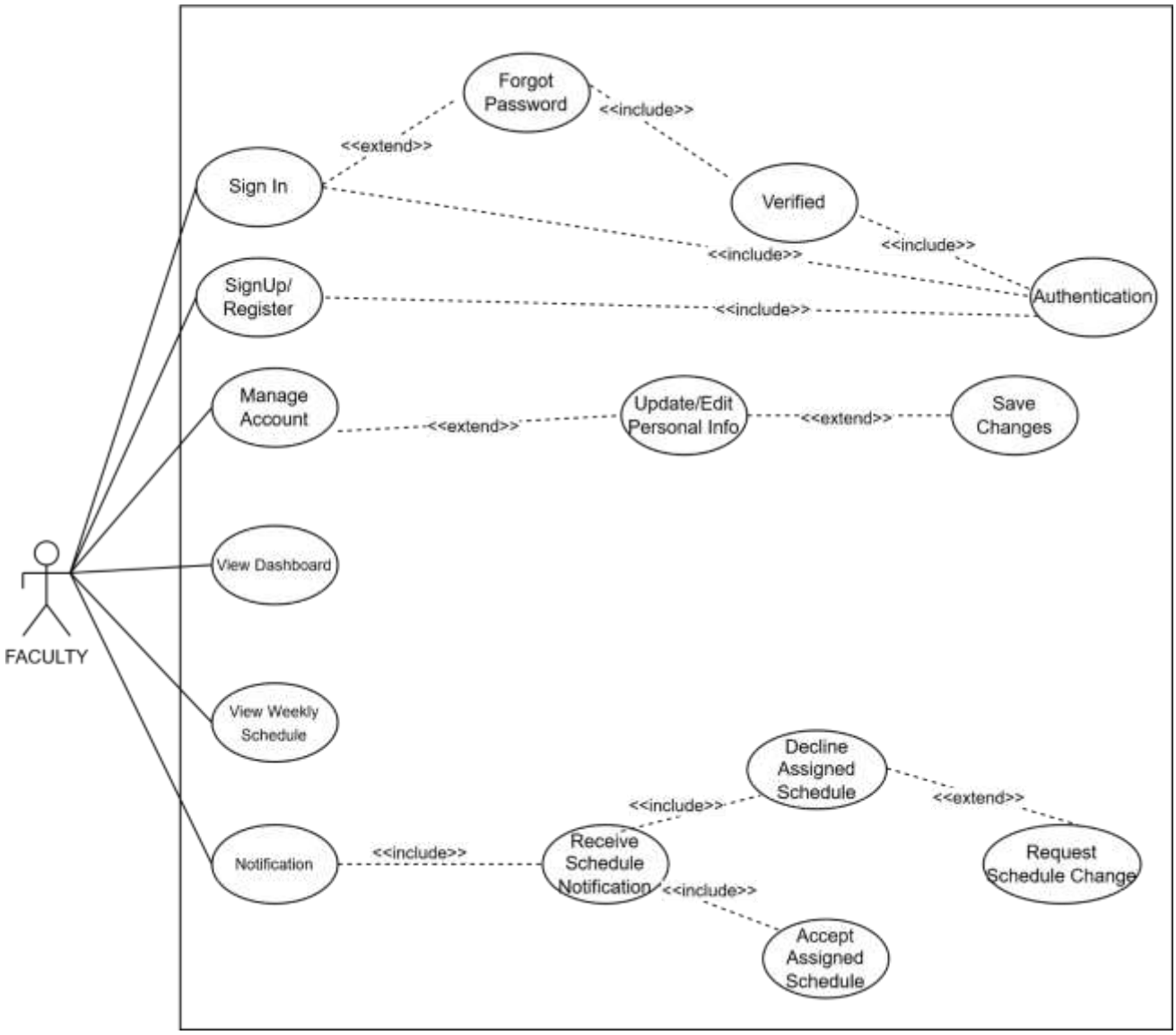

Figure 14: **Use Case Diagram for Faculty**

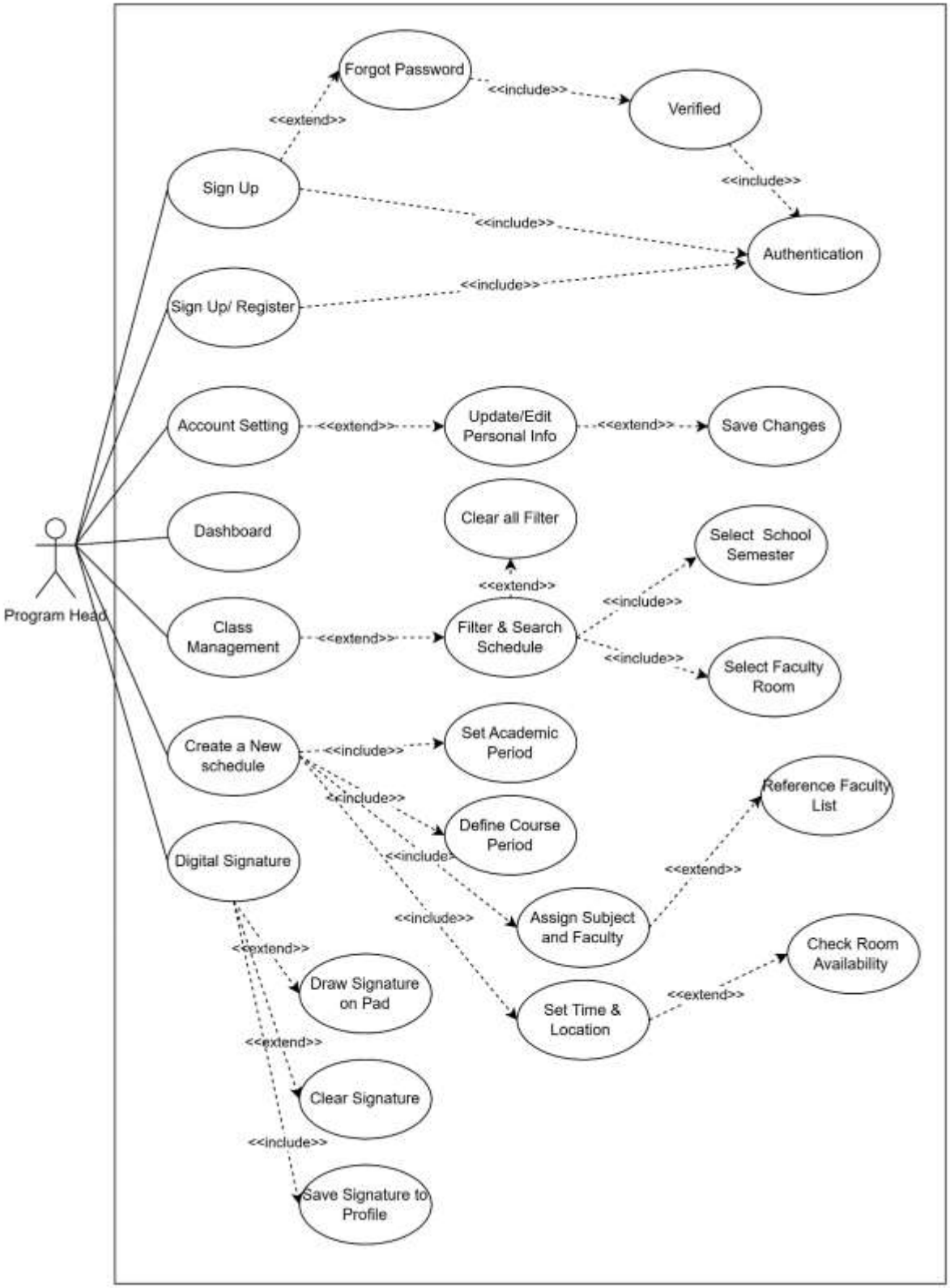

Figure 15: **Use Case Diagram for Program Head**

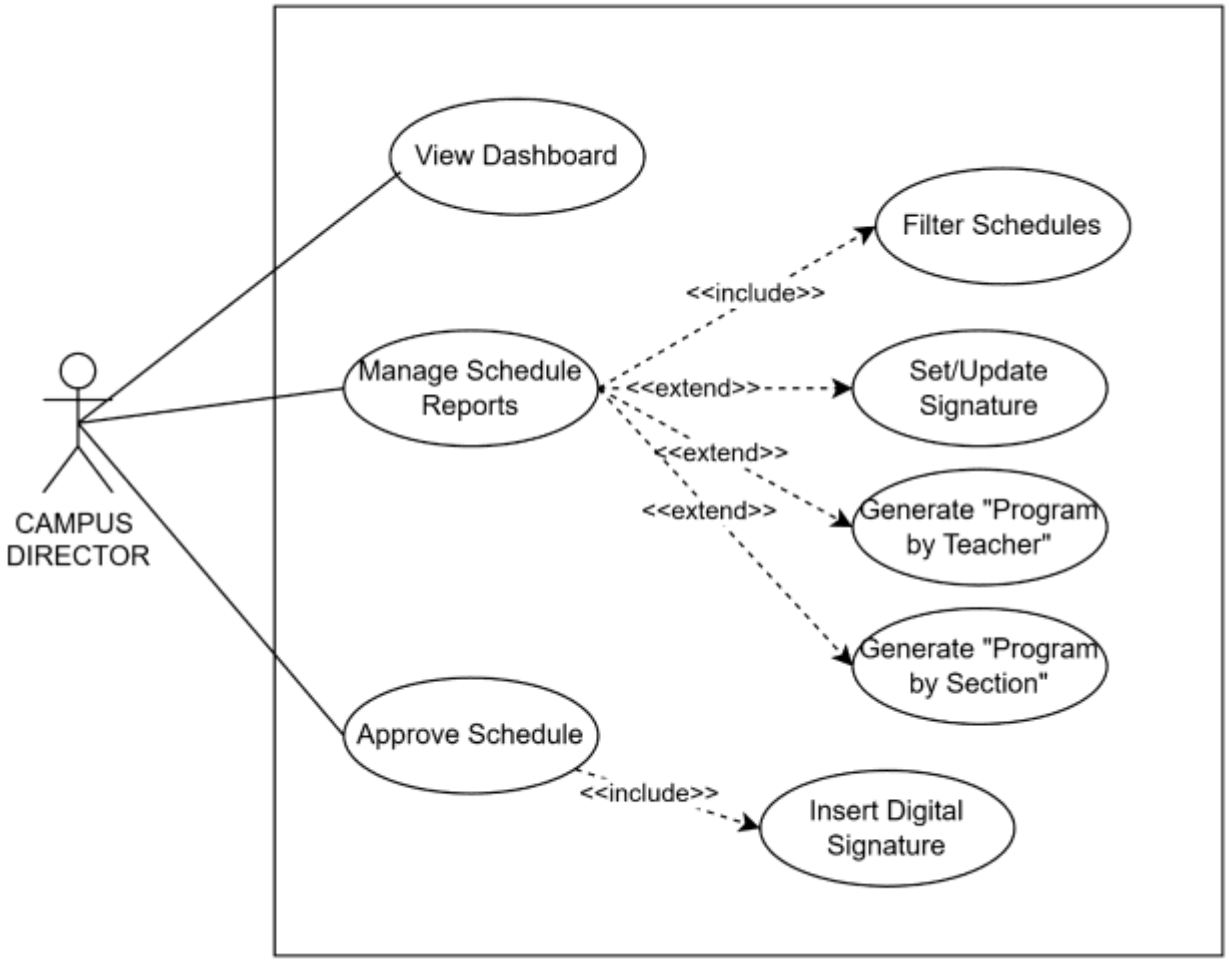

Figure16**: Use Case Diagram for Campus Director**

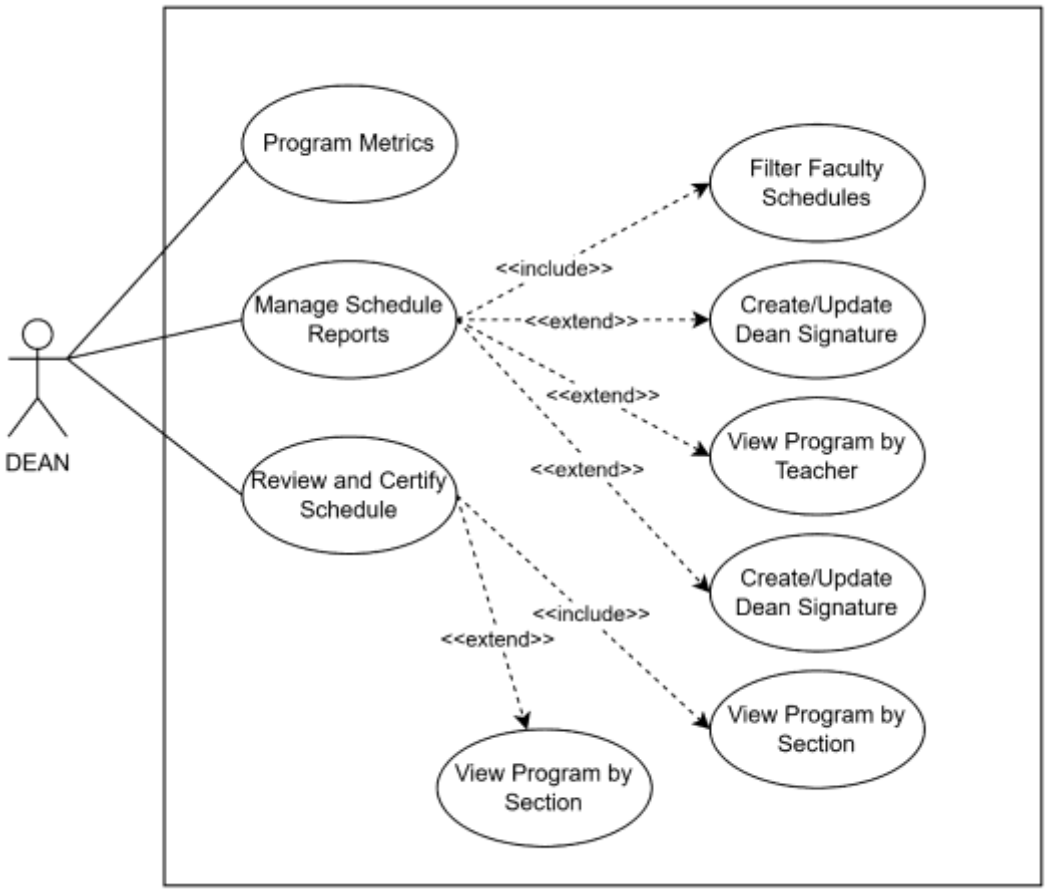

Figure 17: **Use Case Diagram for Dean**

Storyboard (Program Head)

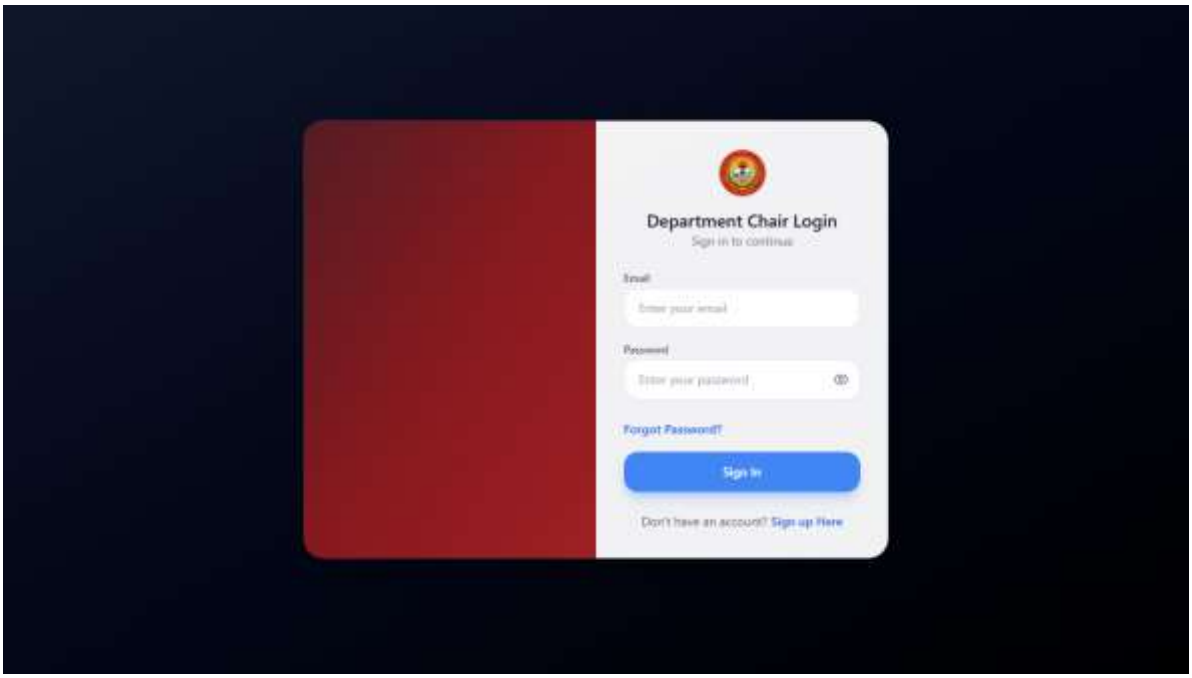

Figure 18: **Program Head Login**

Figure 18: The figure above shows the login page for Program Heads to access the system. This screen is where users enter their email and password to continue.

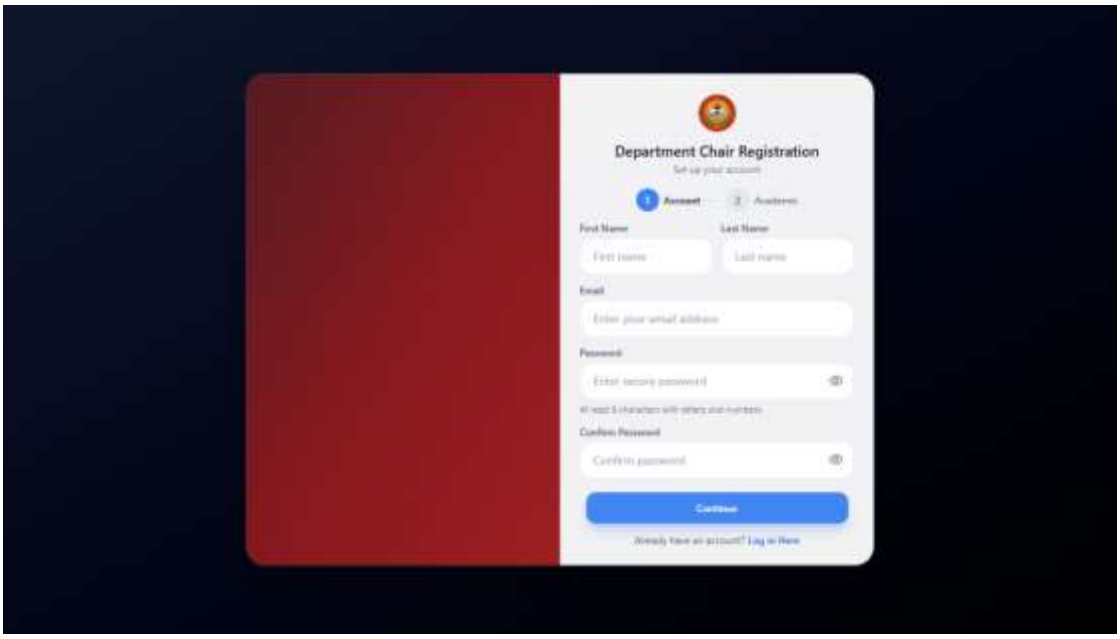

Figure 19**: Program Head Registration - Step 1: Account**

Figure 19: The figure above shows the first step of the registration page for new Program Heads to set up their personal account, requiring their first name, last name, email, and a secure password.

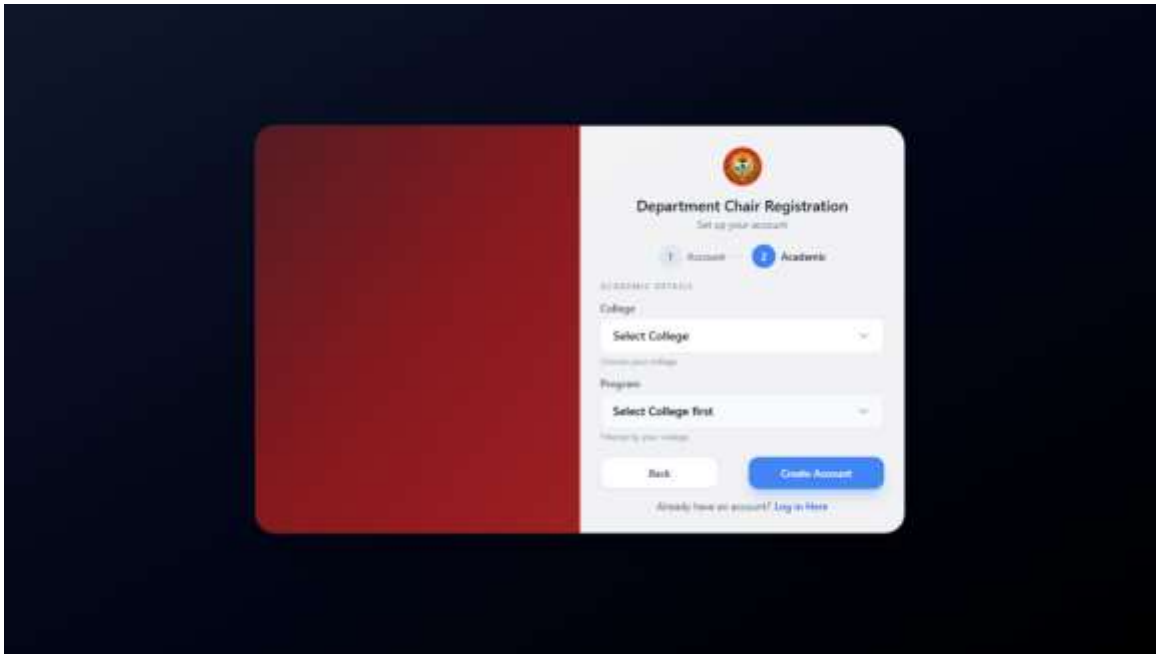

Figure 20**: <u>Program Head Registration - Step 2: Academic</u>**

Figure 20: The figure above shows the second step of the registration page, where the user provides their academic details by selecting their assigned College and Program.

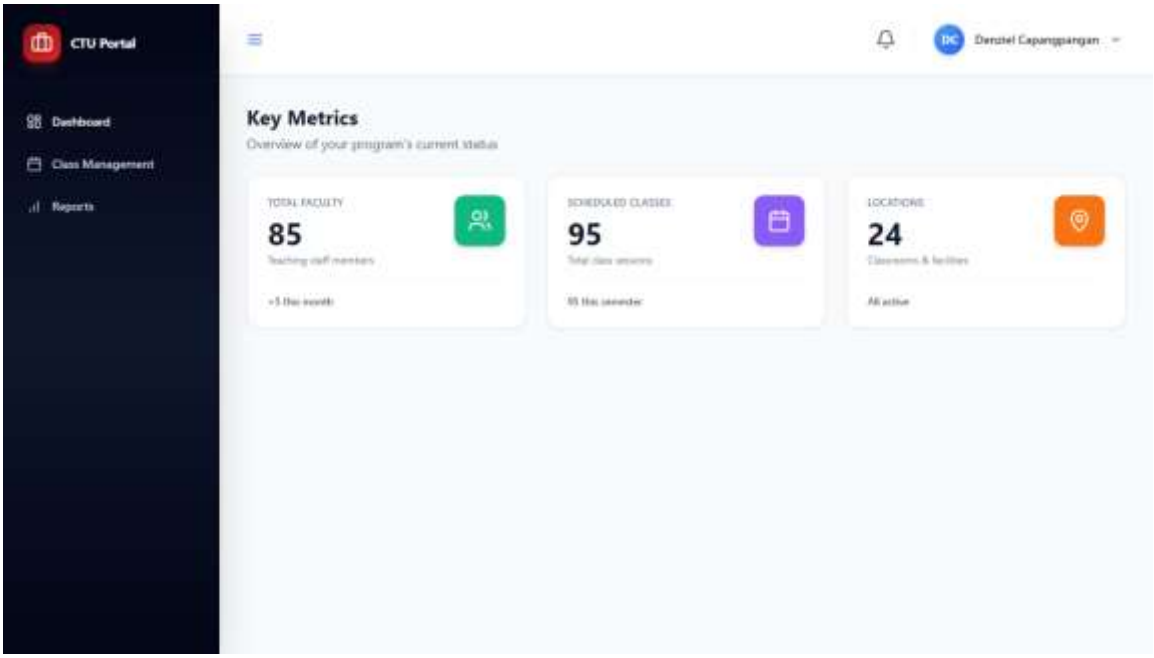

Figure 21: **Dashboard (Key Metrics)**

Figure 21: The figure above shows the Dashboard, providing an overview of the program's current status through key metrics, including Total Faculty, Scheduled Classes, and Locations.

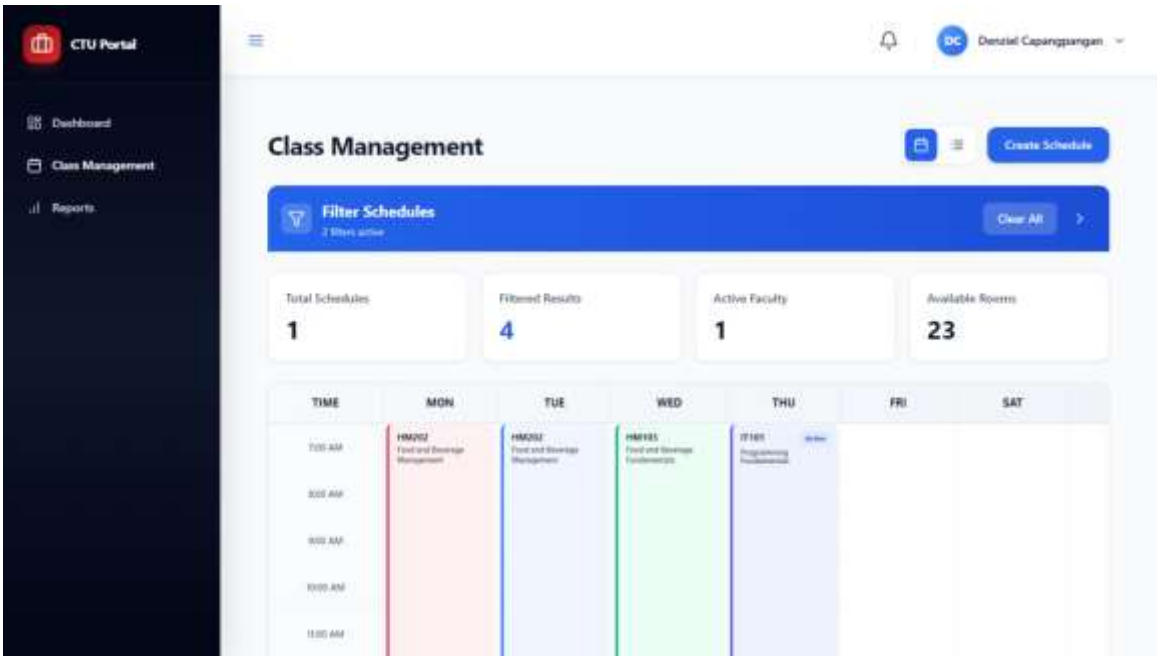

Figure 22**: Class Management (Schedule View)**

Figure 22: The figure above shows the Class Management screen with a weekly schedule grid displaying class sessions, along with summary data for Total Schedules, Filtered Results, Active Faculty, and Available Rooms.

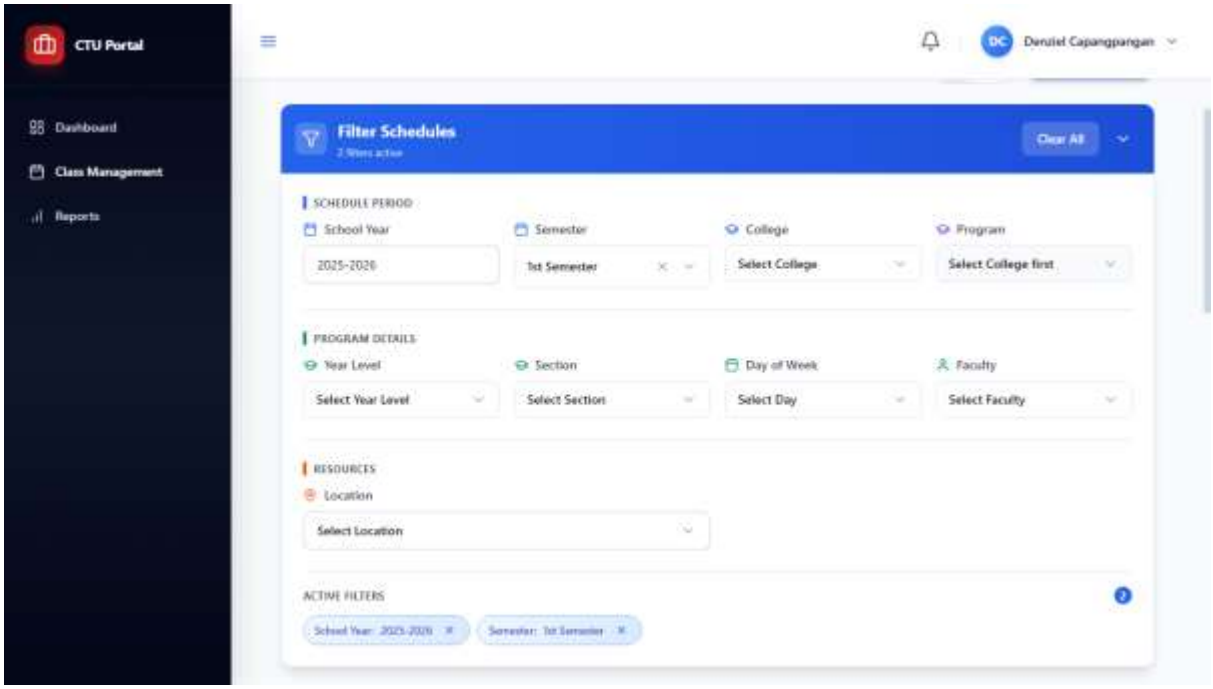

Figure 23: **<u>Class Management (Filter Schedules)</u>**

Figure 23: The figure above shows the Filter Schedules panel within Class Management, allowing users to search and filter schedules based on period, program details, and resources (location).

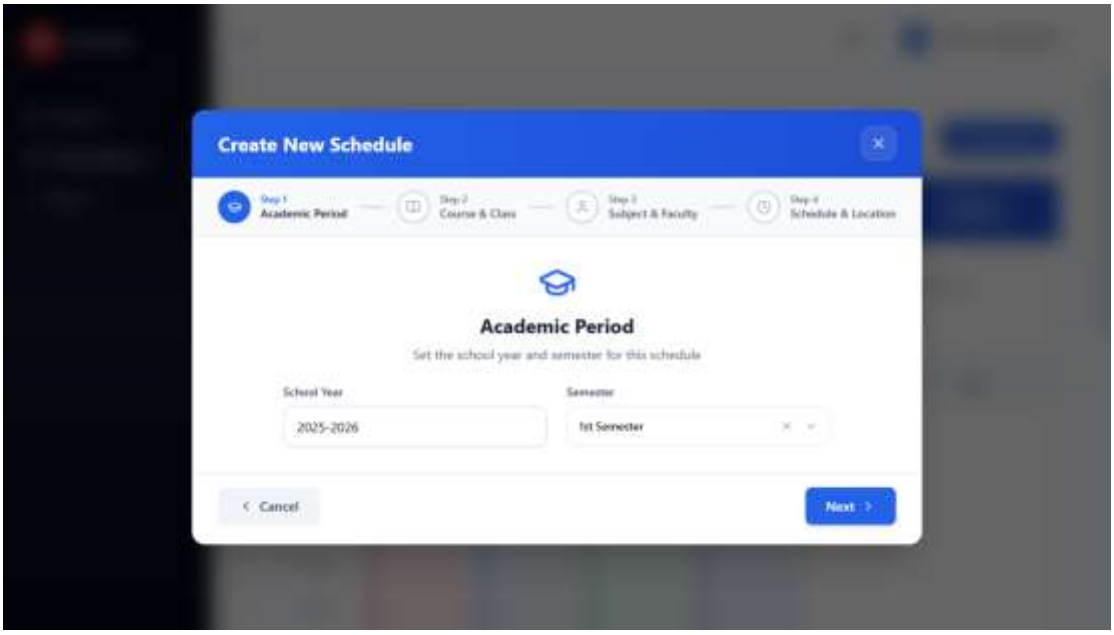

Figure 24**: <u>Create New Schedule - Step 1: Academic Period</u>**

Figure 24: The figure above shows Step 1: Academic Period of the "Create New Schedule" process, where the user sets the School Year and Semester for the new class schedule.

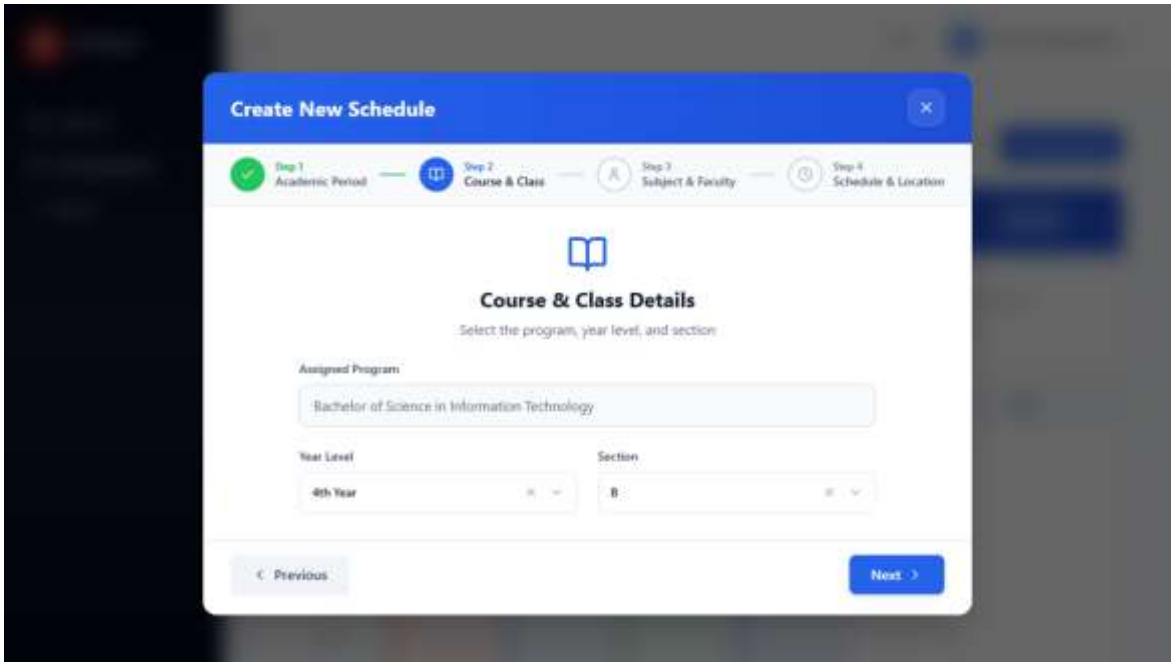

Figure 25**<u>: Create New Schedule - Step 2: Course & Class</u>**

Figure 25: The figure above shows Step 2: Course & Class of the schedule creation process, where the user selects the year level and section for the assigned program.

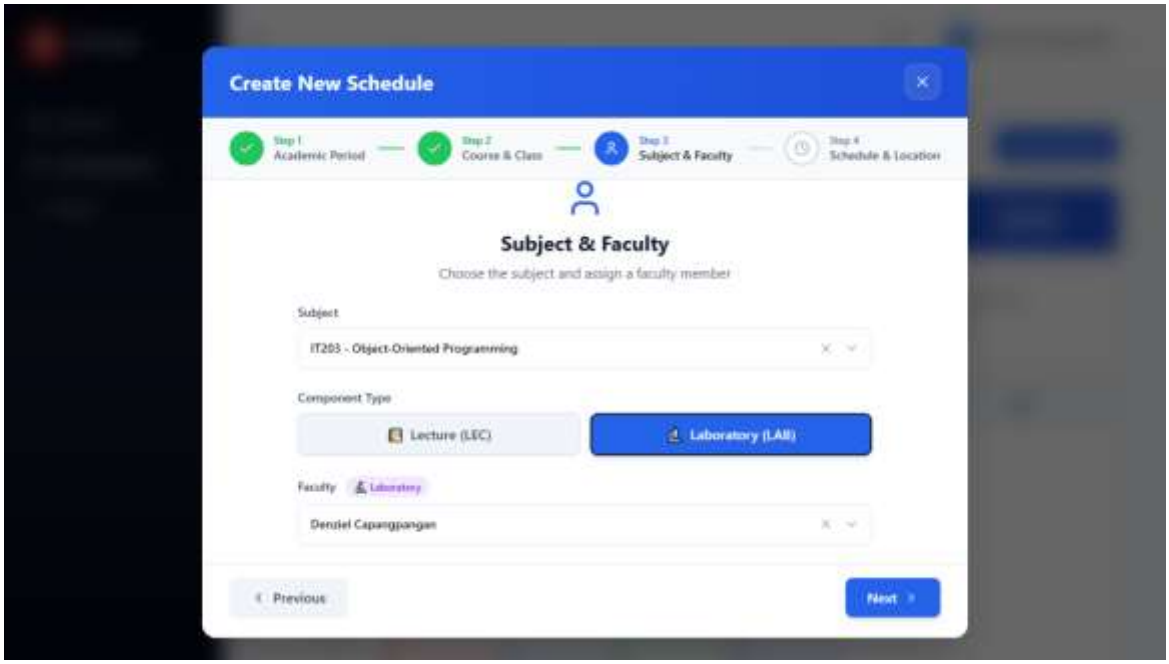

Figure 26**: Create New Schedule - Step 3: Subject & Faculty**

Figure 26: The figure above shows Step 3: Subject & Faculty of the schedule creation process, allowing the user to choose the subject, specify the component type (Lecture or Laboratory), and assign a faculty member.

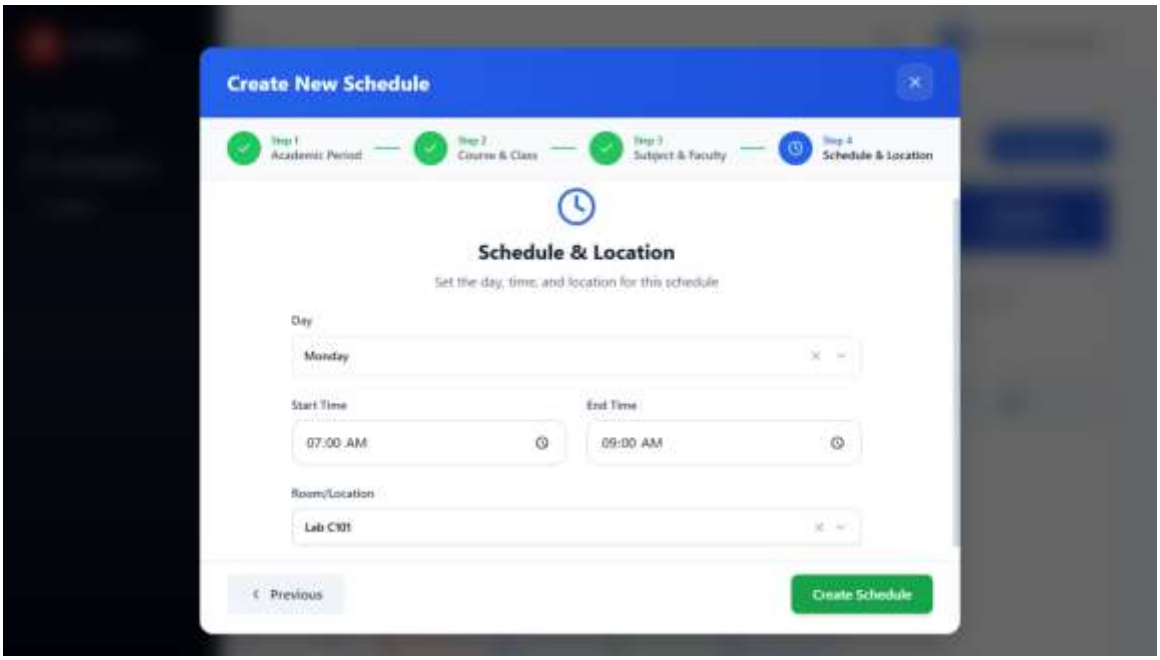

Figure 27**: <u>Create New Schedule - Step 4: Schedule & Location</u>**

Figure 27: The figure above shows Step 4: Schedule & Location, the final step for creating a new schedule, where the user sets the day start time, end time, and room for the class.

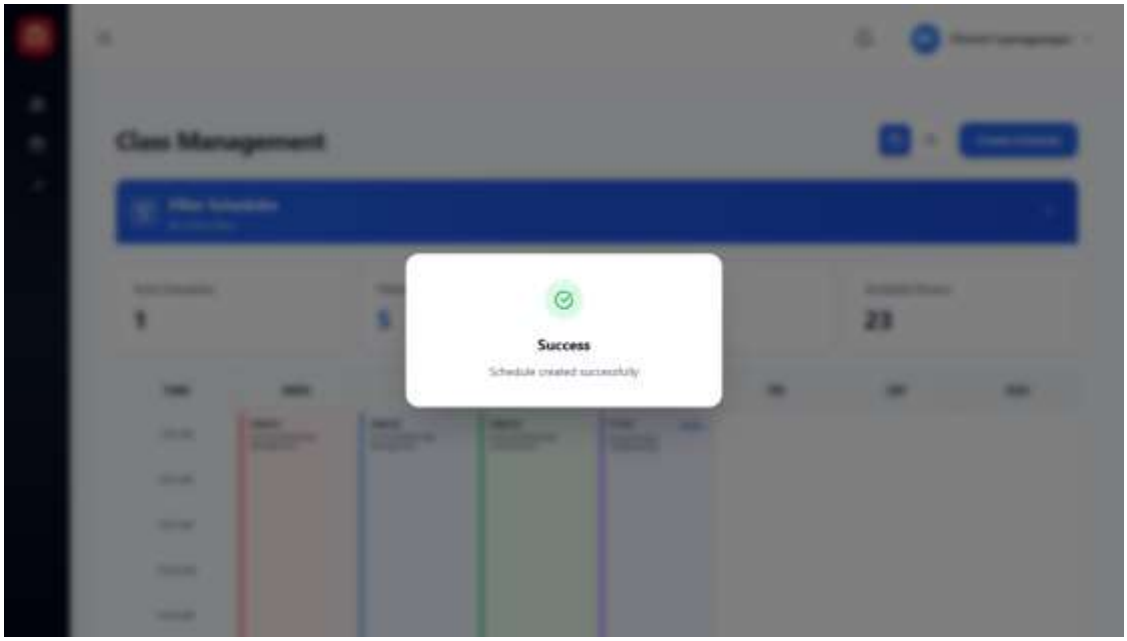

Figure 28**: <u>Schedule Creation Success</u>**

Figure 28: The figure above shows the success message, confirming that a new schedule was created successfully.

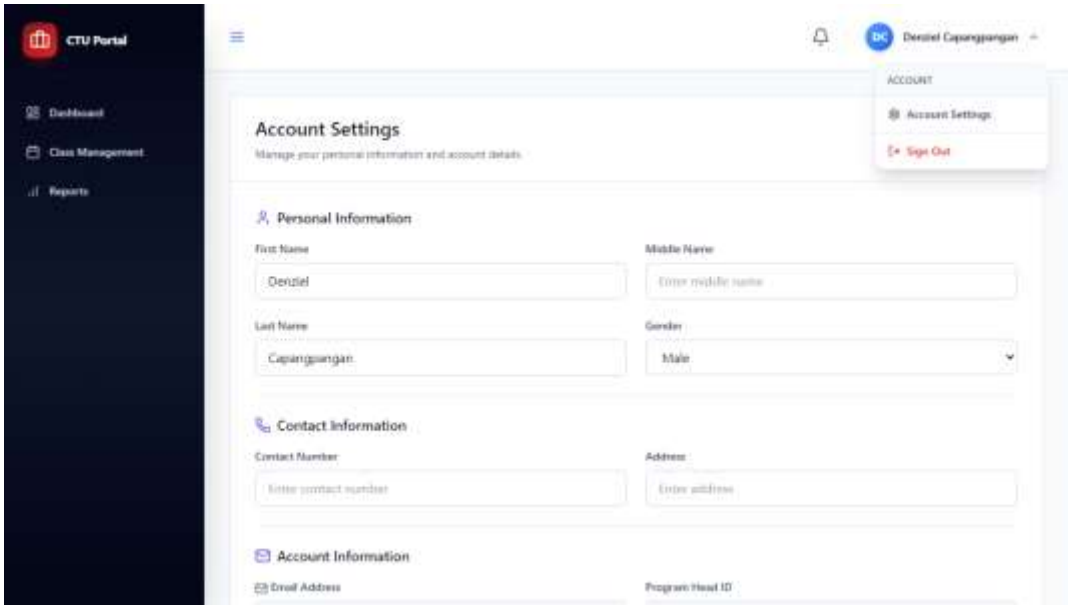

Figure 29**: <u>Account Settings (Personal and Contact Information)</u>**

Figure 29: The figure above shows the Account Settings screen, displaying the user's personal and contact information for management.

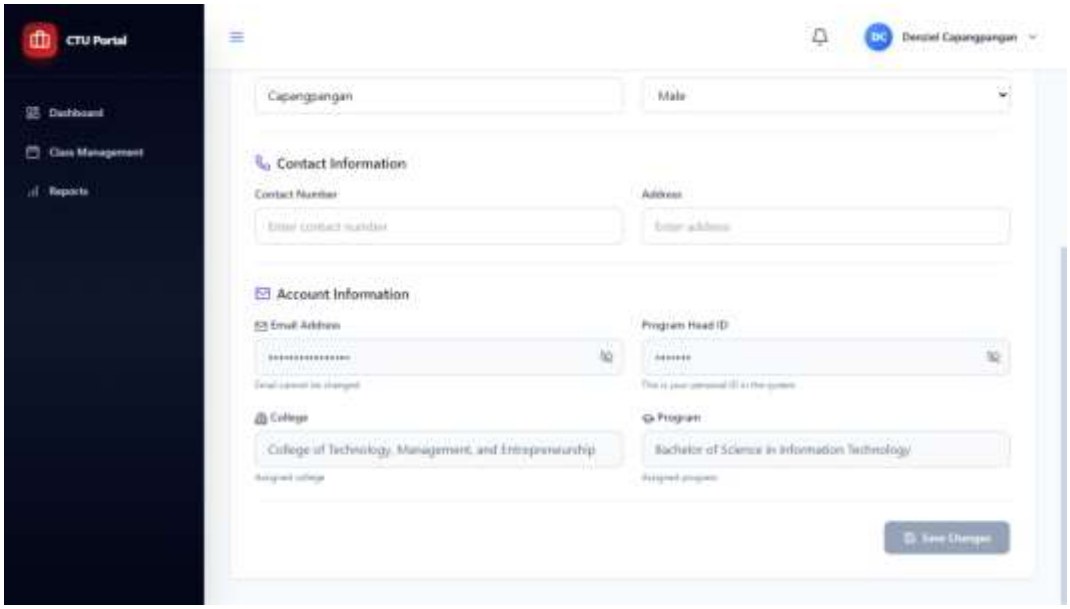

Figure 30**: <u>Account Settings (Account Information)</u>**

Figure 30: The figure above shows the Account Settings screen, displaying the user's account information, including their Email Address, Program Head ID, assigned College and assigned Program.

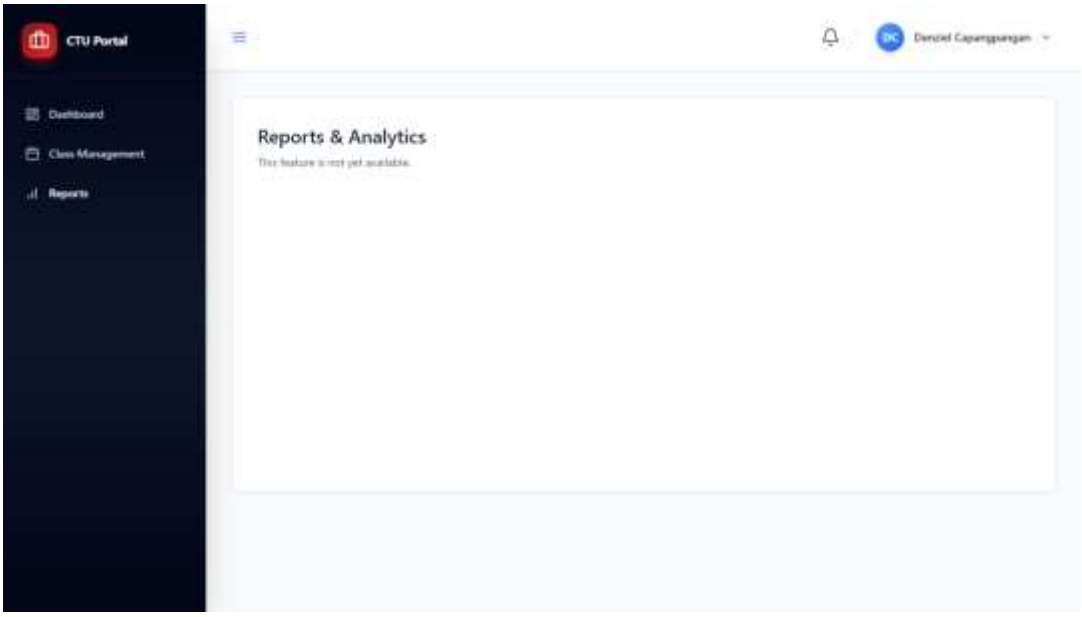

Figure 31**: <u>Reports & Analytics</u>**

Figure 31: The figure above shows the Reports & Analytics screen, indicating that this feature is currently not yet available in the system.

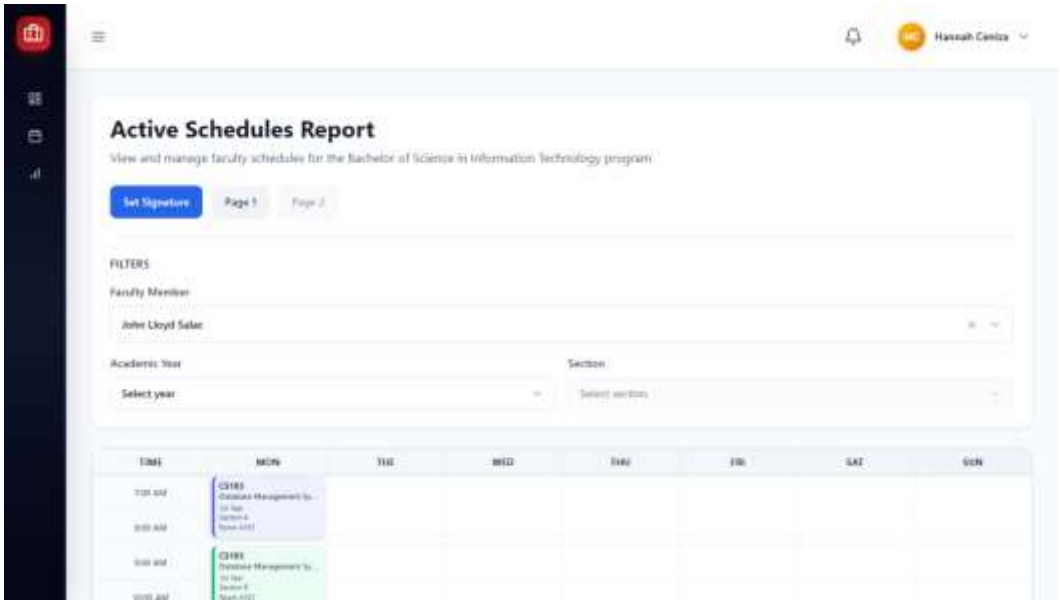

Figure 32**: <u>Active Schedules Report</u>**

Figure 32: The figure above shows the Active Schedules Report dashboard. This interface allows the Program Head to view and manage faculty schedules specifically for the Bachelor of Science in Information Technology program.

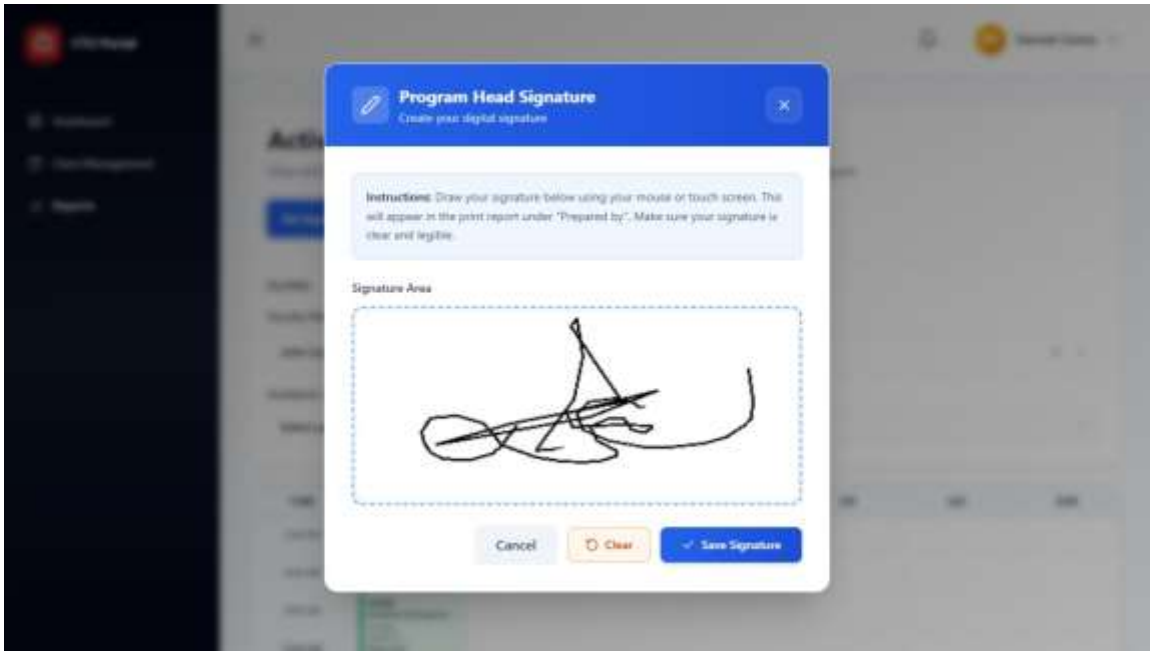

Figure 33: **Digital Signature Setup**

Figure 33 : The figure above illustrates the Program Head Signature modal. Here, the user can draw their digital signature using a mouse or touchscreen. This signature is saved and automatically applied to the "Prepared by" section of printed reports.

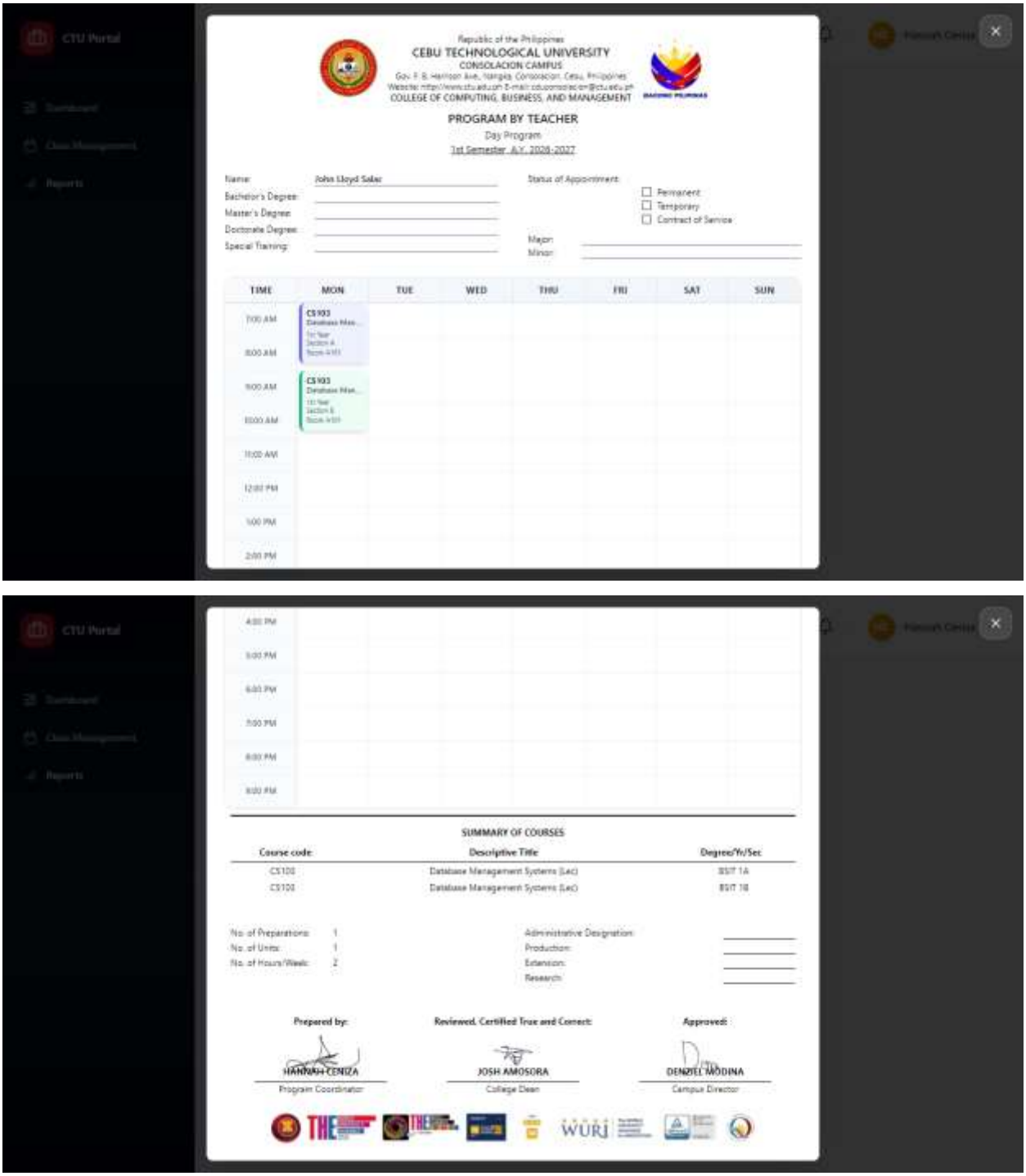

Figure 34**: Program by Teacher Report**

Figure 34: The figure above displays the generated Program by Teacher report for a specific faculty member. It provides a comprehensive view of the teacher's weekly schedule, course loads, and the total number of preparations and units.

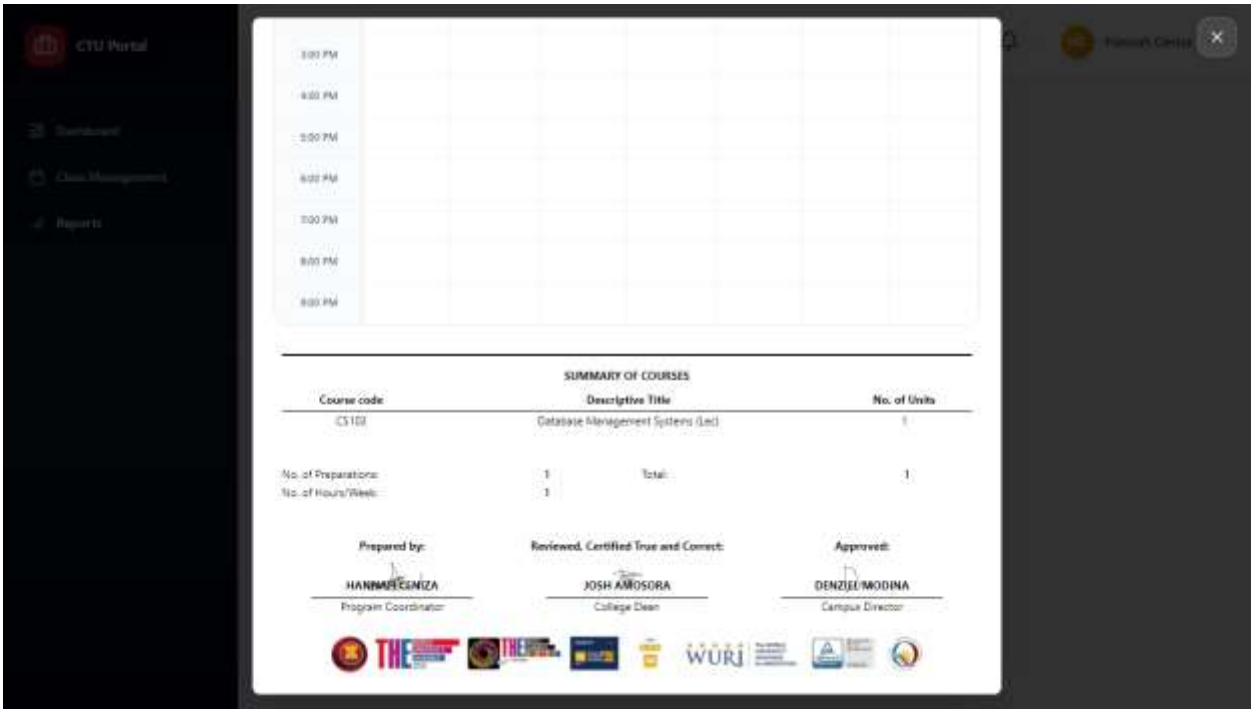

Figure 35**: <u>Program by Section Report</u>**

Figure 35: The figure above shows the Program by Section report. This view filters the schedule by academic year and section listing all courses assigned to that group and identifying the designated class adviser

Administrator

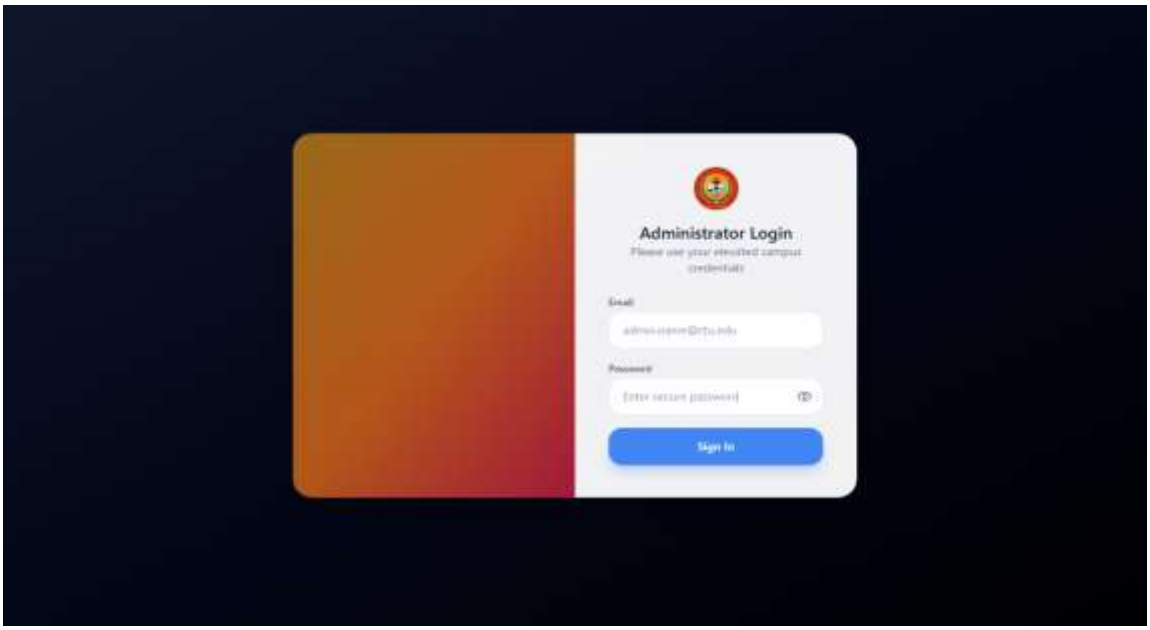

Figure36**: Administrator Login**

Figure 36: The figure above shows the Administrator Login page, which requires users to enter their elevated campus credentials. This screen is where administrators enter their email and password to access the system.

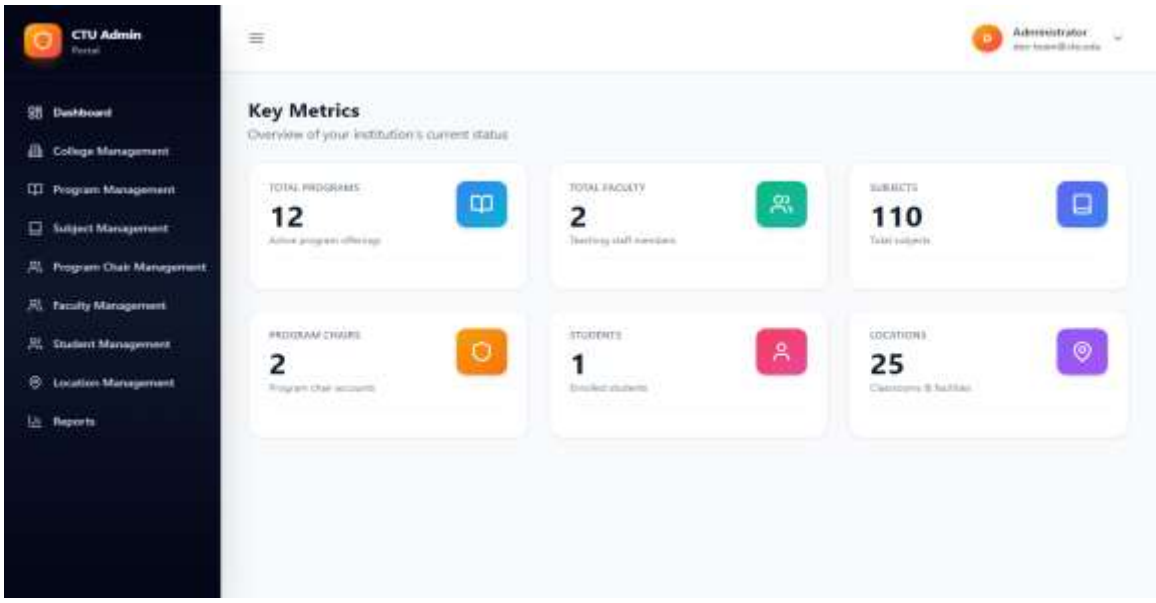

Figure37**: Administrator Dashboard**

Figure 37: The figure above shows the Administrator Dashboard, which provides an overview of the institution's current status using Key Metrics. Metrics include Total Programs, Total Faculty, Subjects, Program Chairs, Students, and Locations.

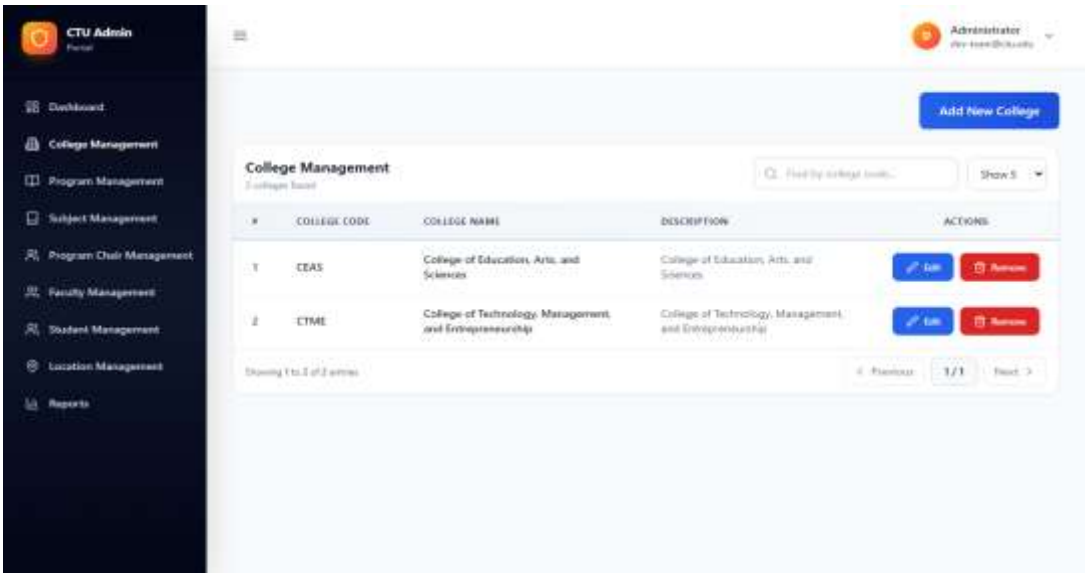

Figure 38**: College Management**

Figure 38: The figure above shows the College Management section, displaying a list of colleges with columns for College Code, Name, and Description. It includes options to Add New College and use Edit/Remove actions for existing entries.

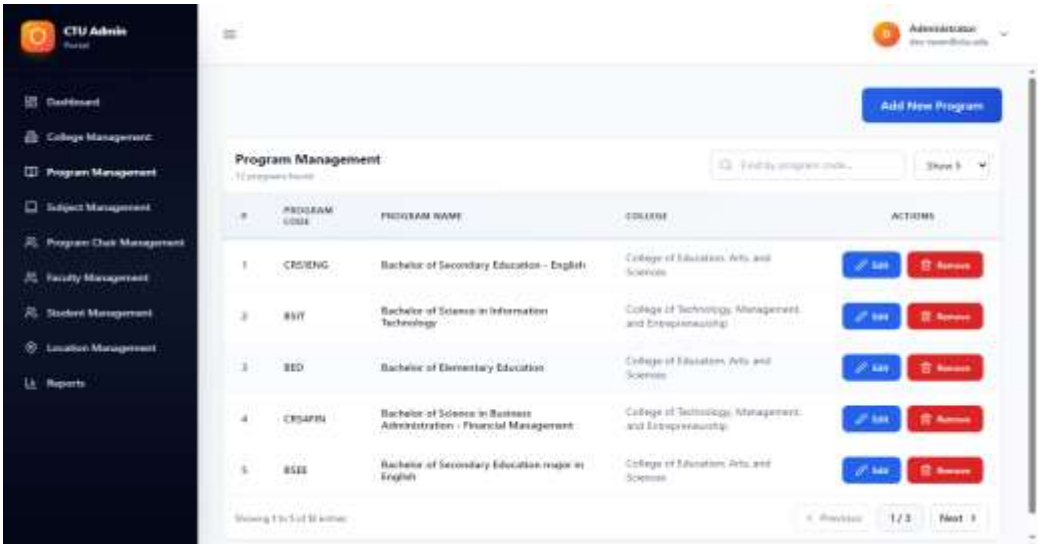

Figure 39**: Program Management**

Figure 39: The figure above shows the Program Management section, which lists all programs with details like Program Code, Name, and the College they belong to. It provides buttons to Add New Program and manage existing programs with Edit/Remove actions.

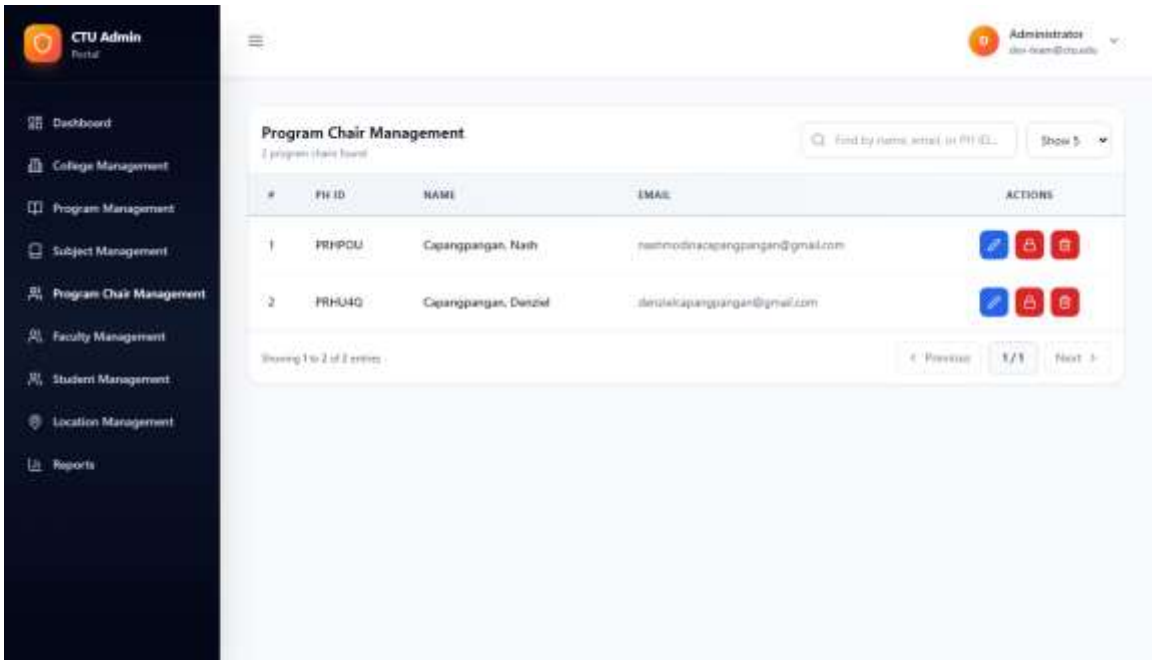

Figure 40**: <u>Program Chair Management</u>**

Figure 40: The figure above shows the Program Chair Management section, listing the two program chair accounts. Details displayed include PH ID, Name, and Email, with options to edit and delete the corresponding accounts.

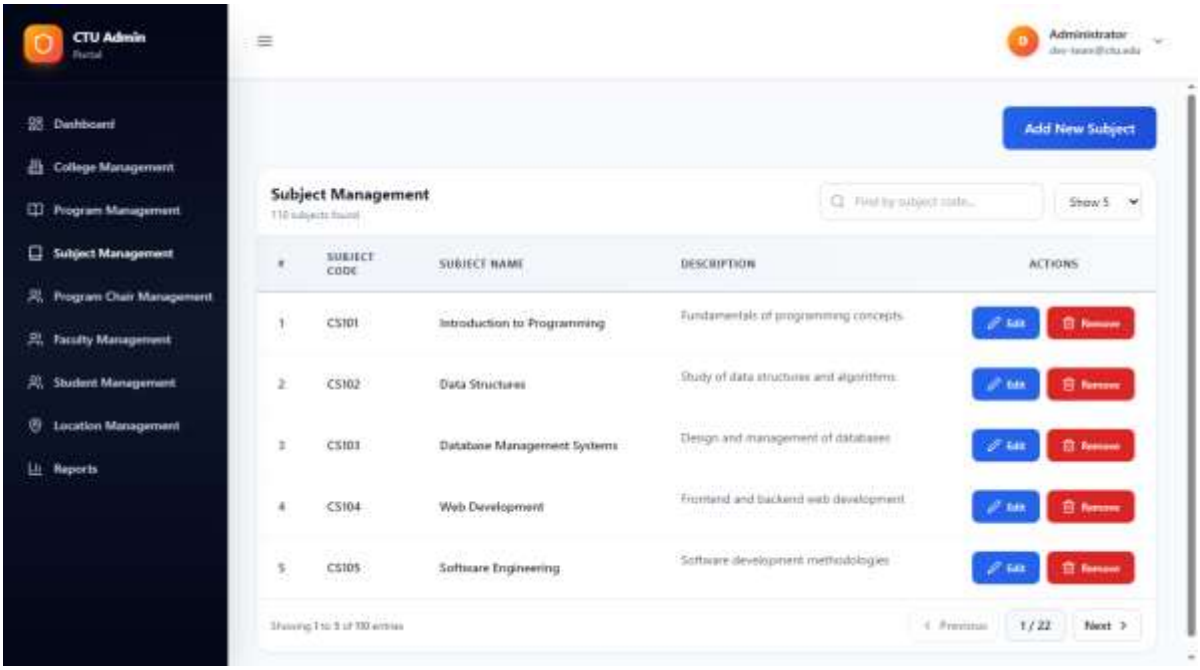

Figure 41**: <u>Subject Management</u>**

Figure 41: The figure above shows the Subject Management section, displaying 110 subjects with details such as Subject Code, Name, and Description. It allows the administrator to Add New Subject and use Edit/Remove actions on the existing subject entries.

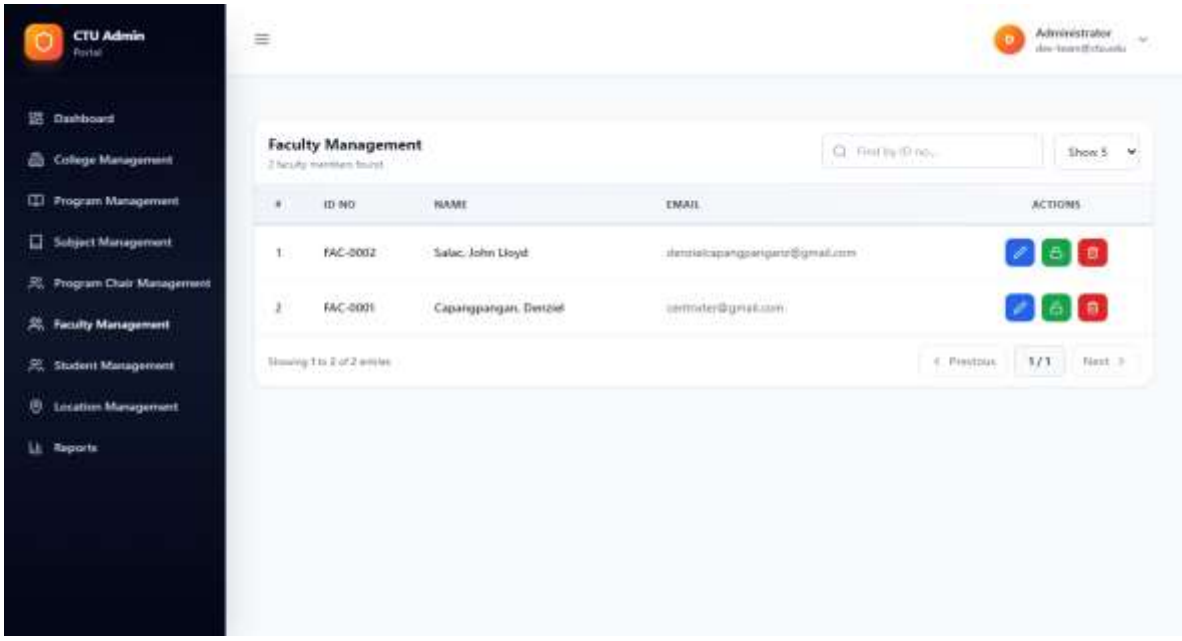

Figure 42**: <u>Faculty Management</u>**

Figure 42: The figure above shows the Faculty Management section, displaying the list of two faculty members. The table includes their ID No, Name, and Email, along with options to edit, view, or remove the faculty member's profile.

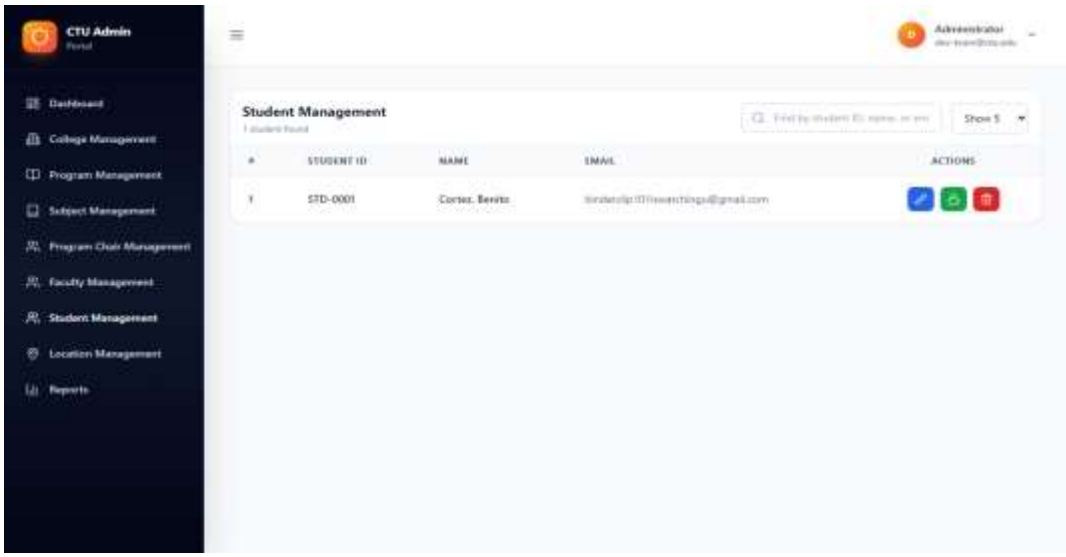

Figure 43**: <u>Student Management</u>**

Figure 43: The figure above shows the Student Management section, which displays the list of enrolled students with their Student ID, Name, and Email. The administrator can use the available actions to manage student records.

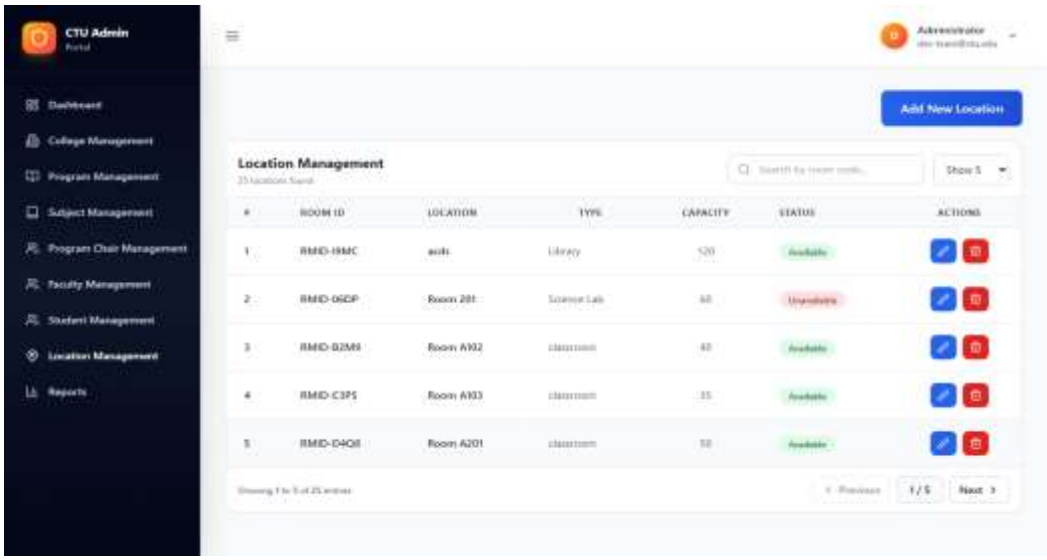

Figure 44**: <u>Location Management</u>**

Figure 44: The figure above shows the Location Management section, displaying 25 rooms or facilities. The list details the Room ID, Location, Type, Capacity, and includes the option to Add New Location.

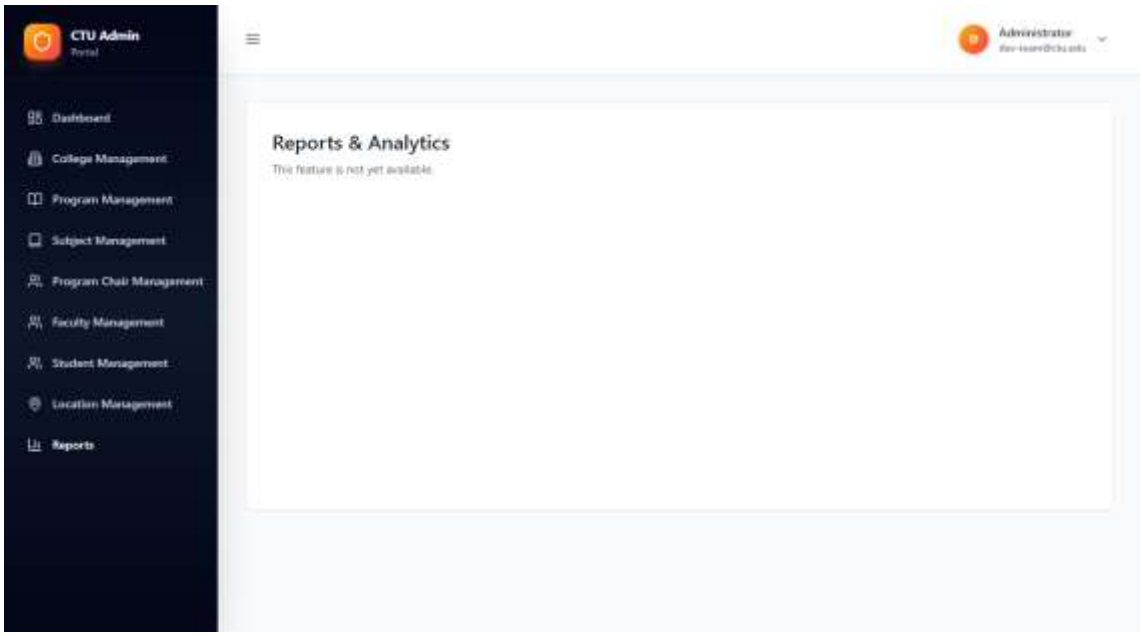

Figure 45**: <u>Reports & Analytics</u>**

Figure 45: The figure above shows the Reports & Analytics section, indicating that this feature is not yet available within the CTU Admin Portal system

Student

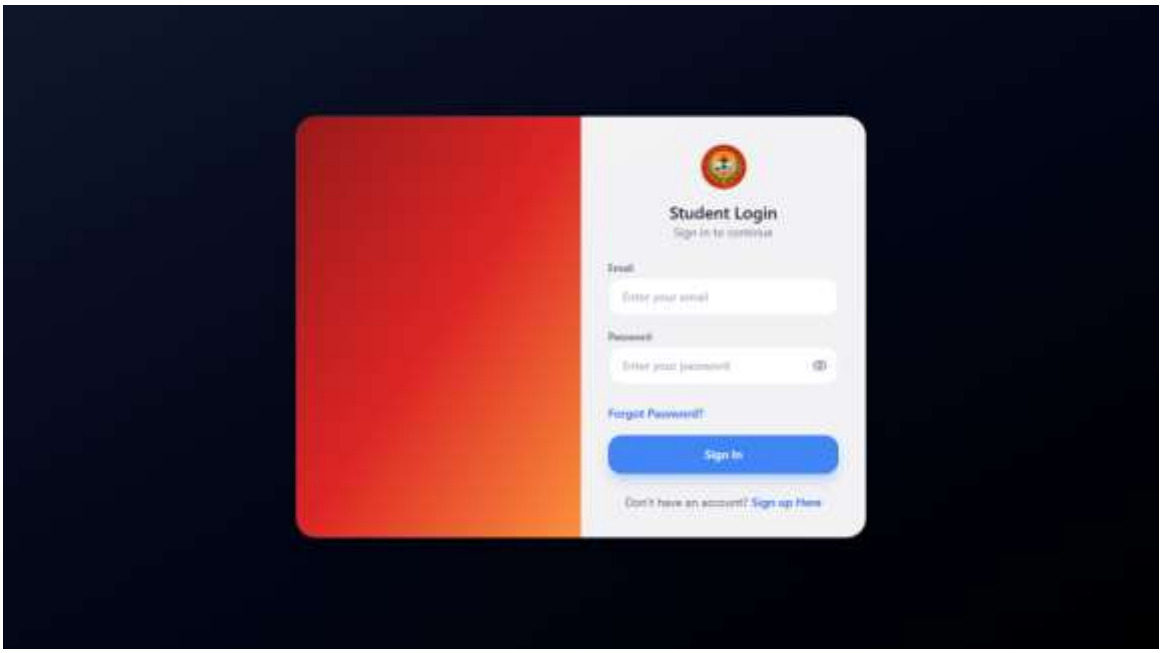

Figure 4**: Student Login**

Figure 46: The figure above shows the login page for Student members to access the system. This screen is where users enter their email and password to continue.

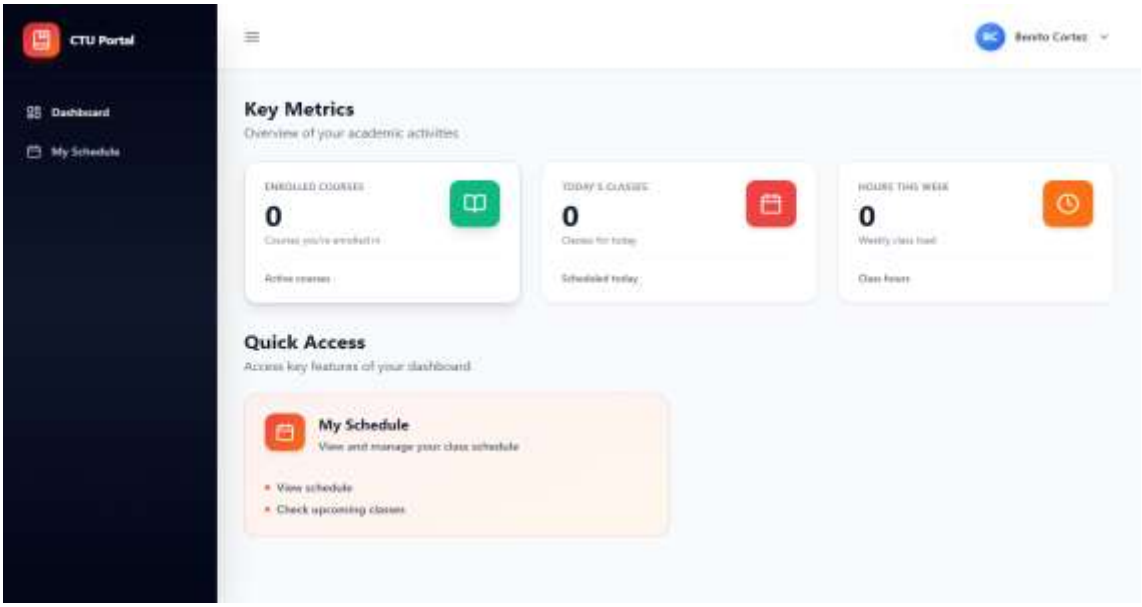

Figure 47**: Student Dashboard**

Figure 47: The figure above shows the Dashboard which provides an overview of the user's academic activities. It displays Key Metrics and a Quick Access card for My Schedule.

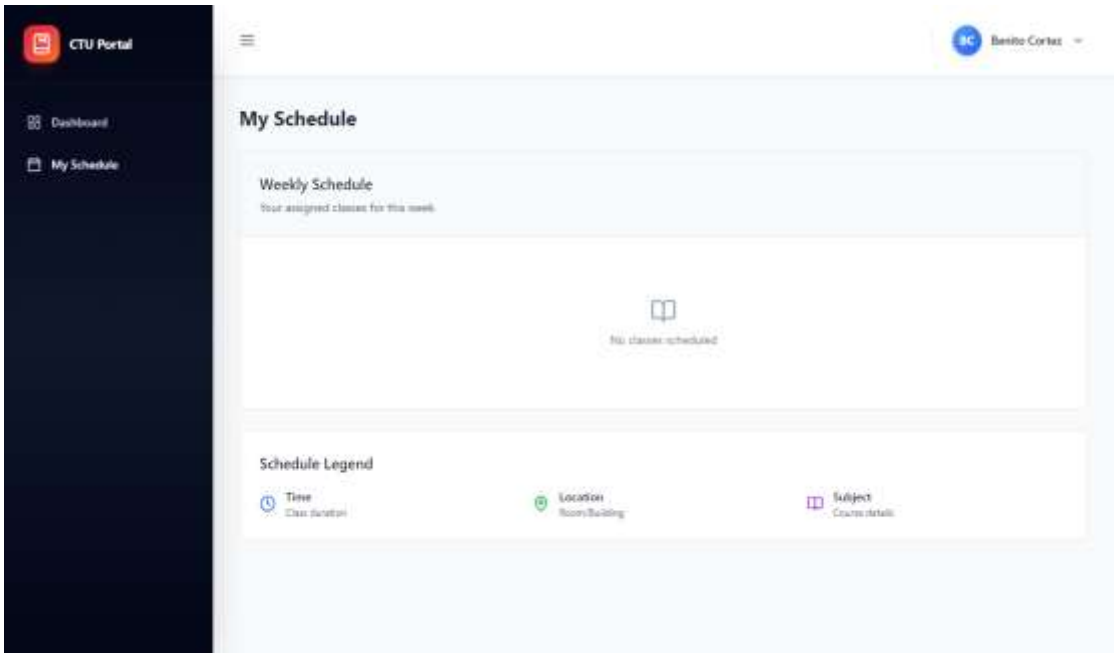

Figure 48**: <u>My Schedule</u>**

Figure 48: The figure above shows the My Schedule page where students can view their Weekly Schedule and check assigned classes. It also includes a Schedule Legend explaining the Time, Location, and Subject details.

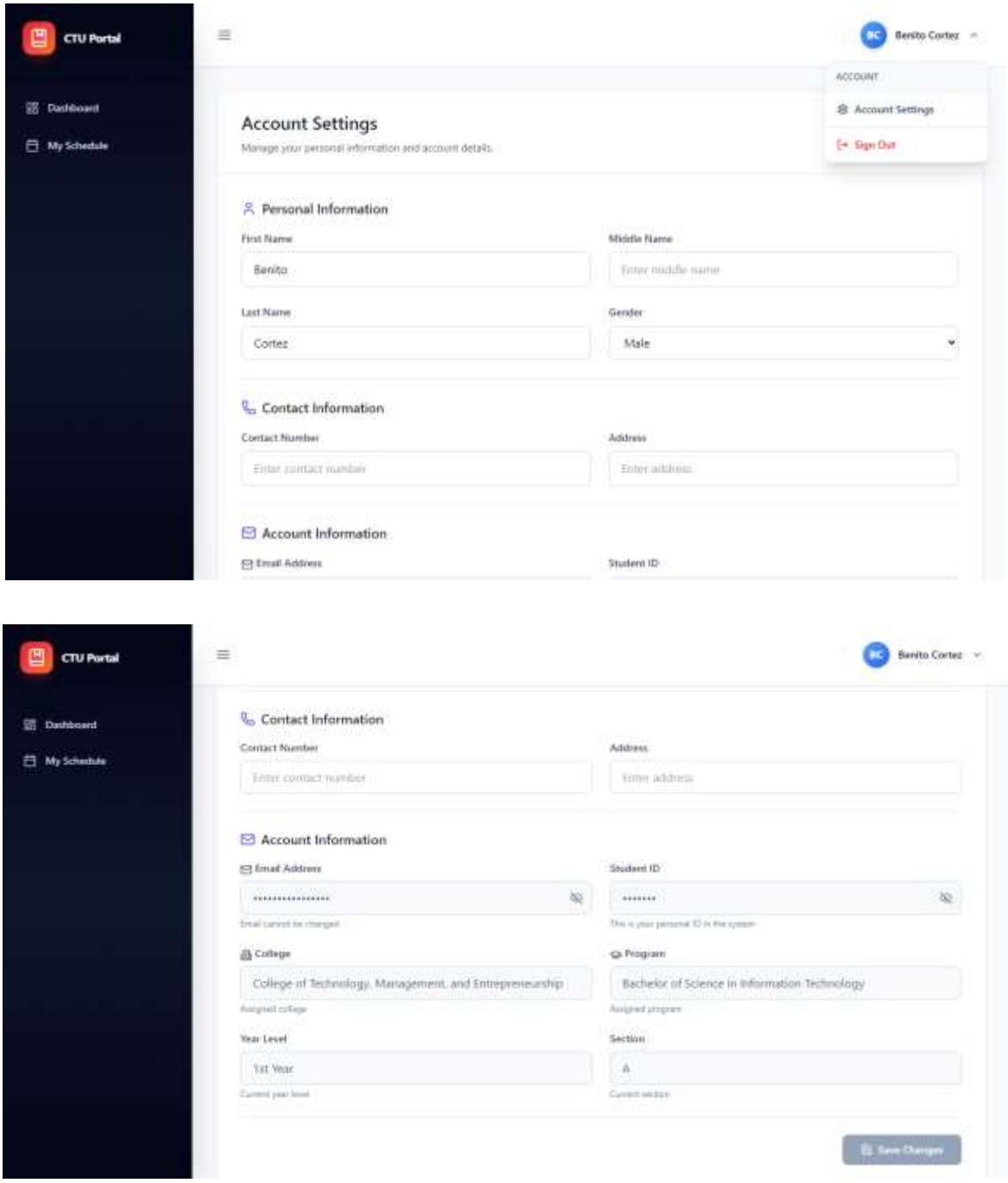

Figure 49**: <u>Account Settings</u>**

Figure 49: The figure above shows the Account Settings page, allowing the user to manage their personal information, contact information, and account details and save any changes.

Faculty

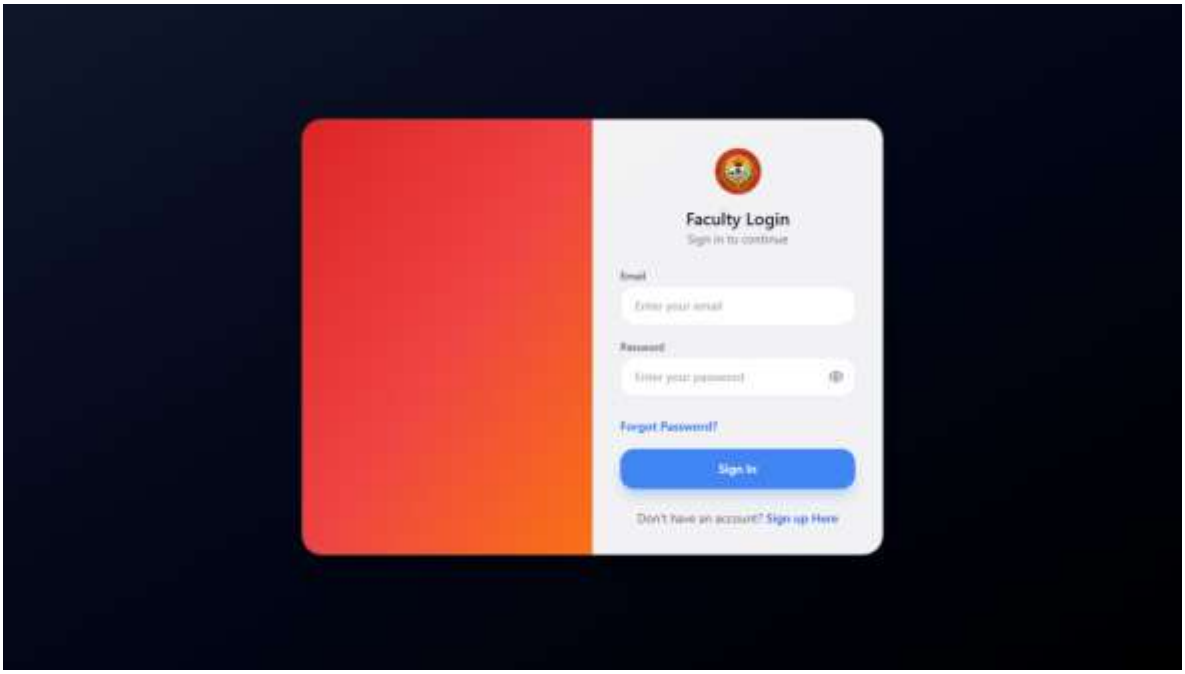

Figure 50**: <u>Faculty Login</u>**

Figure 50: The figure above shows the login page for Faculty members to access the system. This screen is where users enter their email and password to continue.

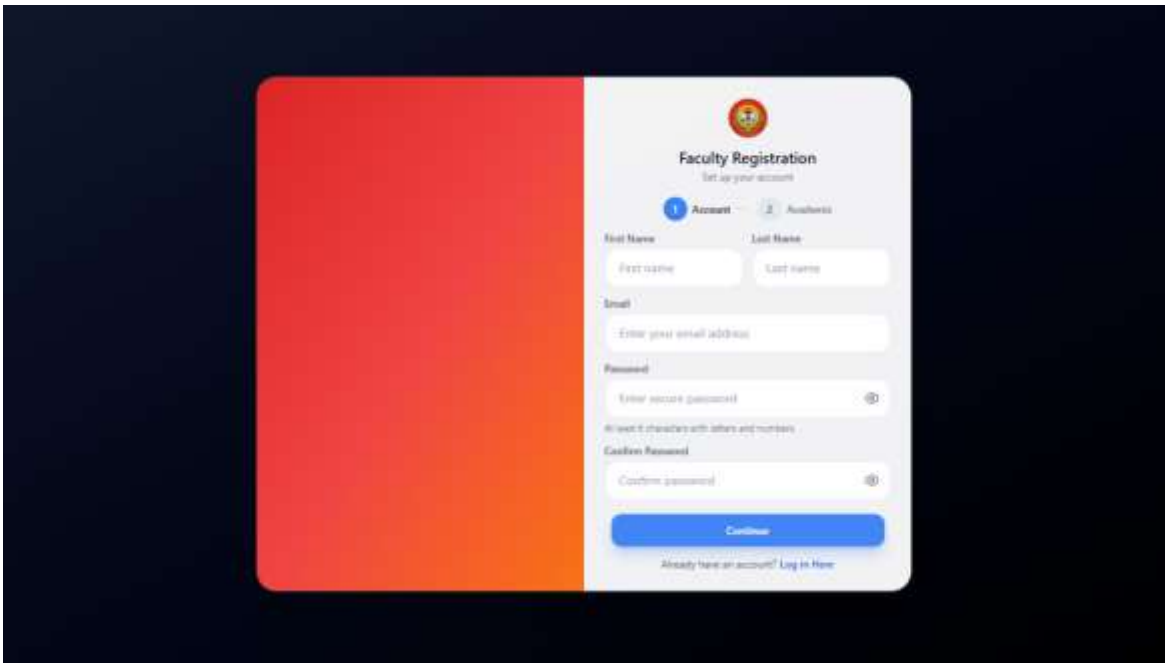

Figure 50:**<u>Faculty Registration - Step 1: Account</u>**

Figure 50: The figure above shows the first step of the registration page for new Faculty members to set up their personal account, requiring their first name, last name, email, and a secure password

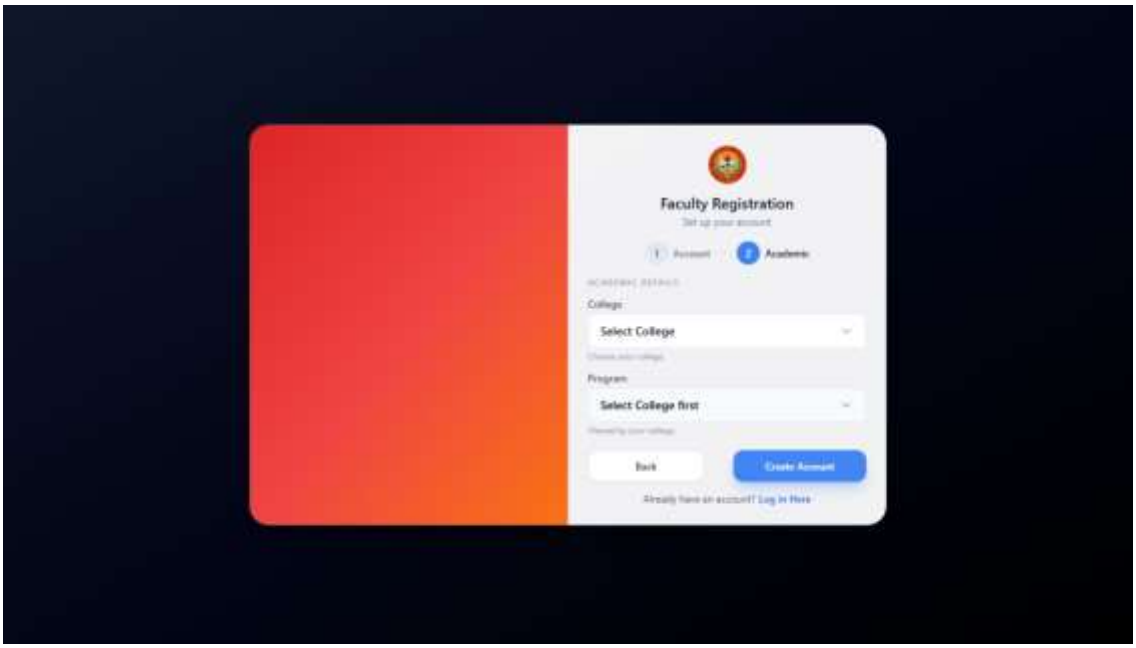

Figure 51**: <u>Faculty Registration - Step 2: Academic</u>**

Figure 51: The figure above shows the second step of the registration page, where the faculty member provides their academic details by selecting their assigned College and Program.

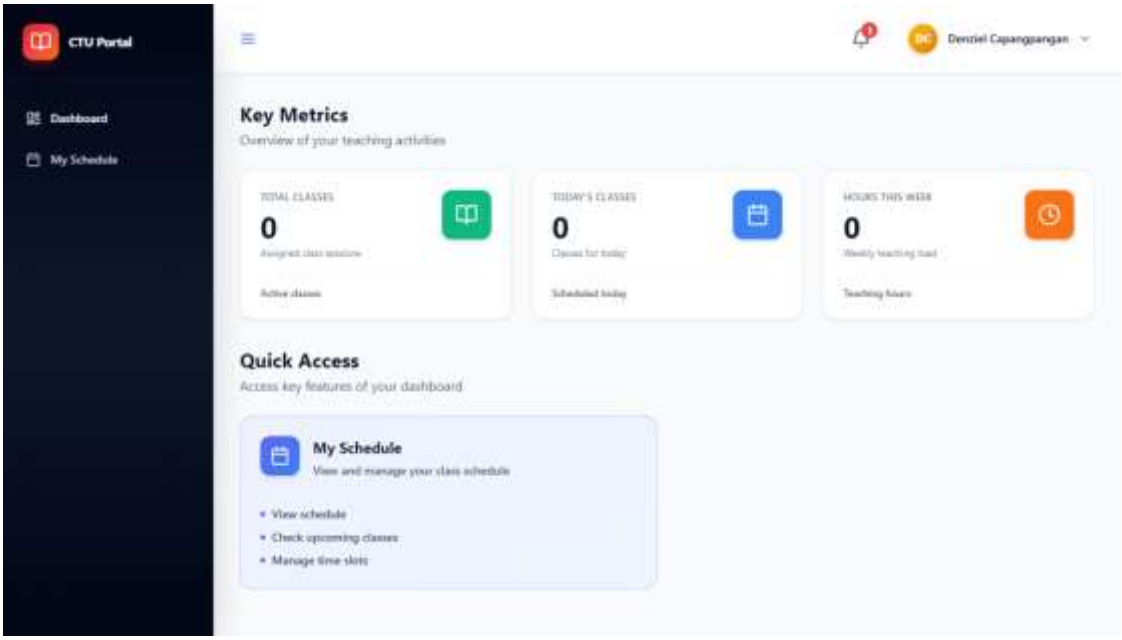

Figure 52: **<u>Faculty Dashboard (Key Metrics)</u>**

Figure 52: The figure above shows the Faculty Dashboard, which provides an overview of the user's teaching activities through key metrics like Total Classes, Today's Classes, and Hours This Week. It also provides Quick Access to the user's schedule.

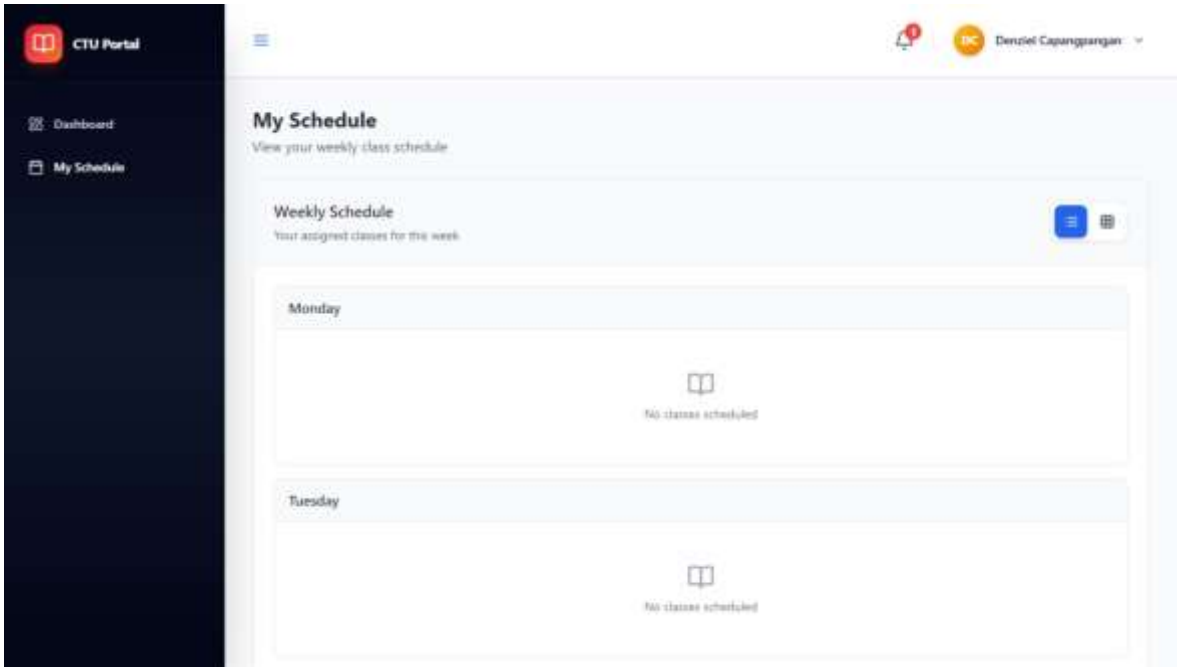

Figure 53**: <u>My Schedule (List View - No Classes & Notification)</u>**

Figure 53: The figure above shows the My Schedule screen in list view format, where the user can view their weekly class schedule. This specific state shows days with "No classes scheduled." When a new schedule is received, the system provides an action flow, likely triggered from the notification. The faculty member can then choose to Accept, Decline, or Request a change to the assigned schedule.

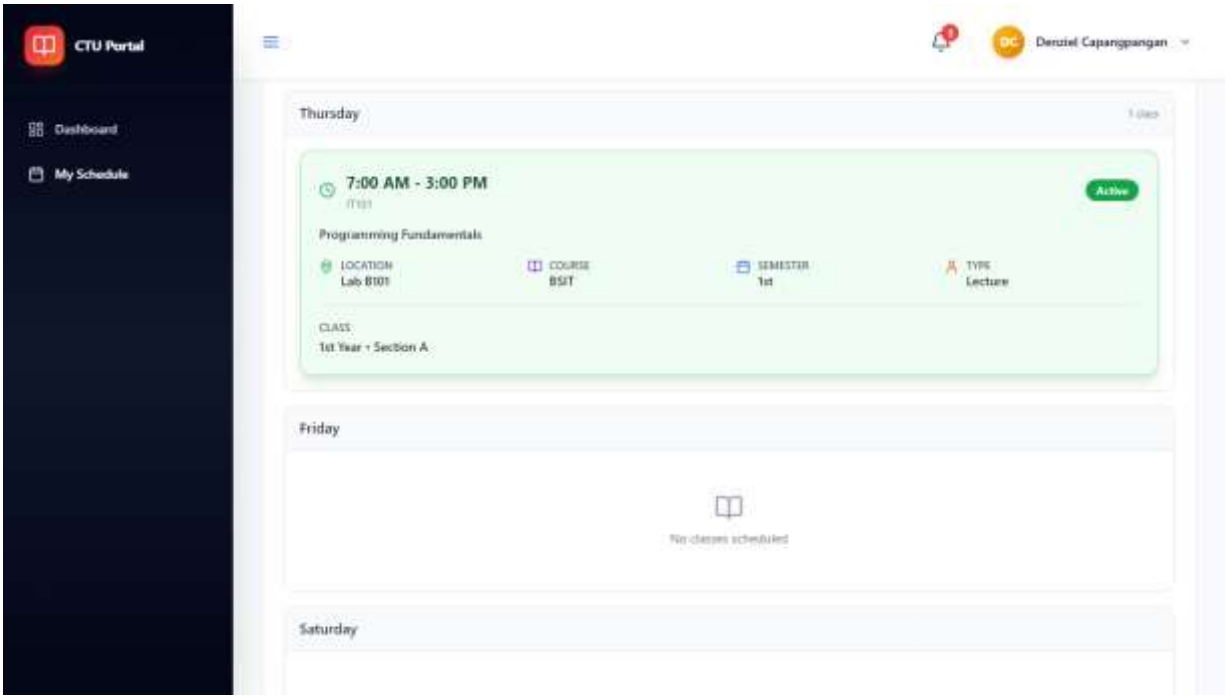

Figure 54: My Schedule (List View - With Class Details)

Figure 54: The figure above shows the **My Schedule** screen displaying detailed information for an assigned class, including the course, time, location, program, semester, type, and class section.

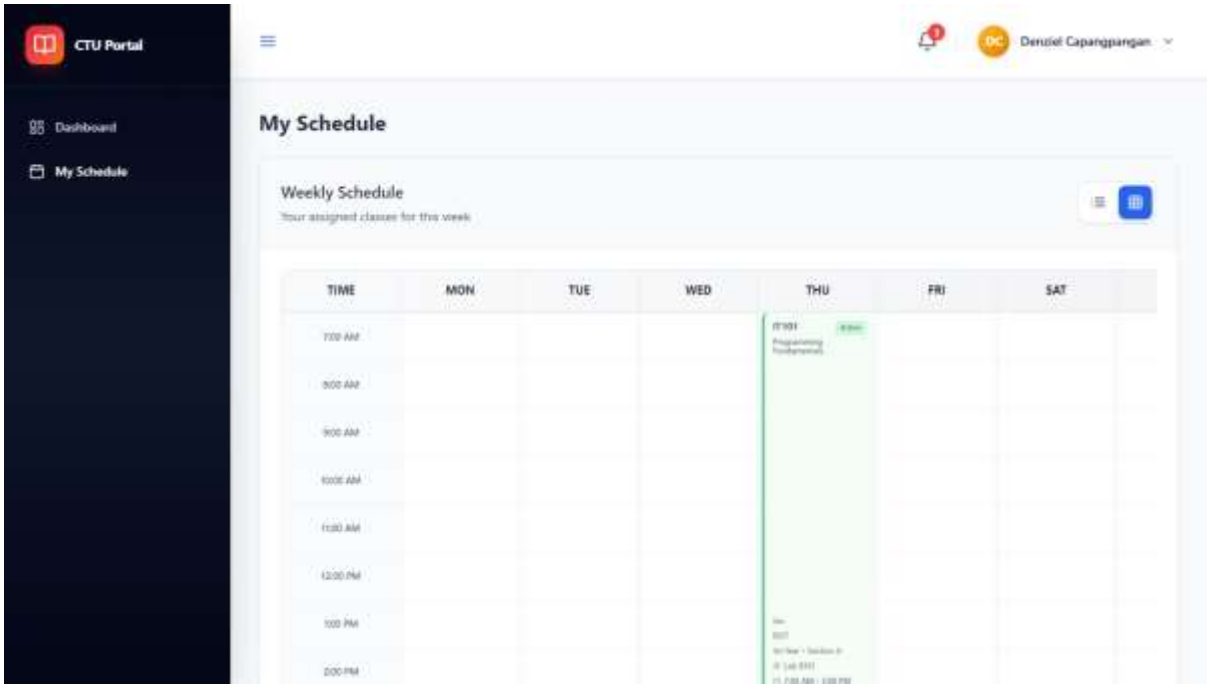

Figure 55**: <u>My Schedule (Grid View)</u>**

Figure 55: The figure above shows the My Schedule screen in a weekly calendar grid view, presenting a visual timeline of assigned classes for the entire week

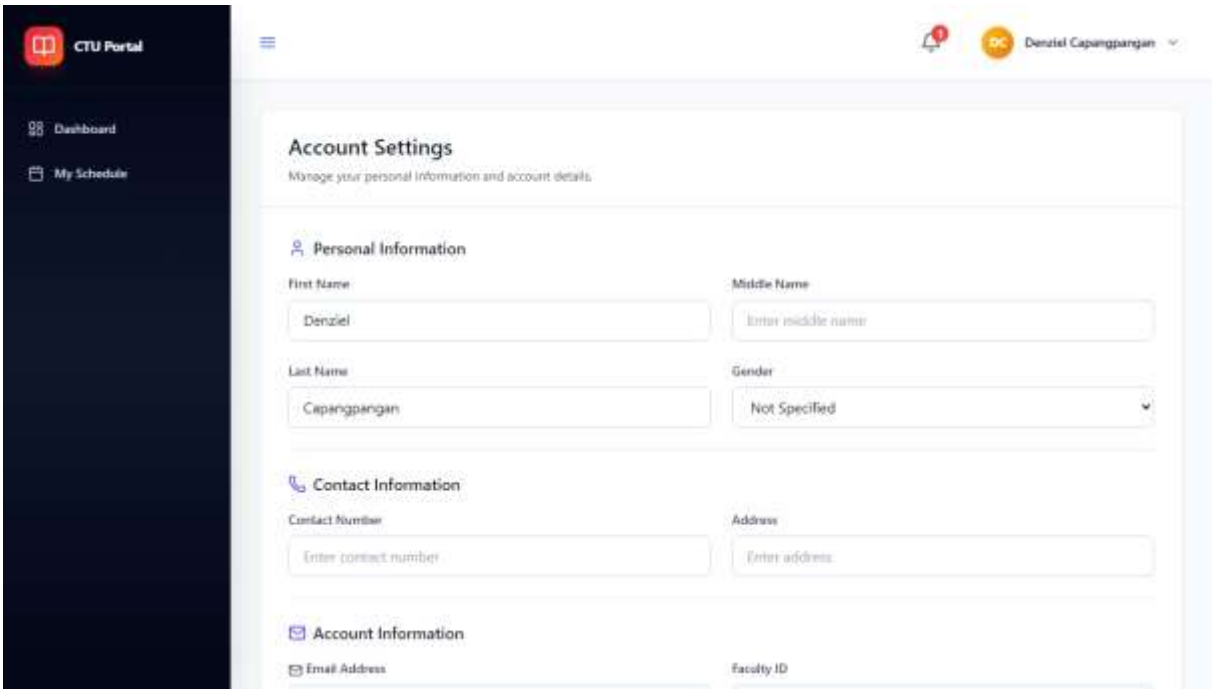

Figure 56**: Account Settings (Faculty Information)**

Figure 56: The figure above shows the Account Settings screen, displaying the faculty member's personal, contact, and account information, including their Email Address, Faculty ID, assigned College and assigned Program.

Dean

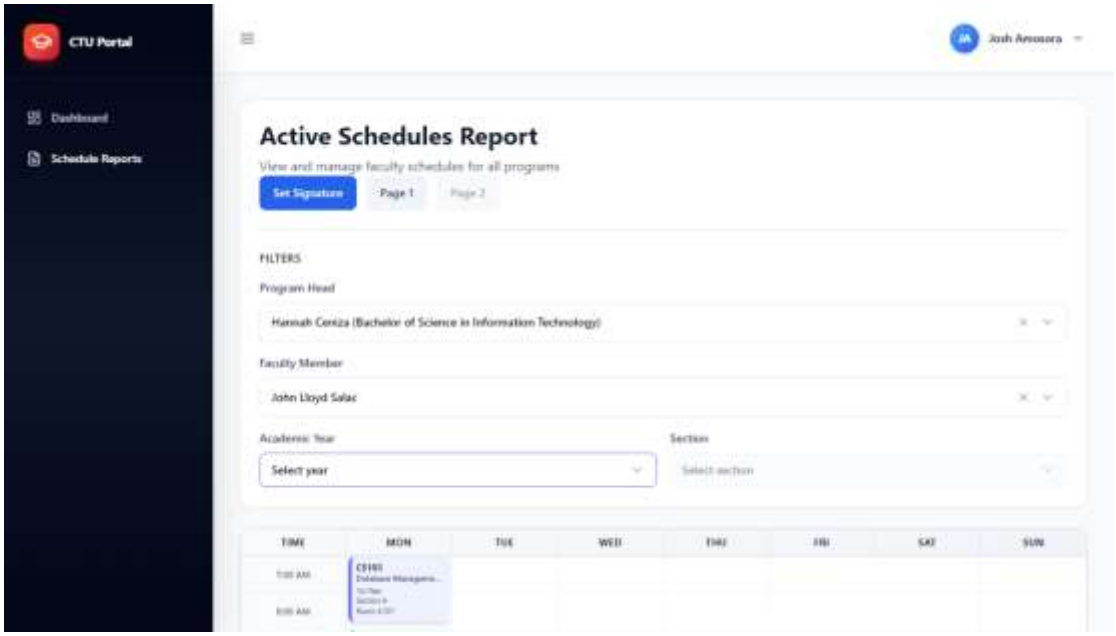

Figure 57**: <u>Active Schedules Report (Dean's View)</u>**

Figure 57: The figure above shows the Active Schedules Report dashboard. The College Dean uses this interface to filter and monitor faculty schedules across various academic programs.

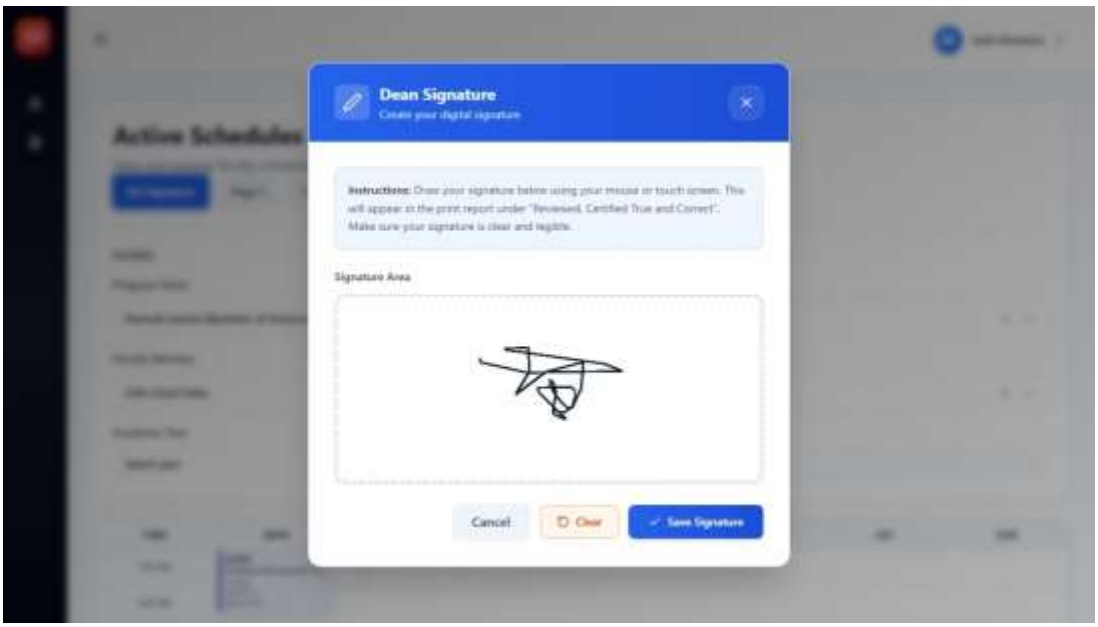

Figure 58**: <u>Digital Signature Setup</u>**

Figure 58: The figure above illustrates the College Dean Signature modal. The Dean draws a digital signature here to be used for the "Reviewed, Certified True and Correct" section of reports

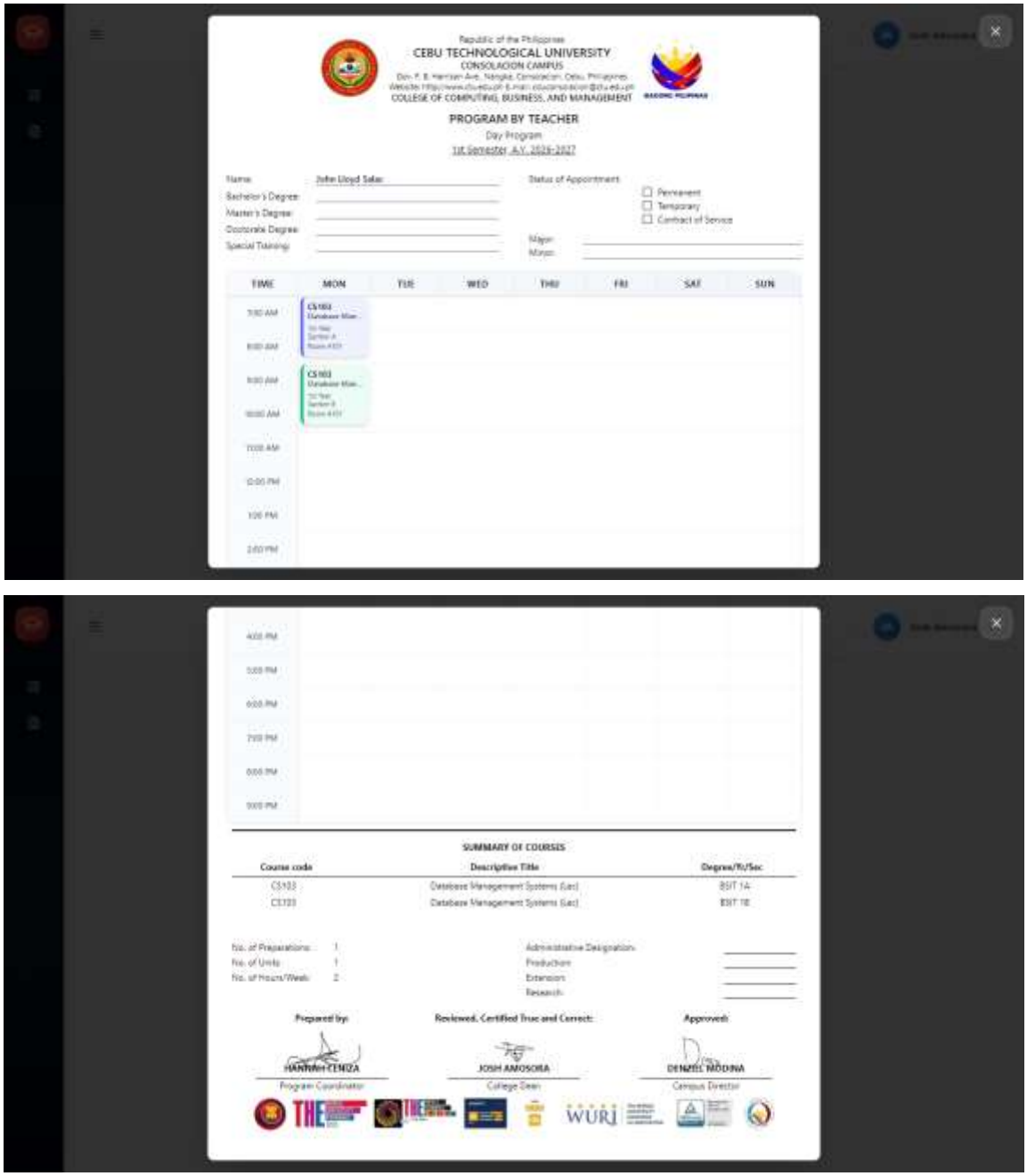

Figure 59**: <u>Teacher Program Review</u>**

Figure 59: The figure above displays the Program by Teacher report. The Dean uses this to verify faculty workloads, including specific courses, units, and total teaching hours.

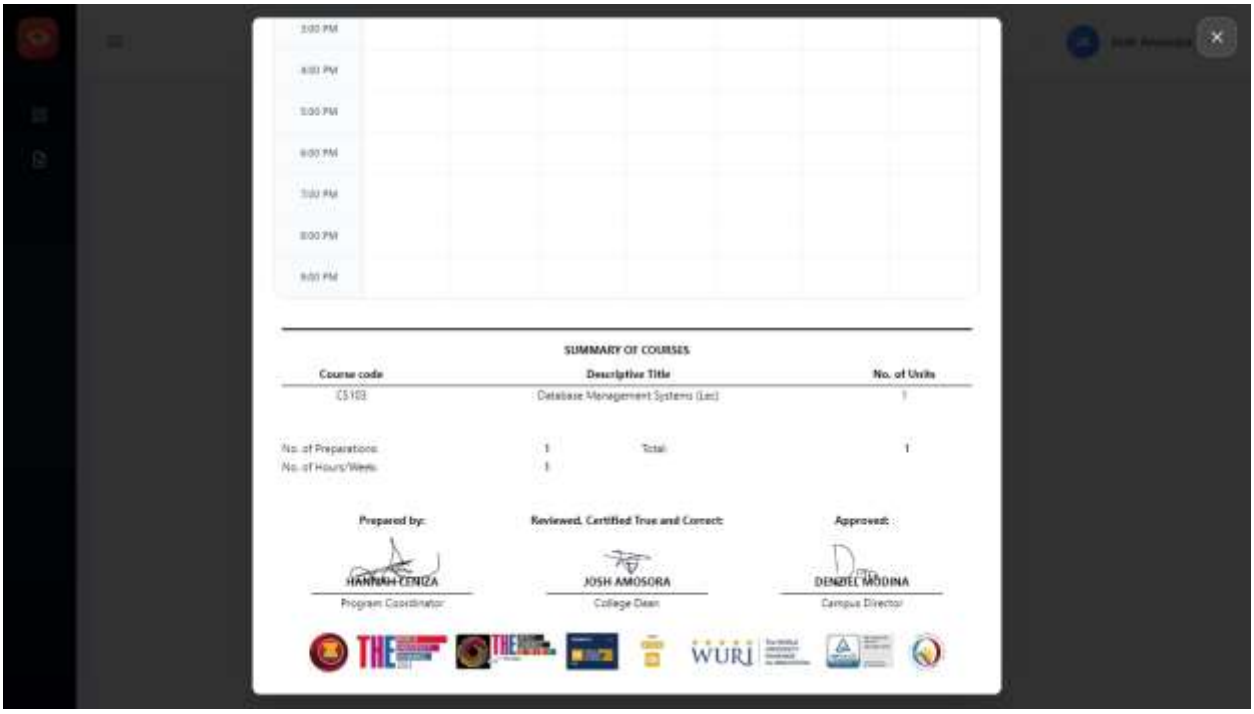

Figure 60**: Section Schedule Oversight**

Figure 60: The figure above shows the Program by Section report. This view allows the Dean to confirm that specific year levels and sections have been assigned an adviser and that their curriculum requirements are being met.

Campus Director

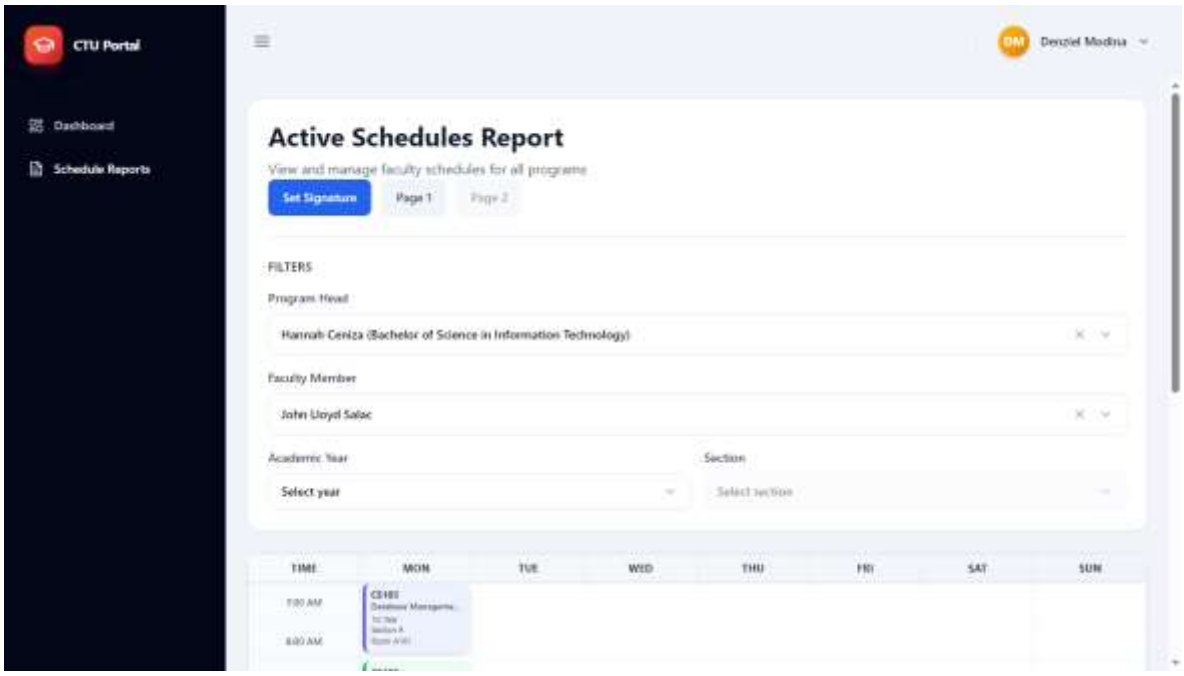

Figure 61**: <u>Active Schedules Report</u>**

Figure 61: The figure above shows the Active Schedules Report dashboard. The Campus Director uses this to oversee faculty schedules for the Bachelor of Science in Information Technology program.

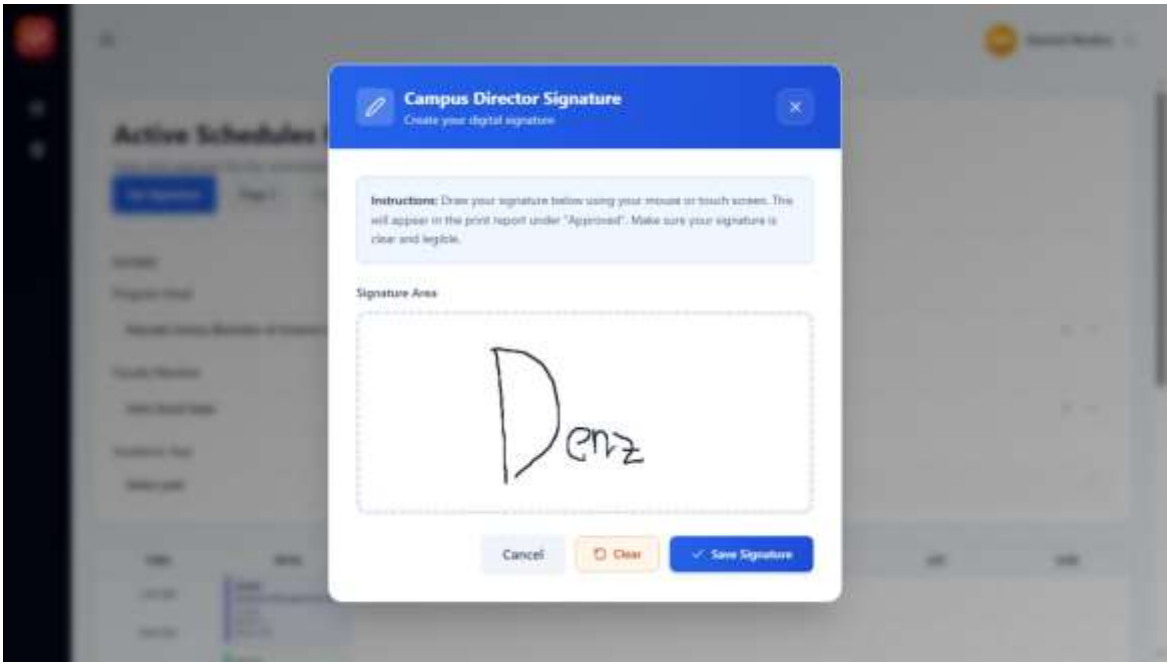

Figure 62**: <u>Digital Signature Setup</u>**

Figure 62: The figure above illustrates the Campus Director Signature modal. The Director draws a digital signature here to be used for the "Approved" section of the official reports.

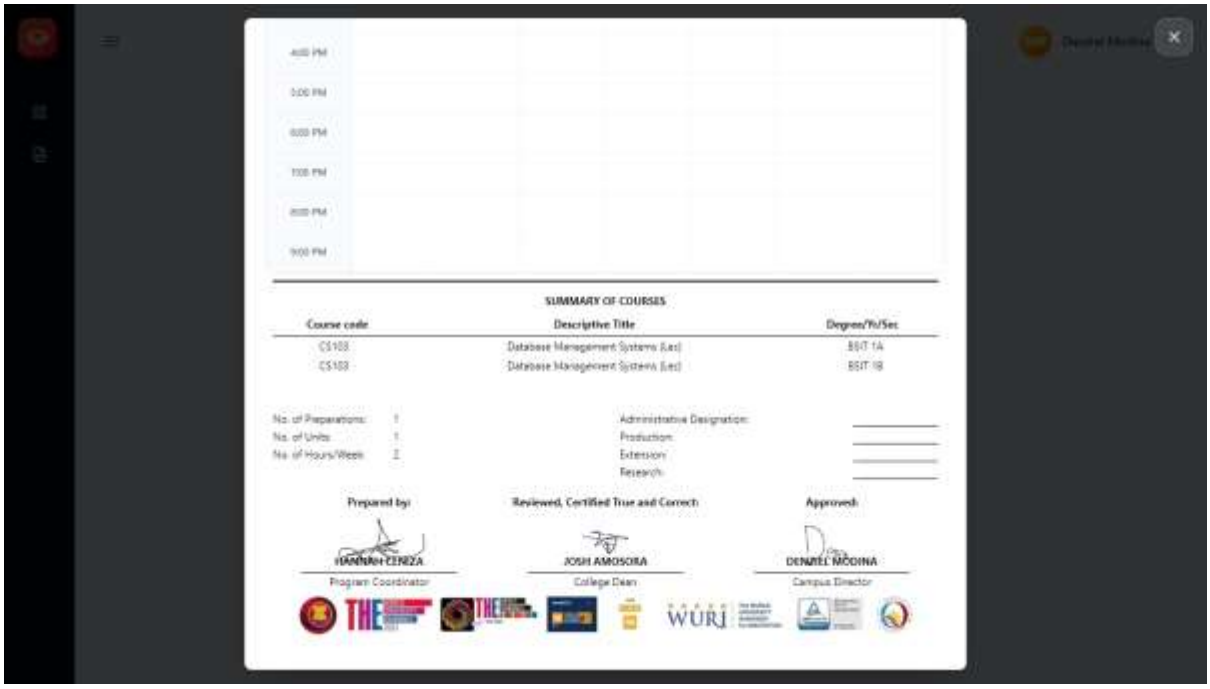

Figure 63**: Teacher Program Review**

Figure 63: The figure above displays the Program by Teacher report. The Director reviews this to verify the faculty's final workload, including course codes, descriptive titles, and total units.

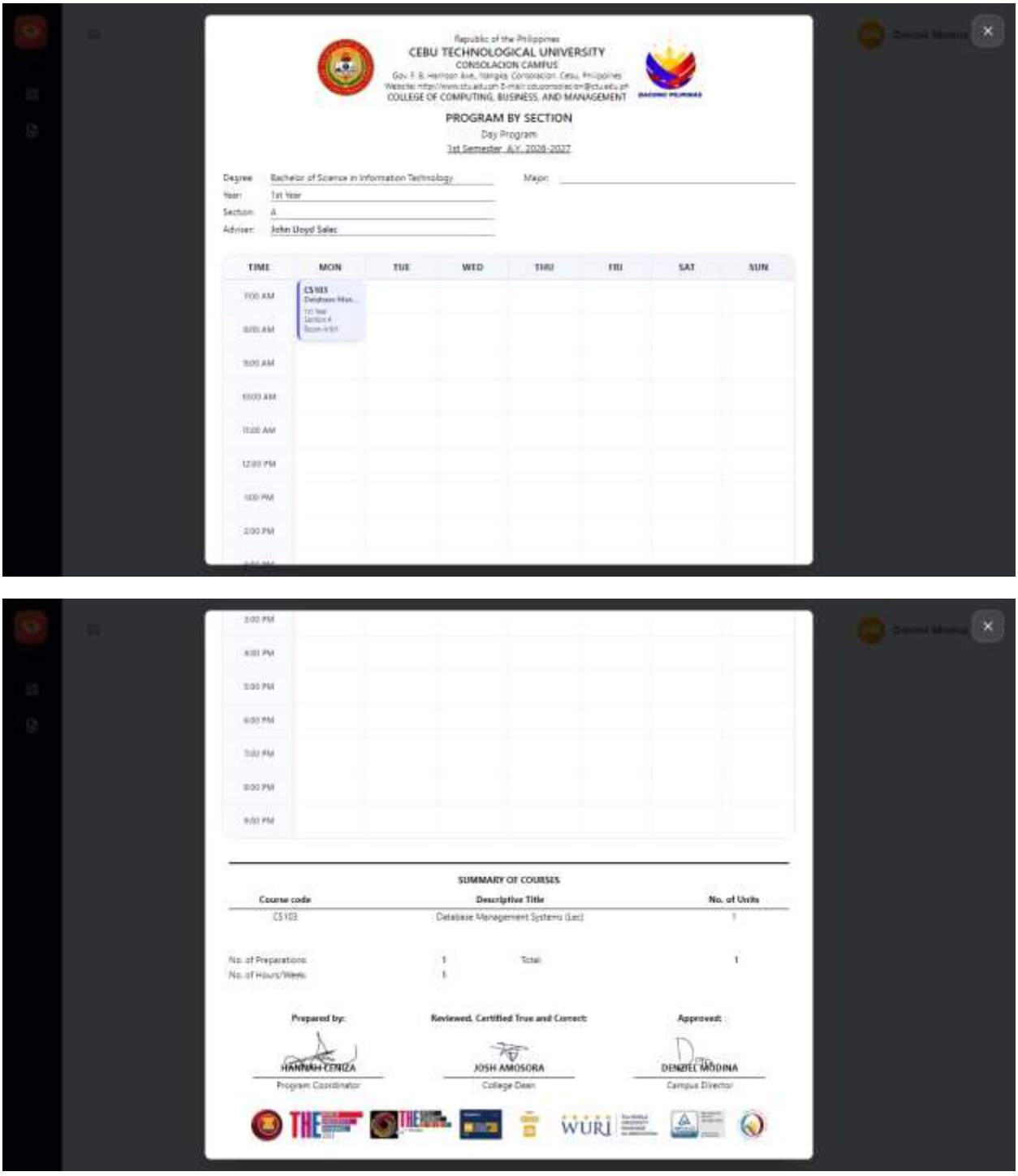

Figure 64**: Section Schedule Oversight**

Figure 64: The figure above shows the Program by Section report. This view allows the Director to confirm that specific sections, such as BSIT 1A, have completed schedules and assigned advisers.

Data Dictionary

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	YES
student_id	uuid	20	YES
auth_id	varchar	--	YES
first_name	varchar	100	NO
middle_name	varchar	100	YES
last_name	varchar	100	NO
email	varchar	225	NO
contact_number	varchar	20	YES
gender	varchar	50	YES
address	text	225	YES
college	varchar	225	YES
program	varchar	255	YES
year_level	varchar	50	YES
section	varchar	100	YES
is_verified	bool	--	YES
status	varchar	50	YES
created_at	timestamp	--	YES
update_at	timestamp	--	YES
is_active	bool	--	YES

Table 4**: Student Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
auth_user_id	uuid	--	YES
first_name	varchar	100	YES
middle_name	varchar	100	NO
last_name	varchar	100	YES
email	varchar	225	NO
contact_number	varchar	20	TES
gender	varchar	50	TES
address	varchar	--	YES
college	varchar	225	YES
program	varchar	225	YES
department	varchar	100	YES
faculty_position	varchar	100	YES
is_verified	bool	--	YES
is_active	bool	--	YES
created_at	timestamp	--	YES
updated_at	timestamp	--	YES
program_id	uuid	--	YES

Table 5**: Faculty Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
auth_user_id	uuid	--	NO
admin_code	varchar	--	NO
first_name	varchar	--	NO
middle_name	varchar	--	YES
last_name	varchar	--	NO
email	varchar	--	NO
gender	varchar	--	YES
contact_number	varchar	--	YES
address	varchar	--	YES
admin_level	varchar	--	YES
is_active	bool	--	YES
create_at	timestamp	--	NO
update_at	timestamp	--	NO

Table 6**: Administrator Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
name	varchar	--	NO
building	varchar	--	YES
room_number	varchar	--	YES
capacity	int	--	YES
type	varchar	--	YES
floor	bool	--	YES
description	int	--	YES
is_available	bool	--	YES
course_id	uuid	--	YES
created_id	timestamp	--	NO
updated_id	timestamp	--	NO
location_code	varchar	--	YES

Table 7: **Location Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
college_id	uuid	--	NO
course_name	text	--	YES
course_code	varchar	--	YES
description	varchar	--	YES
display_order	int	--	YES
created_at	timestamp	--	NO
updated_at	timestamp	--	NO

Table 8**: College Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
course_id	uuid	--	NO
subject_id	uuid	--	NO
offering_type	varchar	--	NO
lecture_units	varchar	--	YES
lab_units	varchar	--	YES
contact_hours	varchar	--	YES
is_active	bool	--	YES
created_at	timestamp	--	NO
update_at	timestamp	--	NO

Table 9**: Course Subject Offering Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
college_id	uuid	--	NO
course_name	varchar	--	NO
course_code	varchar	--	YES
description	varchar	--	YES
display_order	int	--	YES
created_at	timestamp	--	NO
updated_at	timestamp	--	NO

Table 10**: Course Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
recipient_id	uuid	--	NO
recipient_type	varchar	20	NO
notification_type	varchar	50	NO
title	varchar	225	NO
message	varchar	--	NO
related_schedule_id	uuid	--	YES
related_approval_id	uuid	--	YES
is_real	bool	--	YES
read_at	timestamp	--	YES
created_at	timestamp	--	YES
updated_at	timestamp	--	YES
program_id	uuid	--	YES
program_head_id	uuid	--	YES

Table 11**: <u>Notification Table</u>**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
schedule_id	uuid	--	NO
faculty_id	uuid	--	NO
program_head_id	uuid	--	NO
status	varchar	20	YES
faculty_response	varchar	--	YES
faculty_availability	varchar	--	YES
faculty_responded	timestamp	--	YES
program_head_response	varchar	--	YES
program_head_action	timestamp	--	YES
creates_at	timestamp	--	YES
updated_at	timestamp	--	YES
program_id	uuid	--	YES

Table 12**: Schedule Approvals**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
subject_name	varchar	--	NO
subject_code	varchar	--	YES
description	varchar	--	YES
credits	int	--	YES
display_order	int	--	YES
created_at	timestamp	--	NO
updated_at	timestamp	--	NO

Table 13**: Subject Table**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
auth_user_id	varchar	--	NO
program_head_code	varchar	--	NO
first_name	varchar	--	NO
middle_name	varchar	--	YES
last_name	varchar	--	NO
email	varchar	--	NO
college	varchar	--	NO
program	varchar	--	NO
gender	varchar	--	YES
contact_number	varchar	--	YES
address	varchar	--	YES
created_at	timestamp	--	NO
updated_at	timestamp	--	NO
is_active	bool	--	YES

Table 14**: Program Head**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
auth_user_id	uuid	--	NO
campus_director_code	varchar	--	NO
first_name	varchar	--	NO
middle_name	varchar	--	YES
last_name	varchar	--	NO
email	varchar	--	NO
gender	varchar	--	YES
contact_number	varchar	--	YES

address	varchar	--	YES
status	varchar	--	NO
is_active	bool	--	NO
created_at	timestamp	--	NO
updated_at	timestamp	--	NO
signature_data_url	varchar	--	YES

Table 15**: <u>Campus Director</u>**

Field Name	Data Type	Length	Null Characteristic
id	uuid	--	NO
auth_user_id	uuid	--	NO
dean_code	varchar	--	NO
first_name	varchar	--	YES
middle_name	varchar	--	NO
last_name	varchar	--	NO
email	varchar	--	NO
gender	varchar	--	YES
contact_number	varchar	--	YES
address	varchar	--	YES
is_active	bool	--	YES
create_at	timestamp	--	NO
updated_at	timestamp	--	NO
signature_data_url	varchar	--	YES

Table 16**: <u>Dean</u>**

Network Design

Network design refers to the planning and structuring of network connections to enable efficient data exchange among devices. For the CebuTech-Space, the network design ensures reliable access for faculty members and administrators using desktops, and laptops. It supports efficient communication and centralized management of scheduling data.

Network Model

A network model defines how devices communicate and how data flows within a system. The system uses a client–server network model, where users access the CebuTech-Space through the internet. User requests are processed by the server, which interacts with a centralized database to store and retrieve class scheduling information. This model allows centralized control, real-time updates, and secure data management.

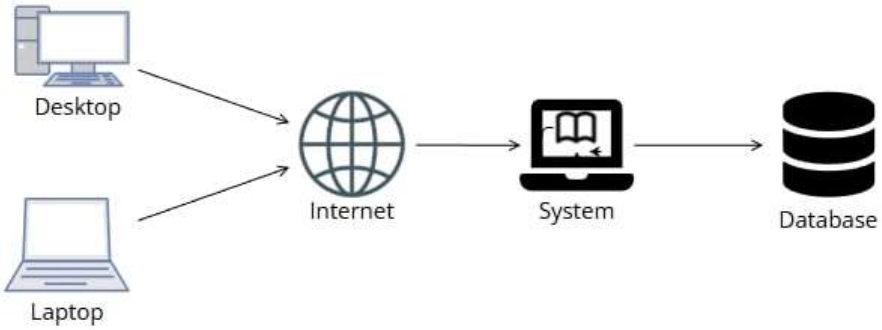

Figure 65**:** **<u>Network Model</u>**

Network Topology

Network topology refers to the arrangement and interconnection of computers, devices, or nodes within a computer network. It defines the network's structure and plays a crucial role in determining how data is shared, accessed, and managed across the system.

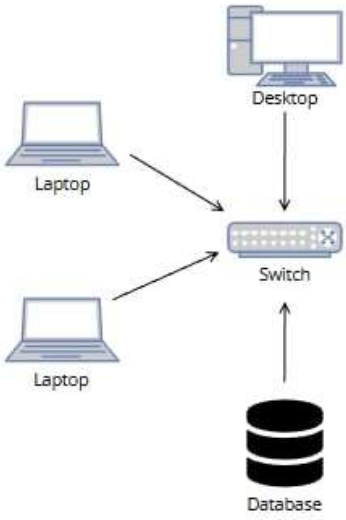

Figure 66: **Network Topology**

For CebuTech-Space, the chosen network topology is the Star Topology, which is ideal for web-based systems utilized by faculty members and administrators. Users can access the system via the internet using devices such as desktops and laptops. In this topology, each device is individually connected to a central switch, which acts as the main hub for communication. The switch, along with a router that provides internet access, manages the flow of data between users and ensures dependable access to the centralized class scheduling system.

Development/Construction/Build Phase

This phase is where the actual coding happens, turning the design into functional software.

Technology Stack Diagram

A Technology Stack Diagram visually depicts the hierarchical arrangement of software components that, when integrated, form the foundation of a solution platform.

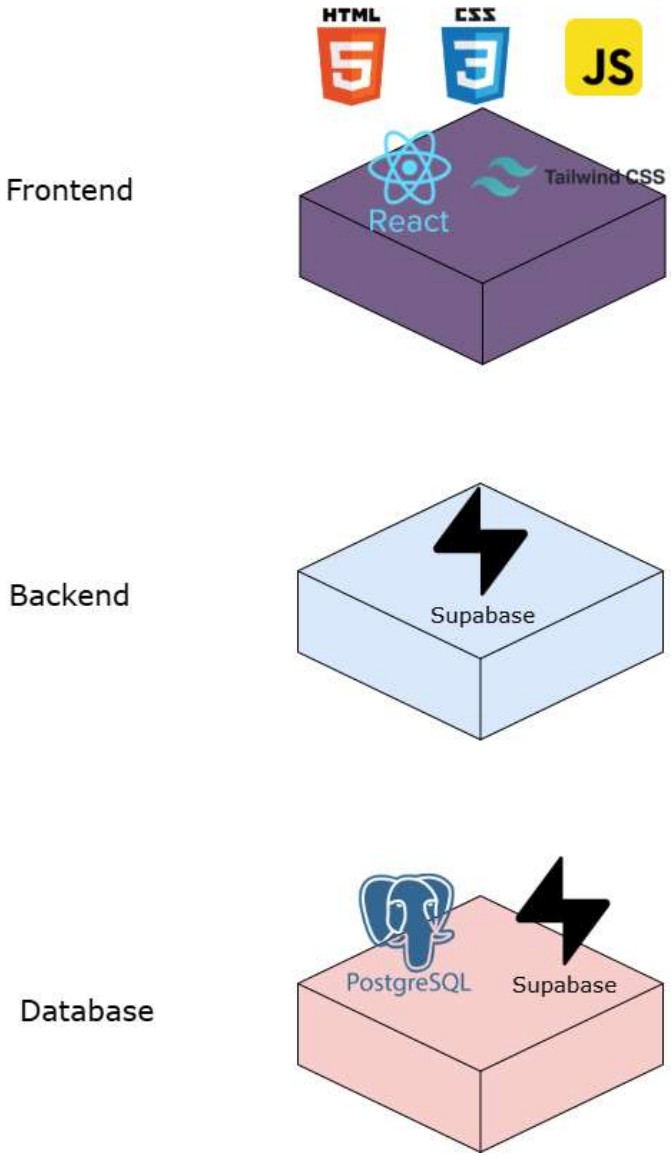

Figure 67: **Technology Stack Diagram**

HTML (HyperText Markup Language) is used to structure the web pages of the Faculty Class Scheduling System. It defines the layout and core elements such as headings, forms, buttons, tables, and other interface components displayed in the browser.

CSS (Cascading Style Sheets) is used to design and style the user interface. In this project, styling is mainly handled through Tailwind CSS, which helps create a responsive, modern, and consistent design across all pages and screen sizes.

JavaScript provides the interactivity and dynamic behavior of the system. It enables actions such as form validation, handling button clicks, showing/hiding modals, and updating the UI without reloading the page.

React.js is the main JavaScript library used to build the user interface. It organizes the system into reusable components (e.g., dashboards, forms, tables) and efficiently updates the UI when data changes.

Supabase is used as the project's Backend-as-a-Service (BaaS). It provides ready-to-use backend features such as authentication, database access, and secured APIs. The frontend communicates with Supabase through the supabase/supabase-js client library over HTTPS.

Supabase Authentication (Auth) manages user login and sessions. It handles sign-in/sign-out, session persistence, and token-based authentication so that only authorized users can access system features.

PostgreSQL (Supabase Database) is the main database used to store and manage system data such as users, colleges, courses, subjects, faculty information, schedules, and approvals.

Row Level Security (RLS) is used to protect data in the database. It ensures that users can only access records they are permitted to see or modify, helping enforce security rules directly at the database level.

Software Specification

Software Specification aims to inform the readers and users about the system and its functions and specifications. It describes the functional and behavioral requirements of the software."

Component	Specification/Details
Operating System	Windows 10/11 (64-bit) (also works on macOS/Linux)
Client Application	Web Browser (Google Chrome / Microsoft Edge / Mozilla Firefox)
Routing	React Router DOM
UI / Styling	Tailwind CSS
Calendar / Scheduling UI	FullCalendar (React)
Development Tools	Visual Studio Code
Programming Language	JavaScript (ES6+)
Frontend Framework/Library	React.js 18
Backend / BaaS	Supabase
Database	PostgreSQL (Supabase Database)
Authentication	Supabase Auth
Runtime / Package Manager	**Node.js (LTS) + npm (or yarn/pnpm)**
Build Tool	**Vite + SWC**

Table 17**: Software Specification**

Hardware Specification

To run the system, the minimum hardware requirements are as follows:

Component	Specifications
Processor	Dual-core 1.6 GHz or higher (Intel i3 / AMD equivalent)
RAM	4 GB (8 GB recommended)
Storage	1 GB free disk space (for browser cache/files)
Display	1366 × 768 minimum (1920 × 1080 recommended)
Internet Connection	Stable broadband connection (at least 5 Mbps recommended)
Peripherals	Keyboard and mouse/trackpad

Table 18**: Hardware Specifications**

Program Specification

A detailed outline guiding the creation of software by specifying its functionalities and requirements.

List of Modules

Systematic breakdown of the different components or sections in software project. Each module focuses on a specific function, making it easier to manage, develop, and maintain the overall system.

1. **Manage Account / Authentication**

Modules	Admin	Program Head	Faculty	Student
Login	*	*	*	*
Logout	*	*	*	*
Registration		*	*	*
Reset Password		*	*	*
Account Settings		*	*	*
Number of points	1	1	1	1

Table 19**: <u>List Of Modules (Manage Account/Authentication)</u>**

2. **Dashboard Module**

Modules	Admin	Program Head	Faculty	Student
Dashboard Overview (summary widgets/cards)	*	*	*	*
Number of points	1	1	1	1

Table 20**: <u>List Of Modules (Dashboard)</u>**

3. College & Program Management

Modules	Admin	Program Head	Faculty	Student
College Management (add/edit/delete colleges)	*			
Program/Course Management (add/edit/delete programs)	*			
Subject Management (add/edit/delete subjects)	*			
Course–Subject Offerings (define subjects per program)	*			
Number of points	1			

Table 21**: <u>List Of Modules (College & Program Management)</u>**

4. User Management

Modules	Admin	Program Head	Faculty	Student
Program Head Management (create/enable/disable)	*			
Faculty Management (create/update/assign/enable/disable)	*	*		
Student Management (create/update/enable/disable)	*			
Number of points	1	1		

Table 22**: <u>List Of Modules (User Management)</u>**

5. Location / Room Management

Modules	Admin	Program Head	Faculty	Student
Location Management (rooms/labs/buildings)	*	*		
Number of points	1	1		

Table 23**: List Of Modules (Location/Room Management)**

6. Scheduling Module

Modules	Admin	Program Head	Faculty	Student
Create Schedule (calendar-based scheduling)		*		
View Schedules (calendar/table)		*	*	*
Update/Edit Schedule		*		
Reset Password		*		
Delete/Cancel Schedule		*		
Number of points		1	1	1

Table 24**: List Of Modules (Scheduling Modules)**

7. Schedule Approval Workflow

Modules	Admin	Program Head	Faculty	Student
Submit Schedule for Approval		*		
Approve / Disapprove Schedule		*	*	
Track Approval Status		*	*	
Number of points		1	1	

Table 25**: List Of Modules (Schedule Approval Workflow)**

8. Notifications Module

Modules	Admin	Program Head	Faculty	Student
View Notifications		*	*	
Schedule-related Notifications		*	*	
Number of points		1	1	

Table 26: **List Of Modules (Notification Module)**

Testing/Quality Assurance Phase

The testing/quality assurance phase explores the system, identifies defects, and meets the expectations of a quality product. This phase includes unit, integration, alpha, and acceptance testing.

Unit Testing

The process of testing individual units or components of a software application to ensure they function correctly. It helps identify and fix errors early in the development phase, promoting code reliability and robustness.

User Testing	Feature to Test	Validation Criteria	Date Tested	Passed or failed	Passed after Retest	Comments
Administrator	Login (Valid Credentials)	Admin can log in using valid email and password	October 12, 2025	1	1	Successfully logged in
Administrator	Login (Invalid Credentials)	Error message must be	October 12, 2025	1	1	Performed as expected

		displayed for invalid login				
Administrator	Logout	Admin session ends and redirects to login page	October 12, 2025	1	1	Successfully logged out
Administrator	College Management (Add College)	Can add a college and it appears in the list	October 16, 2025	1	1	College added successfully
Administrator	College Management (Update College)	Can update college details and changes are saved	October 16, 2025	1	1	Data updated successfully
Administrator	College Management (Delete College)	Can delete a college and it is removed from the list	October 16, 2025	1	1	Deleted successfully
Administrator	Faculty Management (Update faculty)	Can update faculty details and changes are saved	October 18, 2025	1	1	Data successfully saved
Administrator	Faculty Management (Delete Faculty)	Can delete faculty details and changes are saved	October 18, 2025	1	1	Deleted successfully

Administrator	Chairperson Management (Update program head)	Can update program head details and changes are saved	October 19, 2025	1	1	Data successfully saved
Administrator	Chairperson Management (Delete program head)	Can delete program head and it is removed from the list	October 19, 2025	1	1	Delete successfully
Administrator	Student Management (Update Student)	Can update student details and changes are saved	October 20, 2025	1	1	Data successfully saved
Administrator	Student Management (Delete Student)	Can delete a student and it is removed from the list	October 20, 2025	1	1	Student updated / delete successfully
Administrator	Subject Management	Can add /update/delete subject	October 24, 2025	1	1	Data updated successfully
Administrator	Location Management (Add Location)	Can add a room/location and save successfully	October 24, 2025	1	1	Location added successfully

Table 27**: <u>Unit Testing (Admin)</u>**

User Testing	Feature to Test	Validation Criteria	Date Tested	Passed or failed	Passed after Retest	Comments
Program Head	Login (Valid Credentials)	Program Head can log in using valid email/password	October 25, 2025	1	1	Successfully logged in
Program Head	Registration	Account creation succeeds and user can verify email	October 25, 2025	1	1	Registered successfully
Program Head	Login (Invalid Credentials)	Shows error message for invalid credentials	October 25, 2025	1	1	Performed as expected
Program Head	Reset Password	Can request password reset and set a new password	October 25, 2025	1	1	Password reset successfully
Program Head	Scheduling (Create Schedule)	Can create a schedule and save to database	October 27, 2025	1	1	Schedule created successfully
Program Head	Scheduling (Edit Schedule)	Can edit an existing schedule and updates persist	October 27, 2025	1	1	Updated successfully
Program Head	Notifications	Notifications load and redirect to correct section	October 30, 2025	1	1	Navigation works correctly

Table 28**: <u>Unit Testing (Program) Head</u>**

User Testing	Feature to Test	Validation Criteria	Date Tested	Passed or failed	Passed after Retest	Comments
Faculty	Login (Valid Credentials)	Faculty can log in successfully	October 30, 2025	1	1	Successfully logged in
Faculty	Registration	Account creation succeeds and user can verify email	October 30, 2025	1	1	Registered successfully
Faculty	Login (Invalid Credentials)	Shows error message for invalid credentials	October 30, 2025	1	1	Performed as expected
Faculty	View Schedule	Faculty schedule loads and displays correctly	October 30, 2025	1	1	Schedule displayed successfully
Faculty	Approve Schedule	Can approve schedule request and status updates	October 31, 2025	1	1	Approved successfully
Faculty	Disapprove Schedule	Can disapprove schedule	October 31, 2025	1	1	Disapproved successfully

		request with correct status				
Faculty	Account Settings	Can update settings/profile details	November 1, 2025	1	1	Updated successfully
Faculty	Logout	Logs out and redirects to login page	November 2, 2025	1	1	Successfully logged out

Table 29**: Unit Testing (Faculty)**

User Testing	Feature to Test	Validation Criteria	Date Tested	Passed or failed	Passed after Retest	Comments
Student	Login (Valid Credentials)	Student can log in successfully	November 2, 2025	1	1	Successfully logged in
Student	Registration	Student can create an account successfully	November 2, 2025	1	1	Registered successfully
Student	Login (Invalid Credentials)	Shows error message for invalid credentials	November 2, 2025	1	1	Performed as expected
Student	View Schedule	Student schedule loads and displays correctly	November 3, 2025	1	1	Schedule displayed successfully
Student	Notifications (if enabled)	Notifications appear and can be opened	November 3, 2025	1	1	Schedule displayed successfully
Student	Approve Schedule	Can approve schedule request and status updates	November 5, 2025	1	1	Approved successfully
Student	Disapprove Schedule	Can disapprove schedule request with correct status	November 5, 2025	1	1	Disapproved successfully

Student	Account Settings	Can update settings/profile details	November 5, 2025	1	1	Updated successfully
Student	Logout	Logs out and redirects to login page	November 6, 2025	1	1	Successfully logged out

Table 30**: <u>Unit Testing (Student)</u>**

Alpha Testing (Faculty Class Scheduling System)

Alpha testing is the initial phase of software testing where the application is tested internally by the development team before releasing it to a wider audience. It helps uncover bugs, gather feedback, and refine the system for a more stable release.

Test Criteria (GUI)	**Poor**	**Fair**	**Good**	**Very Good**
Consistency (The interface maintains consistent formatting, layout, and uses the same icons throughout the system.)				
Reusability (The system uses reusable UI components such as buttons, forms, tables, and dialogs across pages.)				
Forgiveness and Tolerance (The interface shows confirmations/error messages and prevents critical mistakes when possible.)				
Simplicity (The GUI uses simple navigation and avoids overcrowded screens.)				

Readability (Text size, font style, spacing, and contrast are comfortable for users.)				
Clarity (Labels, helper texts, and messages are clear and easy to understand.)				
User-friendliness (The design is user-friendly and supports smooth navigation for Admin/Program Head/Faculty/Student.)				

Table 31**: Alpha Testing**

Acceptance Testing (Faculty Class Scheduling System)

Acceptance testing is a user-conducted testing activity performed to evaluate the system based onrequired attributes, ensure compliance with specifications, confirm stability, and identify any remaining issues before deployment.

Instruction: Please check/mark only one rating for each attribute.

Attribute being evaluated	**Please check/mark only one rating for each attribute.**
Functionality: (Does the system meet all the required functionalities such as login, managing data, scheduling, approvals, and reports?)	5 4 3 2 1 Comment:
Robustness: (Does the system operate without crashing, and recover properly from invalid inputs or user mistakes?)	5 4 3 2 1 Comment:
Workmanship: (Does the system's interface reflect quality in design, layout consistency, and usability?)	5 4 3 2 1 Comment:
Overall: (General recommendation based on the user's evaluation.)	5 4 3 2 1 Comment:

Table 32**: <u>Unit Testing (Student)</u>**

Legend**:** 5 – Excellent 4 – Good 3 – Good 2 – Fair 1 – Poor

Chapter IV

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

CebuTech-Space provides a practical solution to the existing manual faculty class scheduling process at Cebu Technological University – Consolacion Campus. By utilizing web-based technology and an automated scheduling approach, the system improves how academic schedules are created, managed, and distributed. The system enables centralized access to schedules for different user roles, including Students, Faculty, Administrators, Program Heads, Deans, and the Campus Director.

Findings

Based on the data gathered from interviews and surveys, the researchers found that manual scheduling is time-consuming, prone to errors, and often results in schedule conflicts, unbalanced faculty workloads, and delayed dissemination of schedule updates. Faculty members and students also experience difficulty accessing accurate and timely schedule information. The absence of a centralized system contributes to miscommunication and inefficiency. To analyze the collected data and guide system development, the researchers employed the Agile methodology.

Conclusion

The researchers conclude that CebuTech-Space is an effective and beneficial tool for managing faculty class schedules. The system allows Program Heads and Administrators to efficiently create and update schedules while minimizing conflicts and ensuring proper workload distribution. Faculty members and students benefit from real-time access to accurate schedule information, improving communication and academic preparedness. Overall, CebuTech-Space enhances operational efficiency, supports better decision-making, and contributes to a more organized academic environment at Cebu Technological University – Consolacion Campus.

Recommendations

The researchers recommend further enhancements to improve the functionality and user experience of the CebuTech-Space system. Future developments may include mobile or responsive access to allow users to view schedules on smartphones and tablets. Integration with other institutional systems such as enrollment or attendance systems may also be considered. Additionally, implementing advanced analytics and automated optimization features could further assist administrators in long-term academic planning and resource management.

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